SFU SIMON FRASER UNIVERSITY FACULTY OF APPLIED SCIENCES

MSE 491 – Electrified Transportation Systems – Spring 2026

- Final Project Report -

Design and Evaluation of a Field Oriented Controlled (FOC) Induction Machine Drive

Abstract

This report details the design, implementation, and comprehensive performance evaluation of a Field-Oriented Control (FOC) system for a 2.24 kW three-phase induction machine drive, a vital architecture in electrified transportation systems. Modeled entirely in MATLAB/Simulink, the open-loop FOC drive incorporates a delta-modulated Voltage Source Inverter (VSI), analytical torque and flux estimators, and a quadratic fan-type mechanical load. The study first investigates the switching characteristics of the delta modulator, quantifying the fundamental engineering trade-off between hysteresis band limits, stator current ripple amplitude, and variable switching frequency. Subsequently, the drive's dynamic and steady-state performance is validated across multiple operating scenarios, including nominal rated conditions, flux weakening, and partial torque commands. Simulation results demonstrate the FOC algorithm's ability to successfully decouple torque- and flux-producing currents, yielding exceptionally rapid dynamic torque step responses and near-zero average steady-state tracking error. Furthermore, power quality and conversion efficiency are assessed through Total Harmonic Distortion (THD) and power flow analyses, particularly under a sudden 3 Nm load disturbance. Ultimately, this project validates the FOC induction motor drive's robust transient response, accurate current tracking, and overall efficacy for high-performance variable-speed applications.

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Introduction

Induction machines are widely utilized in various industrial and automotive applications, particularly within electrified transportation systems, due to their robust construction, high reliability, and cost-effectiveness. However, achieving high-performance dynamic control of an alternating current (AC) induction motor—comparable to the straightforward control of a direct current (DC) machine—requires advanced control strategies. Field-Oriented Control (FOC) serves this exact purpose by mathematically decoupling the electromagnetic torque and rotor flux. This decoupling allows the drive to independently and instantaneously regulate the torque- and flux-producing current components, resulting in superior dynamic response and precision.

This project focuses on the design, implementation, and rigorous evaluation of an open-loop FOC induction machine drive. Developed entirely within the MATLAB/Simulink environment , the simulated drive system is modeled around a 2.24 kW, 220 V, 60 Hz four-pole induction motor. The primary objectives of this study are threefold: to thoroughly understand the operating principles of FOC for induction machines used in electric vehicle applications , to design the necessary system configurations and control parameters , and to simulate and analyze the overall performance of the implemented drive.

To achieve these goals, the project is systematically divided into distinct analytical phases. Part A covers the foundational system design, including the implementation of a delta-modulated Voltage-Source Inverter (VSI), current reference generators, flux and torque estimators, and a quadratic fan-type mechanical load model. Part B evaluates the drive's performance across various steady-state and transient operating scenarios—such as rated flux with partial torque, and flux-weakening conditions. This section analyzes critical metrics including dynamic current tracking, torque step responses, Total Harmonic Distortion (THD), and electromechanical power flow. Finally, advanced analyses involving modulator switching characteristics and slip frequency relationships are explored to provide a comprehensive, system-level assessment of the FOC drive.

System Specifications

Table 1: System specifications of a field oriented controlled induction machine

Parameter	Symbol	Value	Unit
Rated Power	P_r	2.24	kW
Rated Voltage (line-line RMS)	V_r	220	V
Number of Poles	p	4	–
Rated Frequency	f_s	60	Hz
Rated Speed	n_r	1710	rpm
Rated Torque	T_r	12.5	N·m
Rated Current	I_r	8.2	A
Stator Resistance	r_s	0.435	Ω
Stator Reactance	X_{ls}	0.754	Ω
Magnetizing	X_M	26.13	Ω

Reactance			
Rotor Reactance	X_{lr}	0.754	Ω
Rotor Resistance	r_r	0.816	Ω
Moment of Inertia	J	0.0089	kg·m^2

Nomenclature and Abbreviations

Table 2: Symbols / Variables

Symbol	Description
$i_d,\ i_q$	Stator current components in dq frame
i_f^*	Flux-producing current reference
i_T^*	Torque-producing current reference
i_s^*	Stator current reference magnitude
θ_s	Synchronous reference frame angle
θ_T	Torque angle
T_e	Electromagnetic torque

T_e^*	Reference electromagnetic torque
λ_r	Rotor flux magnitude
λ_r^*	Reference rotor flux
$v_a,\ v_b,\ v_c$	Three-phase stator voltages
$i_a,\ i_b,\ i_c$	Three-phase stator currents
ω_r	Rotor speed (rad/s)
n	Rotor speed (rpm)
p	Number of poles
J	Moment of inertia
$r_s,\ r_r$	Stator and rotor resistance
$X_{ls},\ X_{lr},\ X_m$	Reactances
V_{dc}	DC bus voltage
$g_1,\ g_3,\ g_5$	Switching signals of inverter
$v_{an},\ v_{bn},\ v_{cn}$	Phase voltages

Table 3: Block Diagram Nomenclature & Abbreviations

Abbreviation	Description
FOC	Field-Oriented Control
VSI	Voltage Source Inverter
DM	Delta Modulator
TCG	Torque Current Generator
FCG	Flux Current Generator
RES	Resolver Block
CRG	Current Reference Generator
FTE	Flux & Torque Estimator
abc–dq	3-phase to dq transformation
dq–abc	dq to 3-phase transformation
IM	Induction Machine
TL	Load Torque
PWM / DM	Switching strategy
i_T^*	Torque current reference output
i_f^*	Flux current reference output

Theoretical Analysis

The primary objective of Field-Oriented Control (FOC) is to govern an alternating current (AC) induction machine with the same decoupled precision as a separately excited direct current (DC) machine. In a DC machine, the field flux and armature current are orthogonal, allowing torque and flux to be controlled independently. FOC achieves this in an induction machine by transforming the three-phase stator currents into two orthogonal components in a synchronously rotating reference frame: a flux-producing current and a torque-producing current.

Based on the open-loop FOC architecture implemented in this project, the system operates through several distinct mathematical stages:

1. Flux Current Generator

The Flux Current Generator block converts the rotor flux reference λ_r^* into the corresponding flux producing current reference i_f^*.

From the rotor flux relation:

$$\lambda_r = L_m i_f \qquad \text{eq.(1)}$$

the flux current reference is obtained as:

$$i_f^* = \frac{\lambda_r^*}{L_m} \qquad \text{eq.(2)}$$

Then the signal is then converted from peak value to RMS value using:

$$i_{f,\mathrm{rms}}^* = \frac{i_{f,\mathrm{peak}}^*}{\sqrt{2}} \qquad \text{eq.(3)}$$

2. Torque Current Generator

The Torque Current Generator block computes the torque-producing current reference i_{qs}^* based on the desired electromagnetic torque T_e^* and the estimated rotor flux magnitude λ_r.

From the induction motor torque equation,

$$T_e = \frac{3}{2}\frac{P}{2}L_m\left(i_{qs}i_{dr} - i_{ds}i_{qr}\right) \qquad \text{eq.(4)}$$

Under field-oriented control (FOC), the rotor flux is aligned with the d-axis, such that:

$$i_{dr} \approx \frac{\lambda_r}{L_r}, \quad i_{qr} \approx 0 \qquad \text{eq.(5)}$$

Thus, the torque expression simplifies to:

$$T_e = \frac{3}{2}\frac{P}{2}\frac{L_m}{L_r}\lambda_r i_{qs} \qquad \text{eq.(6)}$$

Solving for the torque current reference:

$$i_{qs}^* = \frac{T_e^*}{K\lambda_r}, \quad K = \frac{3}{2}\frac{P}{2}\frac{L_m}{L_r} \qquad \text{eq.(7)}$$

3. Resolver Block

Resolver block combines the flux-producing current reference i_f^* and the torque-producing current reference i_t^* into a single stator current vector representation. This vector is expressed in terms of its magnitude and angle in the synchronous reference frame. The outputs are:

- Stator current magnitude:

$$I_s^* = \sqrt{(i_f^*)^2 + (i_t^*)^2} \qquad \text{eq.(8)}$$

- Torque angle:

$$\theta_T^* = \tan^{-1}\left(\frac{i_t^*}{i_f^*}\right) \quad \text{eq.(9)}$$

4. Reference Current Generator

The Reference Current Generator block converts the stator current magnitude I_s^* and the synchronous angle θ_s^* into three-phase current references $i_a^*,\ i_b^*,\ i_c^*$

The dq current components are transformed into three-phase currents using the transformation:

$$\begin{bmatrix} i_{as}^* \\ i_{bs}^* \\ i_{cs}^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} I_s^* \sin\theta_s^* \\ I_s^* \cos\theta_s^* \end{bmatrix} \quad \text{eq.(10)}$$

5. Three Phase VSI (Voltage Source Inverter)

The Three-Phase VSI block converts DC voltage V_{dc} into three-phase AC voltages $V_{dc},\ v_{an},\ v_{bn},\ v_{cn}$ using witching signals $g_1,\ g_3,\ g_5$.

The inverter generates the following voltages:

$$\begin{aligned} v_{aN} &= g_1 V_{dc} \\ v_{bN} &= g_3 V_{dc} \\ v_{cN} &= g_5 V_{dc} \end{aligned} \quad \text{eq.(11)}$$

The neutral point voltage is calculated as:

$$v_{nN} = \frac{v_{aN} + v_{bN} + v_{cN}}{3} \quad \text{eq.(12)}$$

Therefore, the actual three-phase voltages applied to the motor are:

$$\begin{aligned} v_{an} &= v_{aN} - v_{nN} \\ v_{bn} &= v_{bN} - v_{nN} \\ v_{cn} &= v_{cN} - v_{nN} \end{aligned} \quad \text{eq.(13)}$$

6. abc-to-dq Transformation Block

The 3–2 transformation converts three-phase quantities $(x_a,\ x_b,\ x_c)$ into two orthogonal components in the stationary reference frame $(x_{qs},\ x_{ds})$

The 3–2 transformation can be compactly expressed in matrix form as:

$$\begin{bmatrix} x_{qs} \\ x_{ds} \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} \qquad \text{eq.(14)}$$

7. dq-to-abc Transformation Block

The 2–3 transformation converts the stationary two-axis components $(x_{qs},\ x_{ds})$ back into three-phase quantities $(x_a,\ x_b,\ x_c)$.

The 2-3 transformation can be compactly expressed in matrix form as:

$$\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} x_{qs} \\ x_{ds} \end{bmatrix} \qquad \text{eq.(15)}$$

8. Flux and Torque Estimator

The Flux & Torque Estimator block computes the rotor flux magnitude λ_r, flux angle θ_r, and electromagnetic torque T_e using measured stator voltages and currents.

This Estimator block includes the following sub-blocks:

- **abc to dq Transformation (voltage):** The abc to dq transformation converts three-phase stator voltages $(v_a,\ v_b,\ v_c)$, into stationary frame components $(v_{qs},\ v_{ds})$. The 3-2 voltage transformation can be compactly expressed in matrix form as:

$$\begin{bmatrix} v_{qs} \\ v_{ds} \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \qquad \text{eq.(16)}$$

- **abc to dq Transformation (current):** The abc to dq transformation converts three-phase stator currents $(i_a,\ i_b,\ i_c)$ into stationary frame components

$(i_{qs},\ i_{ds})$. The 3-2 current transformation can be compactly expressed in matrix form as:

$$\begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \qquad \text{eq.(17)}$$

- **Rotor Current Calculation:** The Rotor Current Calculation block estimates the rotor currents i_{qr} and i_{dr} using stator voltages and currents based on the induction motor dynamic model.

  Governing equation:

$$\begin{aligned} i_{qr} &= \frac{1}{L_m} \int (v_{qs} - r_s i_{qs})\, dt \;-\; \frac{L_{ss}}{L_m} i_{qs} \\ i_{dr} &= \frac{1}{L_m} \int (v_{ds} - r_s i_{ds})\, dt \;-\; \frac{L_{ss}}{L_m} i_{ds} \end{aligned} \qquad \text{eq.(18)}$$

- **Rotor Flux Estimator:** The Flux Calculation block computes the rotor flux components λ_{qr} and λ_{dr} using the estimated rotor currents and stator currents.

  Governing equation:

$$\begin{aligned} \lambda_{qr} &= L_r i_{qr} + L_m i_{qs} \\ \lambda_{dr} &= L_r i_{dr} + L_m i_{ds} \end{aligned} \qquad \text{eq.(19)}$$

  Therefore the flux can be estimated as:

$$\lambda_r = \sqrt{\lambda_{qr}^2 + \lambda_{dr}^2} \qquad \text{eq.(20)}$$

$$\theta_r = \tan^{-1}\left(\frac{\lambda_{qr}}{\lambda_{dr}}\right) \qquad \text{eq.(21)}$$

- Rotor Torque Estimator: The Rotor Torque Estimator block computes the electromagnetic torque T_e using the dq-axis current components.

  Governing equation:

$$T_e = \frac{3}{2}\frac{P}{2} L_m \left(i_{qs} i_{dr} - i_{ds} i_{qr}\right) \qquad \text{eq.(4)}$$

Calculations, Simulation results and Discussions/explanations

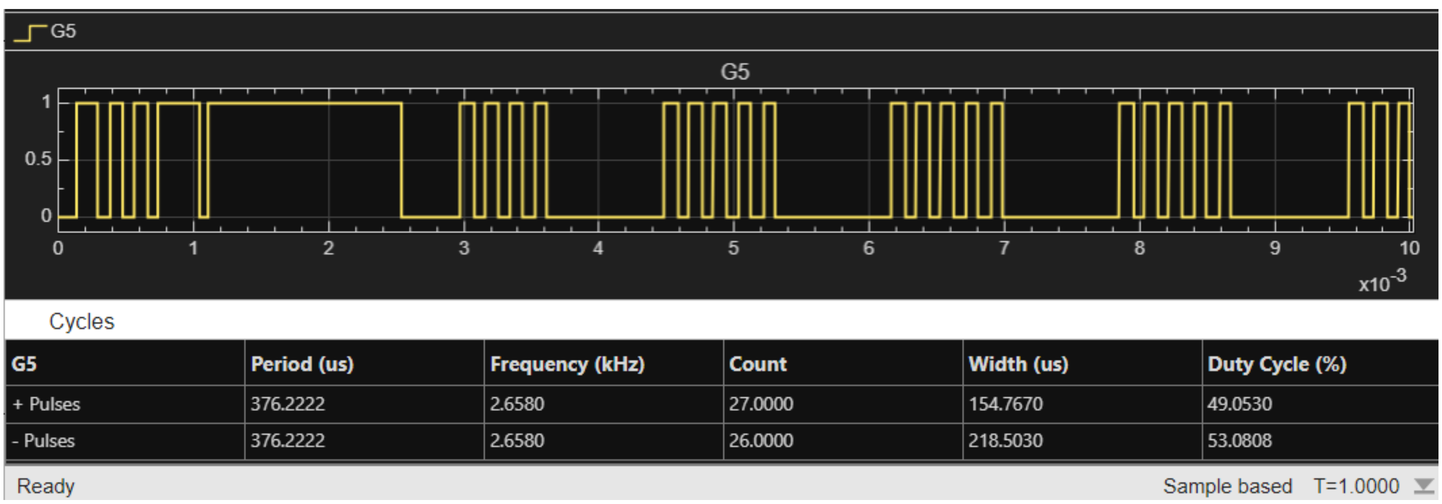
Part A System design and implementation

A.1.1 Switching Behavior of Delta Modulator

- Analyze the switching behavior of the delta modulator by measuring the effective switching frequency under different hysteresis bands. Investigate how the hysteresis band width affects switching frequency and current tracking performance.

To analyze the switching behavior of the delta modulator, the effective switching frequency was measured under three different hysteresis band (h) configurations: a baseline band (h = 0.5), a tight band (h = 0.1), and a loose band (h = 1.0). The purpose of these tests is to investigate how the hysteresis band width affects both the switching frequency and the overall current tracking performance of the drive.

Test 1: Baseline Band ($h = I_{rated} * 0.5$)



G5	Period (us)	Frequency (kHz)	Count	Width (us)	Duty Cycle (%)
+ Pulses	376.2222	2.6580	27.0000	154.7670	49.0530
- Pulses	376.2222	2.6580	26.0000	218.5030	53.0808

Figure 1: Waveform of Gating's Scope (G5) Baseline Band case (h=0.5)

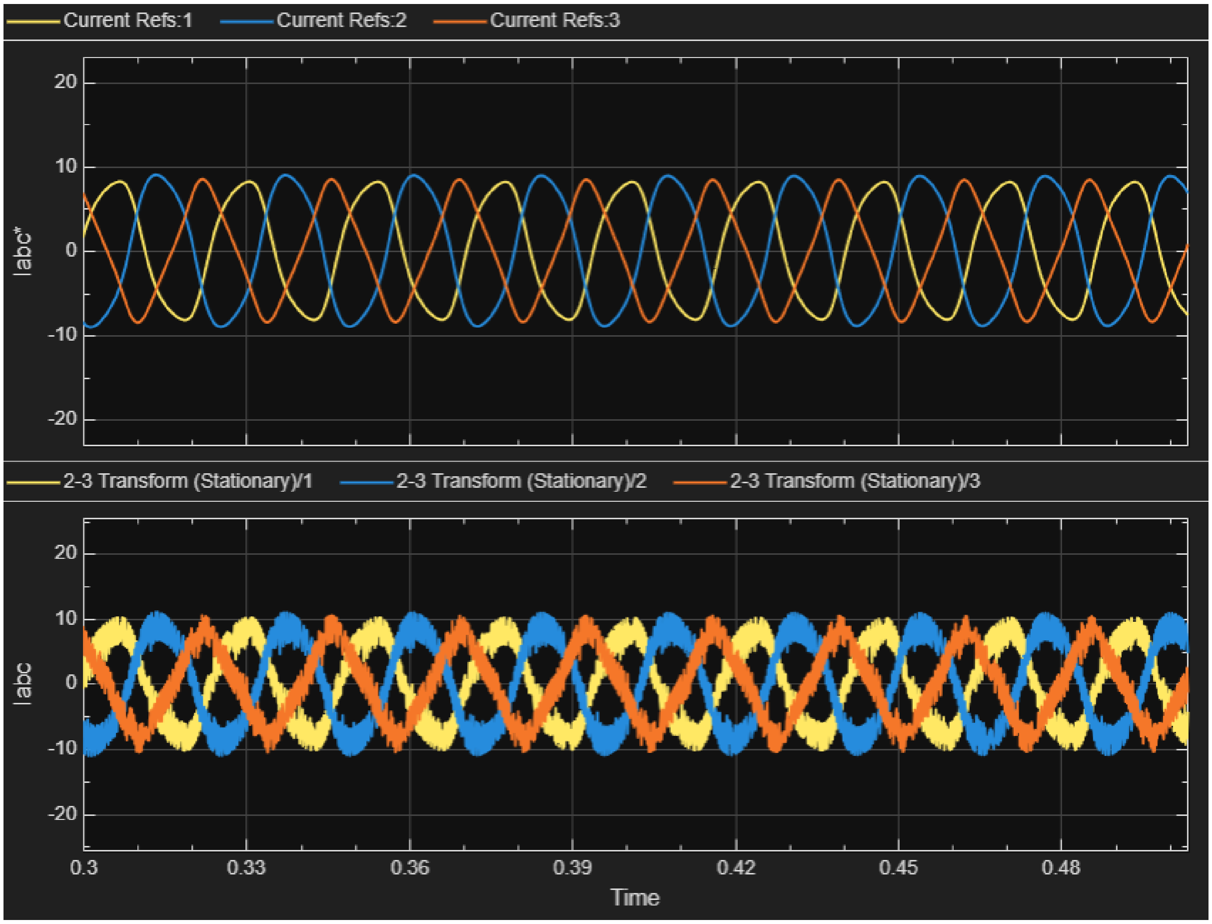
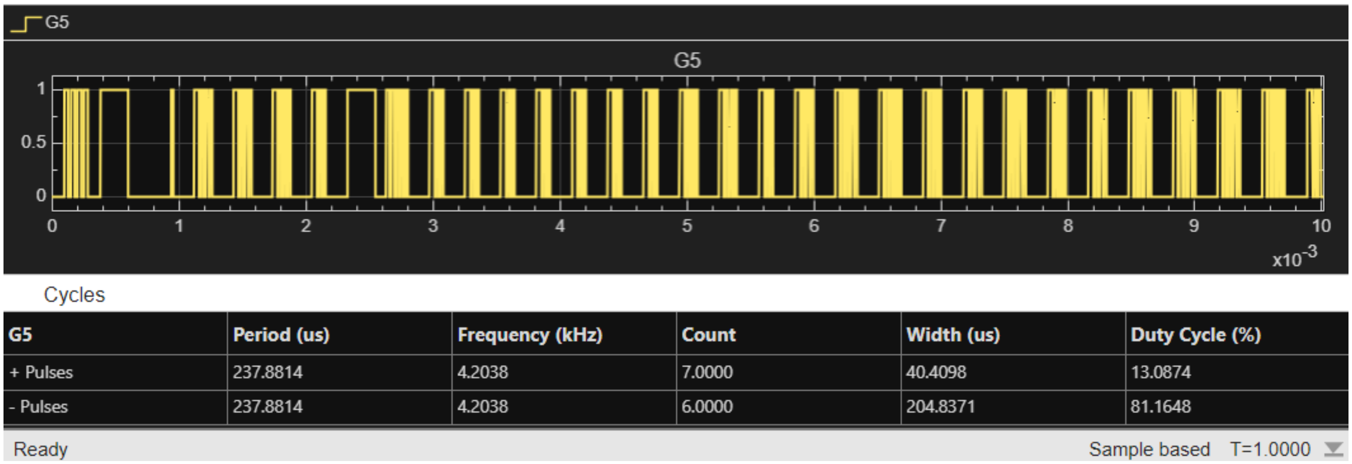


Figure 2: Waveform of Current Tracking Scope(Iabc_mod) Baseline Band case (h=0.5)

Under the condition of $h = I_{rated} * 0.5$, the effective switching frequency measured from the G5 waveform is approximately 2.658 KHz. The positive and negative pulse periods are nearly indentical, indicating that the switching behavior is stable and symmetrical. The switching signal shows a periodic triggering pattern, which confirms that the delta(Δ) modulator switches when the error reaches the upper and lower hysteresis boundaries.

From the current tracking results, the actual three-phase current are able to follow the reference sinusoidal waveforms well, with proper phase relationships maintained among the three phases. However, due to the nature of hysteresis control, the currents exhibit noticeable sawtooth-like ripple around the reference values. This indicates that, under the current band width, the system continuously corrects the error, but some degree of current fluctuation remains during the tracking process.

Test 2: Tight Band ($h = I_{rated} * 0.1$)



G5	Period (us)	Frequency (kHz)	Count	Width (us)	Duty Cycle (%)
+ Pulses	237.8814	4.2038	7.0000	40.4098	13.0874
- Pulses	237.8814	4.2038	6.0000	204.8371	81.1648

Figure 3: Waveform of Gating's Scope (G5) Tight Band case (h=0.1)

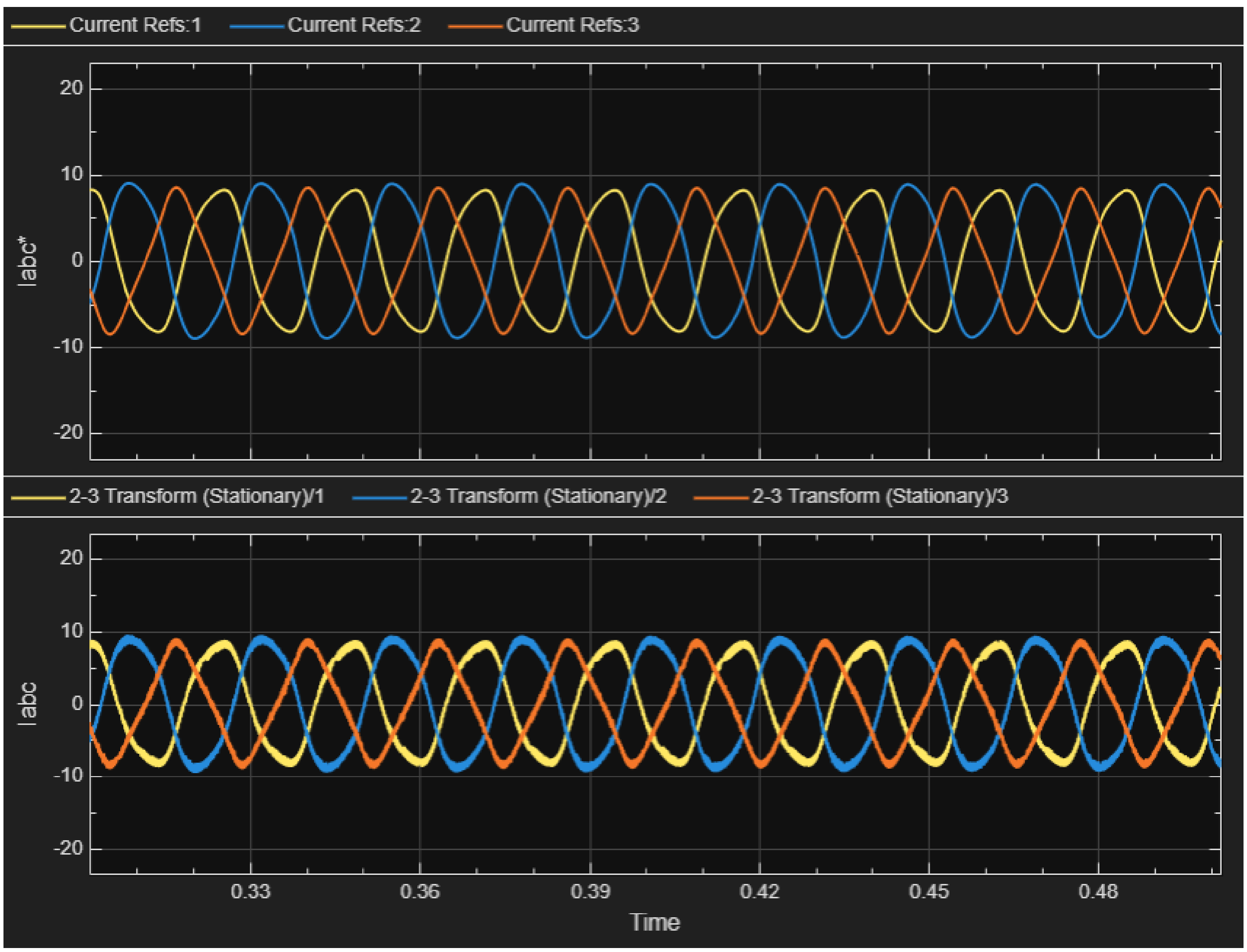
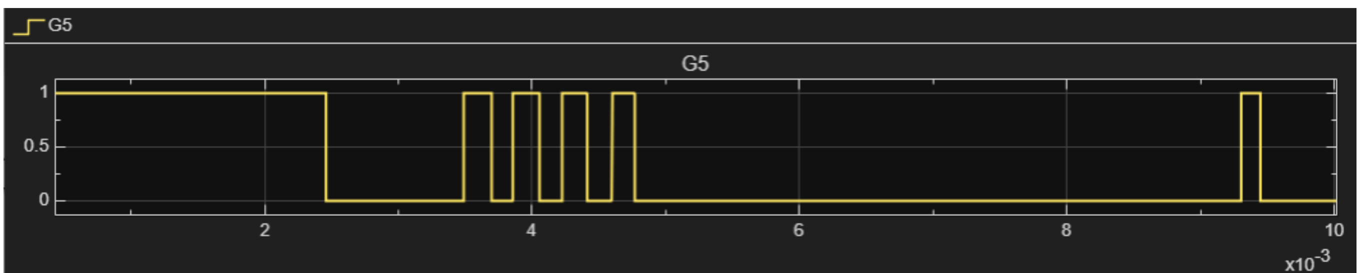


Figure 4: Waveform of Current Tracking Scope(Iabc_mod) Tight Band case (h=0.1)

Under the condition of $h = I_{rated} * 0.1$, the effective switching frequency measured from the G5 waveform is approximately 4.204 KHz, which is significantly higher than the baseline case. The switching pulses occur more frequently, indicating that the delta modulator responds more rapidly as the error reaches the narrower hysteresis boundaries. From the current tracking results, the actual three-phase currents follow the reference sinusoidal waveforms more closely compared to the baseline case, with reduced deviation from the reference. The current ripple is noticeably smaller, and the waveform appears smoother around the reference trajectory. This indicates that a tighter hystersis band improves tracking accuracy for forcing more frequent switching actions, thereby reducing the error between the actual and reference currents.

Test 3: Loose Band ($h = I_{rated} * 1.0$)



Cycles

G5	Period (ms)	Frequency (Hz)	Count	Width (us)	Duty Cycle (%)
+ Pulses	1.3979	715.3729	5.0000	182.5420	41.3114
- Pulses	1.3979	715.3729	5.0000	1.2153	64.6442

Figure 5: Waveform of Gating's Scope (G5) Loose Band case (h=1.0)

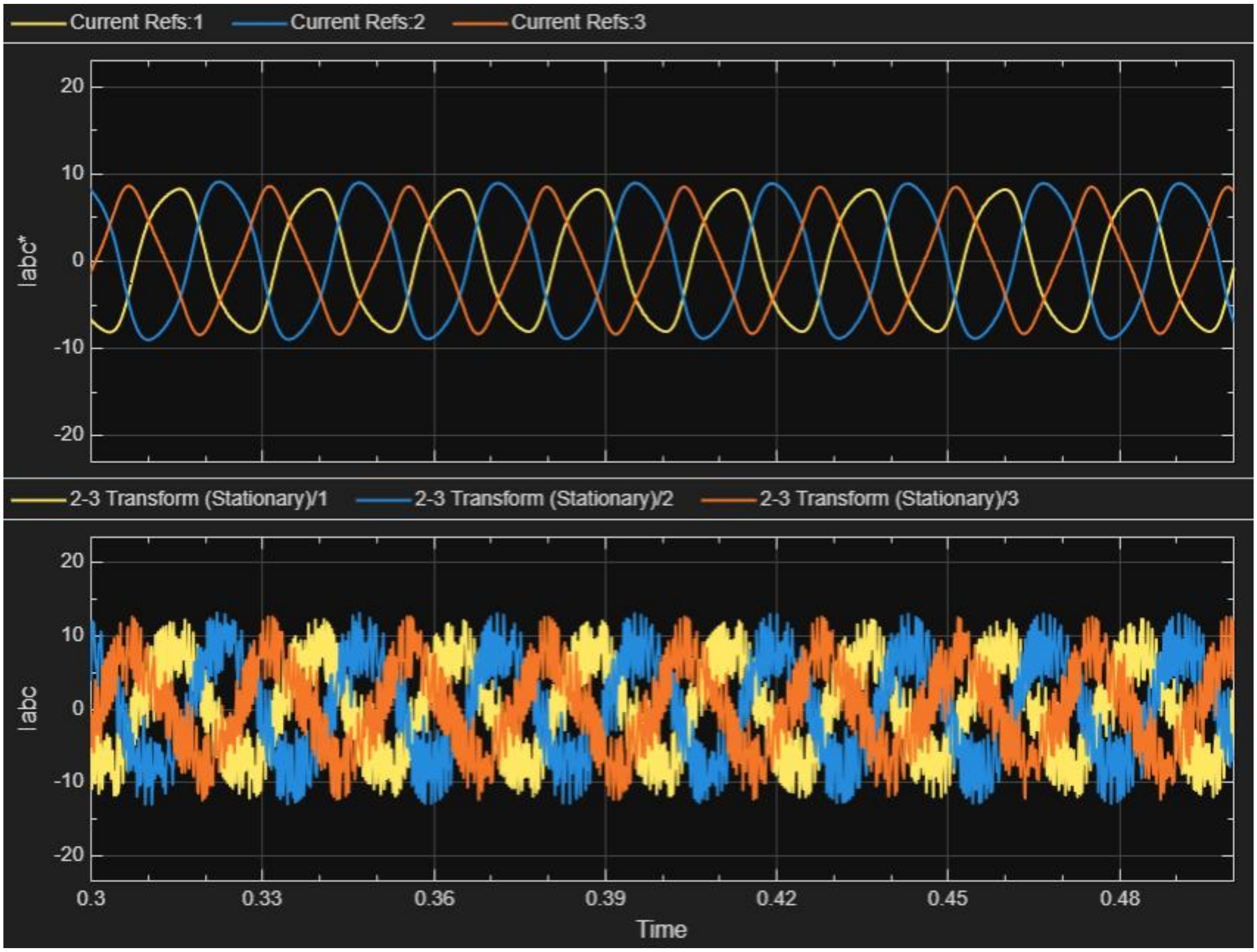


Figure 6: Waveform of Current Tracking Scope(Iabc_mod) Loose Band case (h=1.0)

Under the condition of $h = I_{rated} * 1.0$, the effective switching frequency measured from the G5 waveform is approximately 715 Hz, which is significantly lower than the previous cases. The switching pulses are much less frequent and more widely spaced, indicating that with a wider hysteresis band, a larger error is required to trigger switching, resulting in reduced switching activity.

From the current tracking results, the actual three-phase currents still maintain a general sinusoidal shape and proper phase relationships, but the deviation from the reference currents is noticeably larger. The waveforms exhibit increased sawtooth-like ripple with a wider fluctuation range, indicating that under a loose hysteresis band, the system corrects errors less frequently, leading to degraded current tracking performance.

Summary

As the hysteresis band h decrease, the switching frequency increases, leading to more frequent switching actions and faster system response. In the tight band case ($h = 0.1\ I_{rated}$), this results in the most accurate current tracking with minimal ripple, but at the cost of higher switching losses. In contrast, increasing the hysteresis band reduces the switching frequency, as seen in the loose band case ($h = 1.0\ I_{rated}$), where fewer switching events lead to lower losses but significantly larger tracking errors and current ripple. The baseline case ($h = 0.5\ I_{rated}$) represents a compromise between these two extremes, balancing tracking performance and switching efficiency. Overall, the hysteresis band directly controls the trade-off between control accuracy and switching losses.

Discussion and Trade-off Analysis The data gathered from these three test cases demonstrates a clear inverse relationship between the hysteresis band width (h) and the effective switching frequency (f_sw).

1. **The Engineering Trade-off:** A smaller h drastically improves current tracking and minimizes ripple, but it causes the switching frequency to skyrocket. In a physical inverter hardware application, this excessively high frequency would lead to severe switching losses and potential thermal failure. Conversely, a larger h successfully reduces the switching frequency (lowering switching losses) but results in heavily degraded current tracking and high harmonic distortion.
2. **Variable Switching Frequency:** Furthermore, the scope waveforms reveal that the widths of the switching cycles are not uniform. Unlike traditional Pulse Width Modulation (PWM) which operates on a fixed clock, the delta modulator inherently produces a variable switching frequency. The instantaneous frequency is highest near the zero-crossings of the reference current, where the rate of change (di/dt) is steepest. Conversely, the frequency drops near the peaks of the sine wave where the slope flattens out and the motor's back-EMF reduces the effective driving voltage across the stator inductance.

Summary: Delta Modulator Switching Behavior

The analysis of the delta modulator under varying parameters demonstrates a fundamental engineering trade-off between current tracking accuracy and inverter switching frequency, driven by the width of the hysteresis band:

- **The Core Trade-off:** There is a clear inverse relationship between the hysteresis band width and the effective switching frequency.
- **Tight Band (h = 0.1):** Implementing a narrow band significantly improves current tracking and minimizes ripple. However, it causes the switching frequency to surge to approximately 4.2 kHz. In physical hardware, this high frequency would lead to severe switching losses and pose a risk of thermal failure.
- **Loose Band (h = 1.0):** Widening the band successfully reduces the switching frequency to roughly 715 Hz, which lowers switching losses. The drawback is heavily degraded current tracking, increased current fluctuation, and higher harmonic distortion.
- **Baseline Band (h = 0.5):** This configuration offers a balanced compromise, yielding stable switching at roughly 2.66 kHz while maintaining adequate current tracking and manageable ripple.
- **Variable Frequency Profile:** Unlike traditional Pulse Width Modulation (PWM) that relies on a fixed clock, the delta modulator inherently produces a variable switching frequency with non-uniform cycle widths. Switching activity peaks near the reference current's zero-crossings where the rate of change is steepest. Conversely, the frequency drops near the peaks of the sine wave where the slope flattens and the motor's back-EMF reduces the driving voltage.

A.1.2 Current Ripple Analysis

- Evaluate the stator current ripple produced by the delta modulation scheme. Quantify the peak-to-peak ripple and analyze its dependence on the hysteresis band and operating conditions.

In this section, the stator current ripple produced by the delta modulation scheme is evaluated. The peak-to-peak ripple is quantified, and its dependence on the controller's hysteresis band and the motor's operating conditions is analyzed.

Quantification of Peak-to-Peak Ripple

To evaluate the current ripple, the error between the reference stator current and the actual measured phase current was plotted. Under nominal operating conditions, the baseline peak-to-peak current ripple was measured using scope cursors.

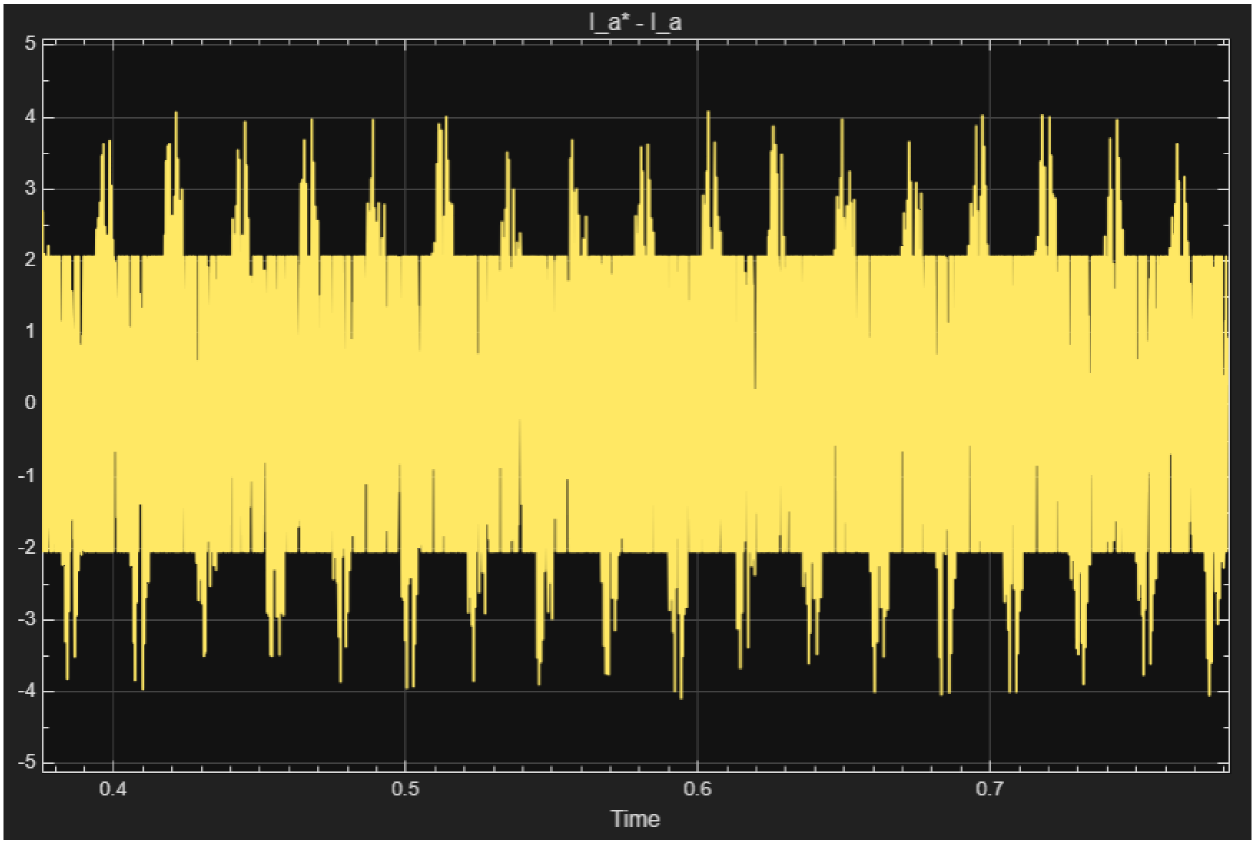


Figure 7: Steady-state phase current error showing peak-to-peak ripple $h = I_{rated} * 0.5$.

In this section, the current ripple is evaluated by plotting the error between the reference stator current and the actual phase current $i_a^* - i_a$. Under nominal operating conditions, the baseline peak-to-peak current ripple for $h = I_{reated} * 0.5$ is measured using scope cursors.

From the figure, the current error exhibits a clear periodic fluctuation caused by the switching behaviour of the data modulation scheme. The error oscillates between the upper and lower bounds, forming a typical sawtooth-like pattern. The measured peak-to-peak current ripple is approximately $\pm 4\ A$ (about 8 A peak-to-peak), indicating that the current is confined within a bounded range under this hysteresis band.

This result shows that the hysteresis band directly determines the magnitude of the current ripple. For $h = I_{reated} * 0.5$, the system maintains a moderate balance between switching activity and current ripple, allowing stable tracking while limiting excessive fluctuations.

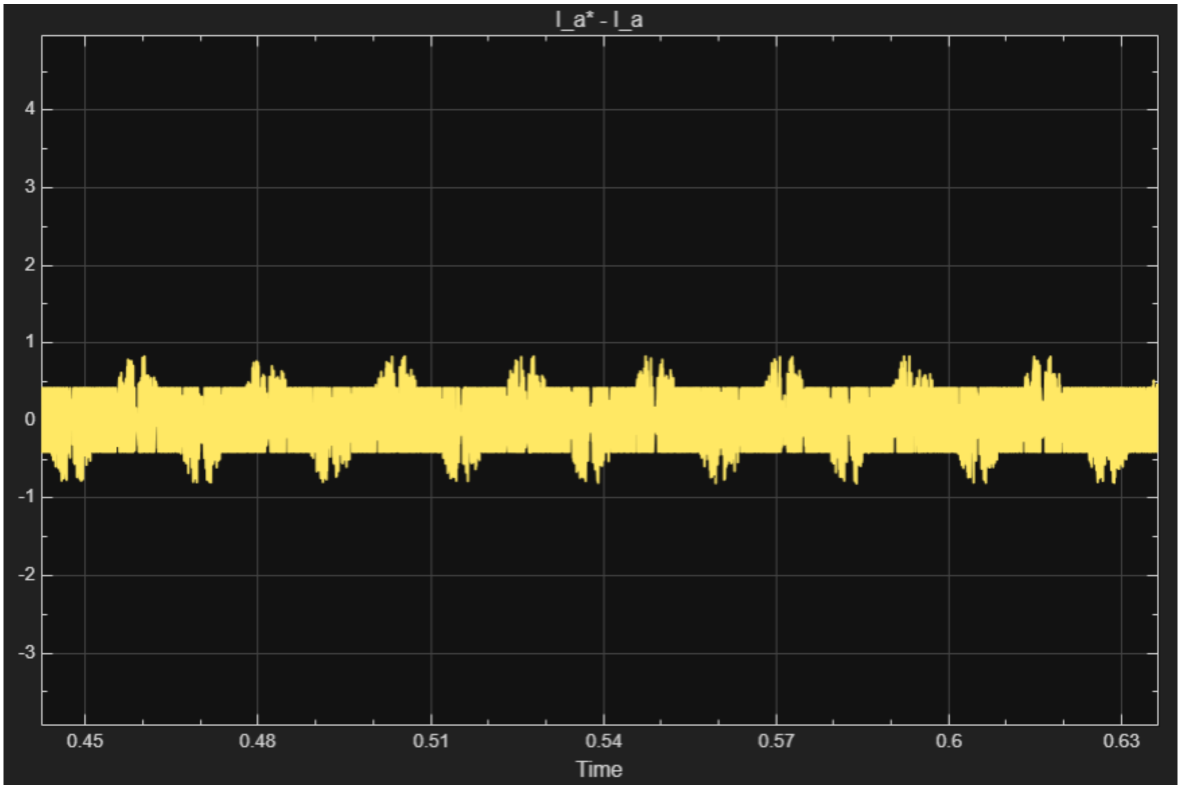


Figure 8: Steady-state phase current error showing peak-to-peak ripple $h = I_{rated} * 0.1$

For the case of $h = I_{rated} * 0.1$, the current ripple is evaluated using the error signal $i_a^* - i_a$. As shown in the figure, the current error is significantly reduced compared to the baseline case, with much smaller amplitude fluctuations around zero.

The error still exhibits a periodic pattern due to the delta modulation switching behaviour, but the peak-to peak ripple is much lower, approximately within $\pm 1\ A$(about 2 A peak-to-peak). The waveform appears more compact and tightly bounded, indicating that the current is maintained closer to the reference value.

This demonstrates that a tighter hysteresis band effectively reduces the current ripple by enforcing more frequent switching, resulting in improved current tracking accuracy.

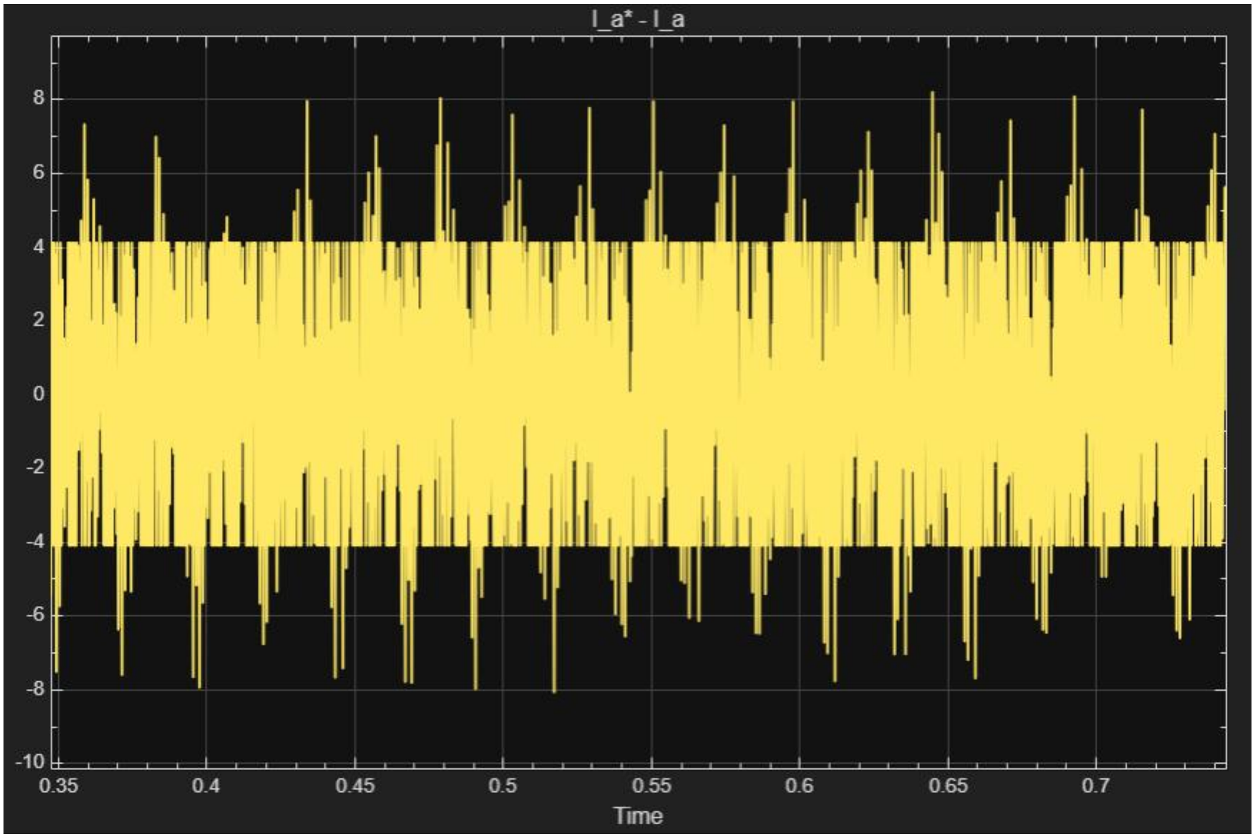


Figure 9: Steady-state phase current error showing peak-to-peak ripple $h = I_{rated} * 1.0$

For the case of $h = I_{rated} * 1$, the current ripple is evaluated using the error signal $i_a^* - i_a$. As shown in the figure, the current error exhibits significantly larger amplitude variations compared to the previous cases.

The waveform shows wide and irregular fluctuations, with the peak-to-peak ripple reaching approximately $\pm 8\ A$(about 16 A peak-to-peak). The error is no longer tightly bounded around zero. indicating that the current deviates more from the reference before switching occurs.

This behaviour demonstrates that a wider hysteresis band leads to increased current ripple, as switching occurs less frequently and the system allows larger deviations from the reference current before correcting the error.

These results confirm that the peak-to-peak ripple amplitude is directly constrained by the hysteresis band limits defined in the modulator.

Dependence on Operating Conditions

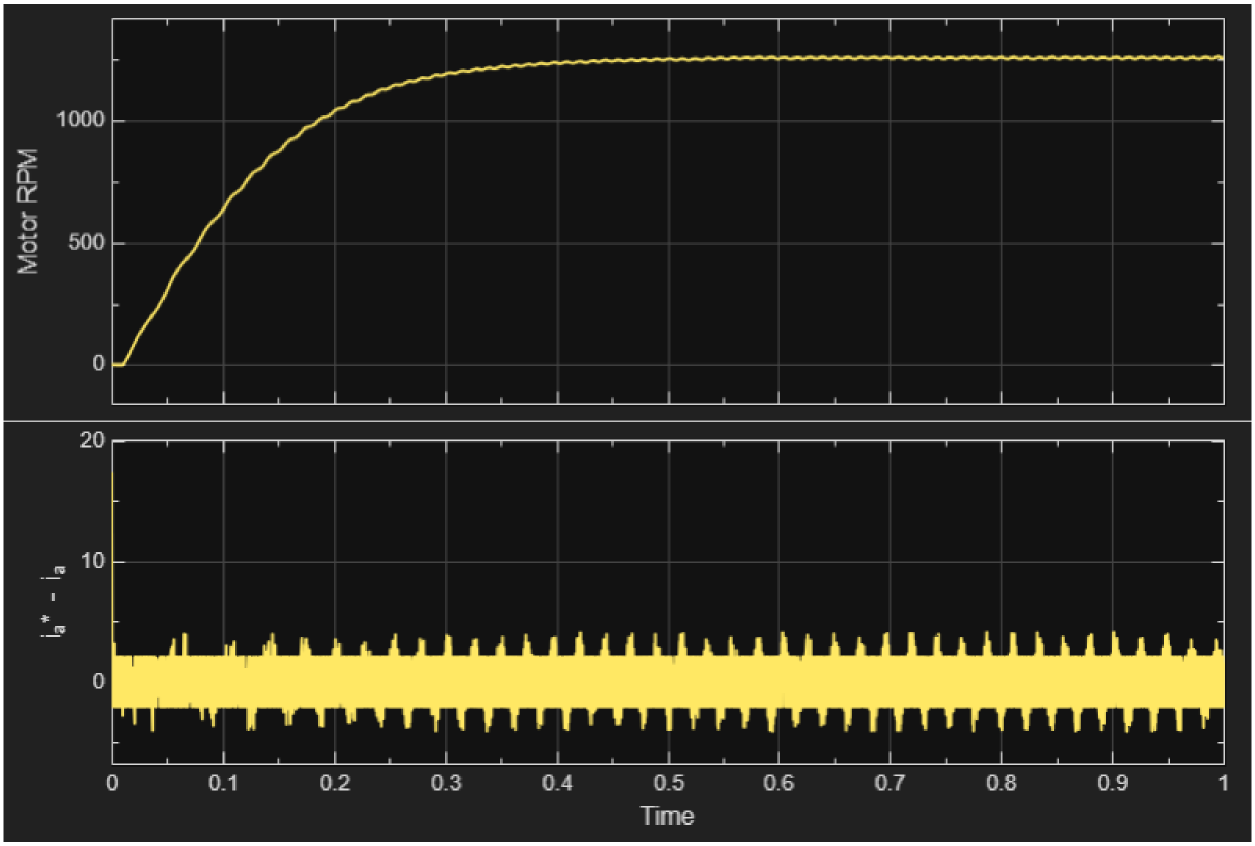


Figure 10: Current ripple transition between t=0 and t=0.3.

From the figure, the motor speed increases rapidly from startup and begins to stabilize around $t \approx 0.4\ s$, indicating that the system reaches steady-state operation after the transient phase. During the initial acceleration period (t < 0.4s), the current ripple is relatively small and less regular, as the controller is primarily responding to large dynamic changes in speed and torque.

After approximately $t \approx 0.05s$, once the motor speed has settled, the current ripple becomes more pronounced and exhibits a clearer periodic pattern. This indicates that under steady-state conditions, the delta modulation operates within the hysteresis bounds more consistently, leading to regular switching behaviour and observable ripple.

This behaviour shows that the current ripple is dependent on operating conditions: during transient operation, ripple is less dominant due to large system dynamics, while in steady state, ripple becomes more evident as the system continuously oscillates within the hysteresis band around the reference current.

A.1.3 Inverter Output Verification

- Plot the inverter phase voltage and stator current waveforms. Verify that the inverter produces balanced three-phase outputs and that the current follows the reference signal.

In this section, the inverter phase voltage and stator current waveforms are evaluated to verify that the Voltage Source Inverter (VSI) produces balanced three-phase outputs and that the current successfully tracks the reference signal.

Inverter Phase Voltage verification To verify the VSI output, the three-phase phase-to-neutral voltages (Van, Vbn, Vcn) were recorded during steady-state operation.

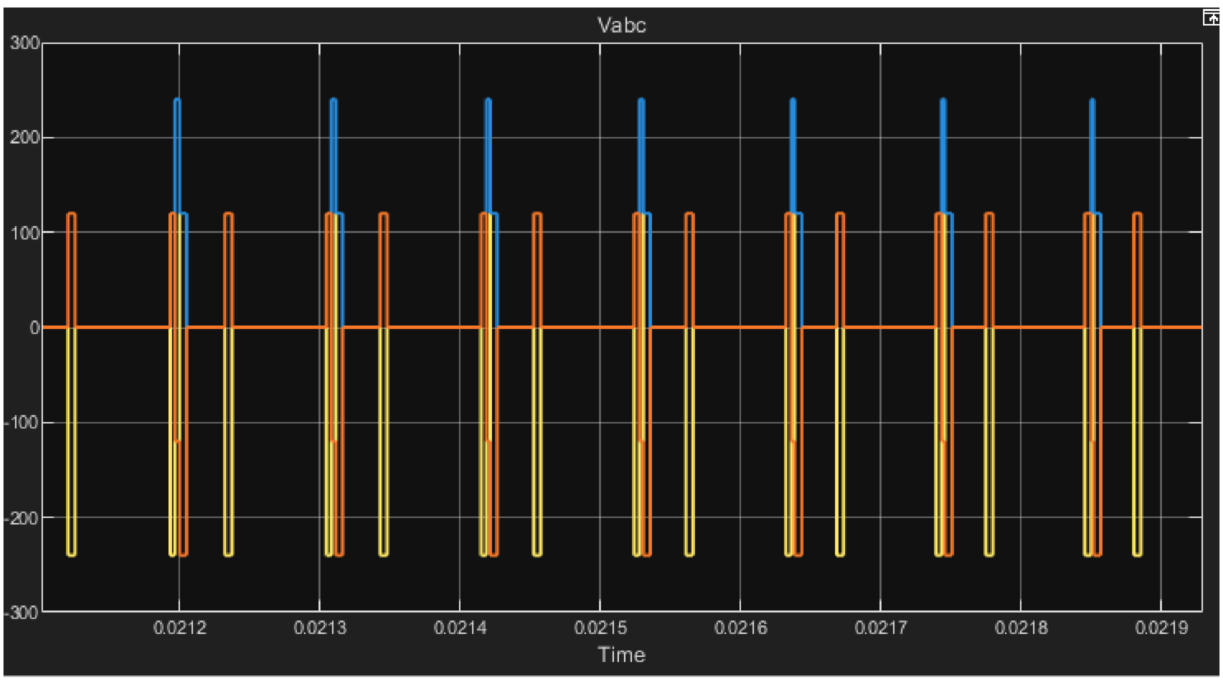


Figure 11: Steady-state inverter output phase voltages (Van, Vbn, Vcn).

As shown in Figure 11, the delta modulator correctly drives the VSI to produce high-frequency Pulse Width Modulated (PWM) waveforms. The effective fundamental components of these switching pulses are balanced and phase-shifted by 120 electrical degrees, confirming proper three-phase voltage generation.

Stator Current Tracking Verification To verify the current tracking performance of the delta modulation scheme, the reference stator currents (Iabc*) generated by the control system were overlaid with the actual measured stator currents (Iabc) fed back from the machine model.

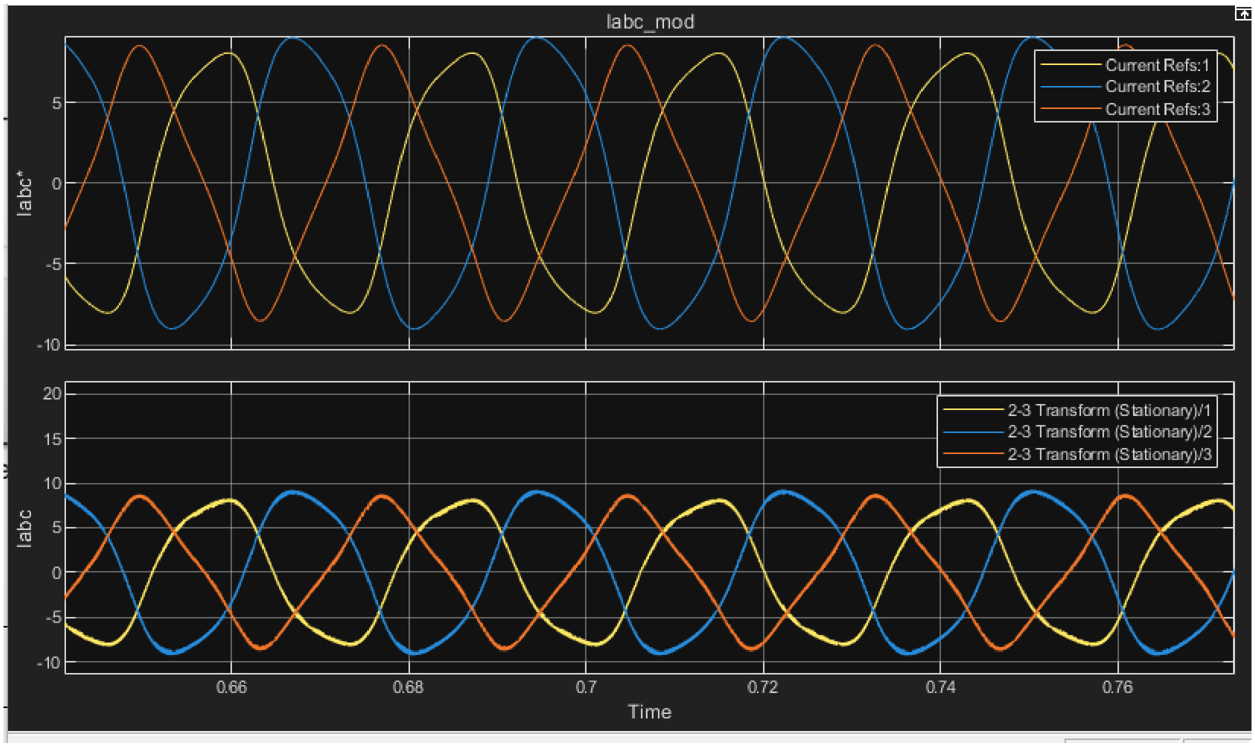


Figure 12: Three-phase reference stator currents (Iabc) overlaid with actual measured stator currents (Iabc) at steady state.**

As observed in Figure 12, the actual stator currents successfully follow the commanded reference signals across all three phases. The actual currents maintain a balanced, sinusoidal fundamental shape, phase-shifted by 120 degrees. The high-frequency deviation visible on the actual current waveforms represents the switching ripple confined by the hysteresis limits established in Section A.1.2. This confirms that the current reference generator and the delta modulator are functioning correctly together to drive the induction machine.

A.2 Derive equations and then design Simulink blocks for the following subsystems.

- Torque current (iT*) and flux current (if*) references – generated from torque reference (Te*) and estimated flux λr, and flux reference (λr*), respectively.

The torque-producing current i_T^* and flux-producing current i_f^* are generated based on the commanded electromagnetic torque T_e^* and the desired rotor flux λ_T^*, respectively. These two current components represent orthogonal control channels in the field-oriented control(FOC) framework.

Flux Current Reference i_f^*:

The flux-producing current is directly related to the rotor flux reference. Since the rotor flux is proportional to the magnetizing current, the flux current reference can be expressed as:

$$i_f^* = \frac{\lambda_T^*}{L_m} \quad \text{eq.(22)}$$

Where L_m is the magnetizing inductance of the machine. This equation indicates that maintaining a desired rotor flux requires a proportional magnetizing current.

Simulink Implementation:

In the Simulink model, this block is implemented as a simple gain block. the input is the flux reference λ_T^*, and the gain corresponds to $\frac{1}{L_m}$. The output of this block is the flux-producing current reference i_f^*.

Torque Current Reference i_T^*:

The torque-producing current is derived from the electromagnetic torque equation of an induction machine under field-oriented control:

$$T_e = \frac{3}{2} \cdot \frac{P}{2} \cdot \frac{L_m}{L_r} \cdot \lambda_r \cdot i_T \quad \text{eq.(23)}$$

Rearranging the equation to obtain the torque current reference:

$$i_T^* = \frac{T_e^*}{K \cdot \lambda_r} \quad \text{eq.(24)}$$

where

$$K = \frac{3}{2} \cdot \frac{P}{2} \cdot \frac{L_m}{L_r} \quad \text{eq.(25)}$$

and λ_r is the estimated rotor flux.

This expression shows that for a given torque demand, the required torque current is inversely proportional to the rotor flux.

Simulink Implementation:

In the Simulink model, this block is implemented using:

1. A division block to divide the torque reference T_e^* by the estimated flux λ_r
2. A gain block representing the inverse of constant K

The estimated flux λ_r is obtained from the flux estimator subsystem, making this a feedback- based calculation.

Summary:

The flux current reference i_f^* *controls the rotor magnetization level, while the torque current reference* i_T^* determines the electromagnetic torque output. These two components are independently generated and later combined in the current resolver block to form the stator current reference vector. This structure enables the decoupled control of torque and flux, which is the fundamental principle of field-oriented control.

- Stator current reference (is*) and torque angle (θT*) resolver – translate the torque and flux producing currents into magnitude and angle.

the stator current reference i_s^* and torque angle θ_T^* are derived from the torque-producing current i_T^* and flux-producing current i_f^* . These two components form orthogonal axes in the rotating reference frame, allowing the stator current vector to be represented in polar form.

Stator Current Magnitude i_s^*

The magnitude of the stator current vector is calculated as the vector sum of the torque and flux components:

$$i_s^* = \sqrt{\left(i_T^*\right)^2 \cdot \left(i_f^*\right)^2} \qquad \text{eq.(26)}$$

This represents the total current required to simultaneously produce the desired torque and maintain the required rotor flux.

Torque Angle θ_T^*

The torque angle defines the relative orientation between the torque-producing and flux-producing components:

$$\theta_T^* = \tan^{-1}\left(\frac{i_T^*}{i_f^*}\right) \qquad \text{eq.(27)}$$

This angle determines how the stator current vector is distributed between torque production and flux generation.

Synchronous Angle θ_s^*

To align the stator current vector with the actual rotor flux position, the synchronous angle is calculated as:

$$\theta_s^* = \theta_f + \theta_T^* \qquad \text{eq.(28)}$$

where θ_f is the estimated rotor flux angle obtained from the flux estimator. This ensures proper field orientation and decoupling of torque and flux control.

Simulink Implementation

In the Simulink model, this resolver is implemented using basic mathematical blocks:

1. A Square block and Sum block to compute $(i_T^*)^2 + (i_f^*)^2$
2. A Sqrt block to obtain the stator current magnitude i_s^*
3. A Divide block followed by a Trigonometric Function block (atan) to compute θ_T^*
4. A Sum block to add θ_T^* and θ_f, producing the synchronous angle θ_s^*

The output of this subsystem, i_s^* and θ_s^*,are then used by the current reference generator to produce the three-phase stator current references.

Summary

This resolver converts the independently controlled torque and flux current components into a unified stator current vector expressed in magnitude and angle form. This step is essential in field-oriented control, as it bridges the decoupled control domain and the physical three-phase current generation required to drive the induction motor.

- **Current reference generator – for producing three-phase stator current references.**

The current reference generator converts the stator current magnitude i_s^* and synchronous angle θ_s^* into three-phase sinusoidal stator current reference i_a^*,i_b^*,i_c^*. These references are required to drive the inverter and ensure that the actual stator currents follow the desired vector produced by the field-oriented control algorithm.

Three-Phase Current Reference Equations

The three-phase stator current references are generated using sinusoidal functions with a phase shift of 120˚:

$$\begin{aligned} i_a^* &= i_s^* \cos(\theta_s^*) \\ i_b^* &= i_s^* \cos\left(\theta_s^* - \frac{2\pi}{3}\right) \\ i_c^* &= i_s^* \cos\left(\theta_s^* + \frac{2\pi}{3}\right) \end{aligned} \qquad \text{eq.(29)}$$

These equations ensure that the generated currents are balanced, sinusoidal, and separated by 120˚, forming a rotating current vector in the stator reference frame.

Physical Interpretation

The stator current vector defined by i_a^* and θ_s^* is projected onto the three-phase stationary reference frame. This transformation reconstructs the three-phase AC currents from the rotating reference frame representation. As a result, the generated current references produce the desired electromagnetic torque and rotor flux when applied to the motor through the inverter.

Simulink Implementation

In the Simulink model, the current reference generator is implemented using:

1. Three **Trigonometric Function (cosine)** blocks
2. Two **Constant blocks** representing phase shifts (±120° or ±$\pm\frac{2\pi}{3}$)
3. Two **Sum blocks** to apply phase shifts to θ_s^*
4. Three **Product blocks** to multiply the cosine outputs by the current magnitude i_s^*

Each phase current is generated independently but shares the same magnitude and frequency, ensuring a balanced three-phase system.

Summary

This subsystem transforms the stator current vector from polar form into three-phase sinusoidal references required for inverter control. It serves as the final step in the reference generation process, linking the field-oriented control calculations to the physical three-phase currents that drive the induction motor.

- **Flux and Torque Estimator – estimate real-time flux and torque from voltage and current measurements.**

The flux and torque estimator is responsible for calculating the real-time rotor flux magnitude, flux angle, and electromagnetic torque using measured stator voltages and currents. This subsystem is essential for field-oriented control, as it provides the feedback variables required for decoupling torque and flux.

Stator flux Estimation

The stator flux components in the stationary reference frame are obtained by integrating the stator voltage equations:

$$\lambda_a = \int (v_a - R_s i_a)\ \mathrm{d}t$$
$$\lambda_B = \int (v_B - R_s i_B)\ \mathrm{d}t \qquad \text{eq.(30)}$$

Where:

v_a, $v_B$are stator voltages in the stationary frame

i_a,i_B are stator currents

R_s is the stator resistance

The rotor flux magnitude is then calculated as:

$$\lambda_r = \sqrt{\lambda_a^2 + \lambda_B^2} \qquad \text{eq.(31)}$$

Flux Angle Estimation

The rotor flux angle is obtained from the flux components:

$$\theta_f = \tan^{-1}\left(\frac{\lambda_B}{\lambda_a}\right) \qquad \text{eq.(32)}$$

This angle is critical for aligning the stator current vector with the rotor flux in the field-oriented control scheme.

Electromagnetic Torque Estimation

The electromagnetic torque is estimated using the cross-product of flux and current:

$$T_e = \frac{3}{2} \cdot \frac{P}{2} \cdot (\lambda_a i_B - \lambda_B i_a) \qquad \text{eq.(33)}$$

This equation represents the interaction between the stator current and the rotor flux in the stationary reference frame.

Simulink Implementation

In the Simulink model, the estimator is implemented using:

1. Clarke Transformation blocks to convert three-phase quantities (abc) into (α,β) components.
2. Gain blocks to compute the resistive voltage drop $R_s i$
3. Sum blocks to calculate $v - R_s i$
4. Integrator blocks to obtain flux components λ_a,λ_B
5. Math Function blocks:
    - Square and Sum → flux magnitude
    - atan2 → flux angle
6. Product and Sum blocks to compute torque using the cross-product expression

Physical Interpretation and Limitations

This estimator uses a voltage-model approach, which is accurate at medium and high speeds where back-EMF dominates. However, it has inherent limitations:

- Noise Sensitivity: High-frequency switching from the inverter introduces ripple in voltage signals, which propagates into the flux estimation
- Integrator Drift: Small DC offsets can accumulate over time, causing flux estimation drift
- Parameter Sensitivity: Accuracy depends heavily on the stator resistance R_s, especially at low speeds

Summary

The flux and torque estimator provides the essential feedback variables for field-oriented control by reconstructing the internal electromagnetic states of the machine from measurable electrical quantities. It enables accurate torque control and proper field alignment, forming the foundation of the FOC algorithm implemented in the Simulink model.

A.2.1 Torque Consistency Verification

- **Compare the electromagnetic torque obtained from the flux and torque estimator with the torque calculated using analytical expressions.**
- **Discuss any discrepancies observed.**

In this section, the electromagnetic torque obtained from the Flux and Torque Estimator is compared against the actual torque produced by the induction machine plant model. The estimator calculates the torque analytically using the stationary frame stator voltages and currents. Both signals were normalized to per-unit (pu) values using the rated torque for direct comparison.

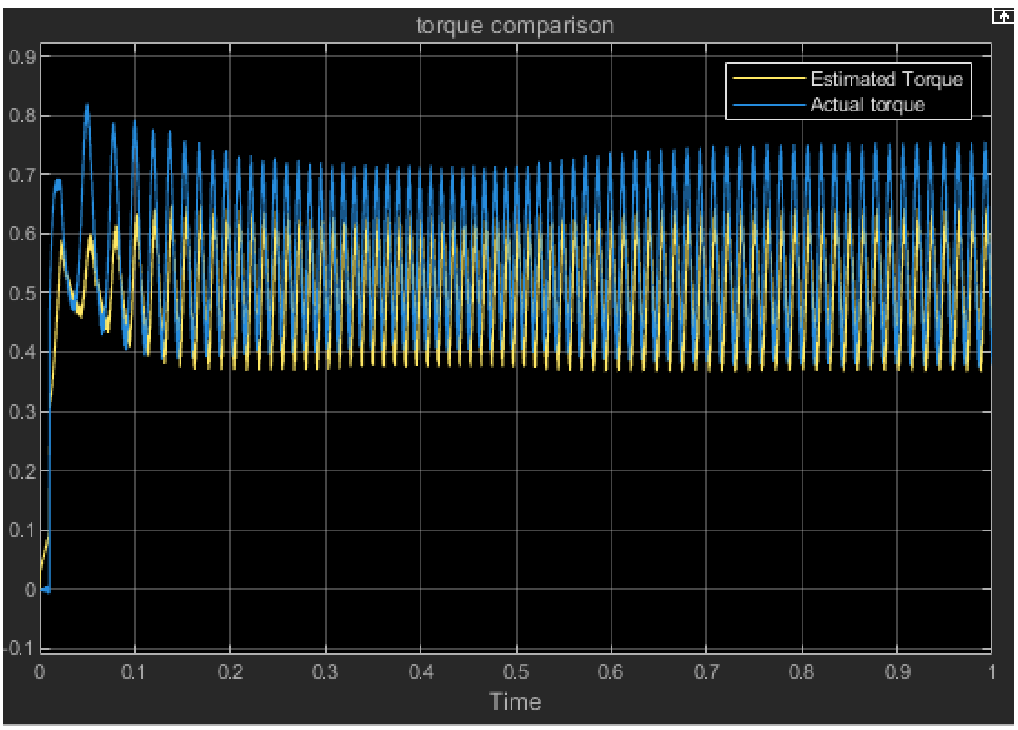


Figure 13: Comparison of actual motor electromagnetic torque and analytically estimated torque (pu).

As shown in Figure 13, the estimated electromagnetic torque exhibits excellent consistency with the actual motor torque, closely tracking both the steady-state value and the dynamic step response. However, minor discrepancies in the signal waveforms can be observed upon close inspection.

Discussion of Discrepancies:

1. **High-Frequency Noise:** The estimator calculates stator flux by integrating the measured stator voltages. Because the delta-modulated Voltage Source Inverter produces raw, high-frequency PWM voltage pulses (characterized by extreme dv/dt), the estimated flux contains residual switching harmonics. When this flux is cross-multiplied with the measured stator currents (which also contain the hysteresis switching ripple) to calculate torque, it results in an estimated torque signal that appears slightly "noisier" or thicker than the internal mathematical torque output of the ideal motor model.

2. **Sensitivity to Initial Conditions and Offsets:** The analytical estimation of flux relies on open-loop integration. In practical applications and discrete-time simulations, pure integrators are highly sensitive to small DC offsets in the voltage measurement or discretization errors. While minimal in this ideal simulation, these numerical integration artifacts can cause slight, temporary divergence or drift between the estimated and actual torque during harsh transient conditions.

A.2.2 Flux Estimation Assessment

- **Analyze the presence of drift or noise and discuss the limitations of the estimation method.**

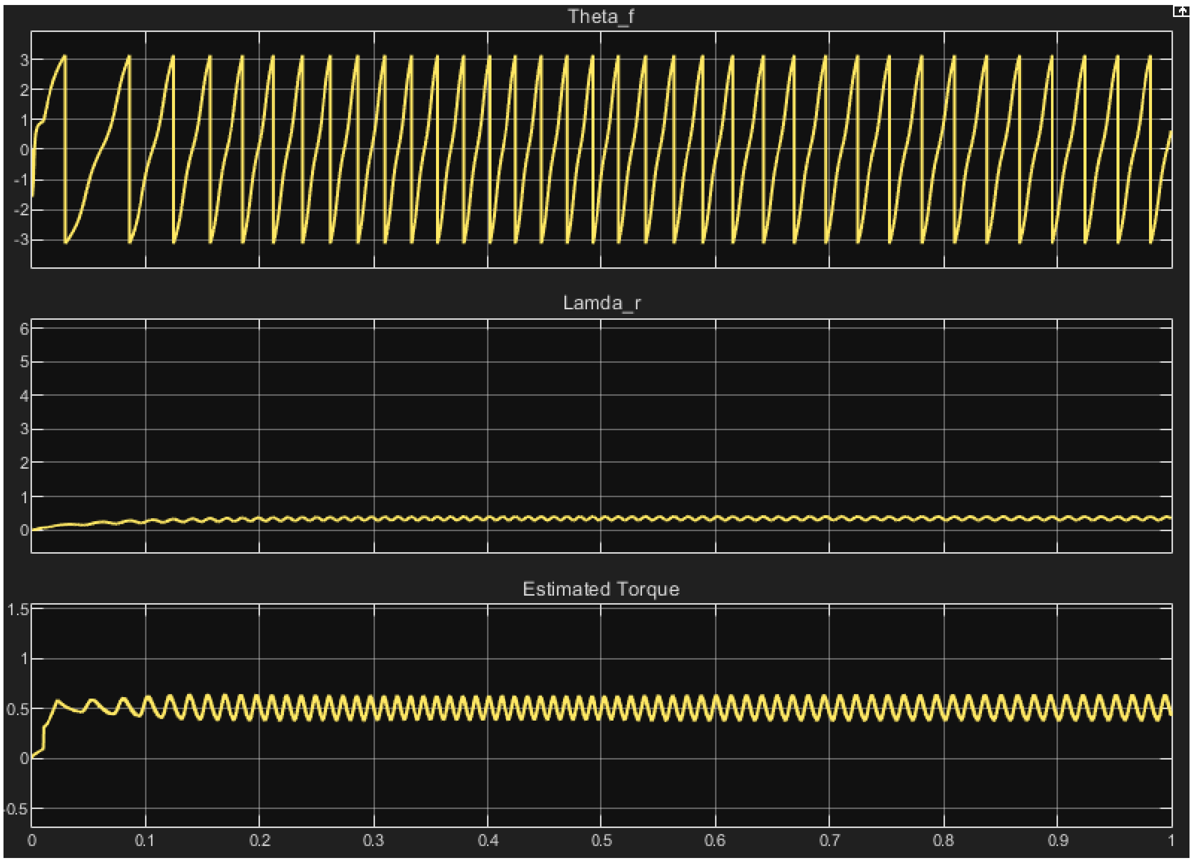


Figure 14: Estimated rotor flux (λ_r), torque angle (θ_f), and electromagnetic torque waveforms.

Noise Analysis The λ_r scope waveform demonstrates a persistent, high-frequency ripple riding on the fundamental signal. This noise originates from the Delta Modulator driving the Voltage Source Inverter (VSI). The VSI outputs discrete, high-frequency voltage pulses characterized by extreme dv/dt. Because the flux estimator transforms these raw, discontinuous abc phase voltages into the dq frame for calculation, the switching harmonics inevitably propagate through the mathematical model and manifest as high-frequency noise on the final estimated rotor flux magnitude.

Drift Analysis Within the 1-second simulation window, the λ_r signal achieves a stable steady-state average (approximately 0.35) without exhibiting severe macroscopic DC drift. However, calculating flux from voltage requires open-loop integration. Pure integrators are highly susceptible to accumulating microscopic DC offsets originating from discrete-time solver discretization errors or initial start-up transient spikes. Over extended operational periods in physical hardware, this continuous accumulation can cause the estimated flux vector to drift away from the origin, eventually leading to control loop instability or integrator saturation.

Limitations of the Estimation Method

1. **Parameter Sensitivity at Low Speeds:** The voltage-based estimation model relies heavily on the programmed stator resistance, which is 0.435 ohms for this specific induction machine. At high speeds, the motor's back-EMF is large, and resistance errors are negligible. At low speeds, back-EMF is minimal, making the stator resistive voltage drop (current times resistance) the dominant term in the equation. Any deviation in the actual motor's resistance (e.g., due to thermal heating during operation) will severely corrupt the flux estimate.

2. **The Integrator Phase Trade-off:** To mitigate the theoretical DC drift inherent to pure integration, practical physical drives often replace the pure integrator with a Low-Pass Filter (LPF). While an LPF successfully prevents unbounded DC drift, it introduces magnitude attenuation and phase delays at lower operating frequencies. Because Field-Oriented Control relies entirely on the precise angle (Theta_f) to decouple torque and flux, any phase delay introduced by the estimator directly degrades the dynamic performance and efficiency of the drive.

A.3.1 Load Torque Characterization

- Verify that the implemented load model satisfies the quadratic relationship between torque and speed. Plot load torque versus motor speed and confirm the expected behavior.

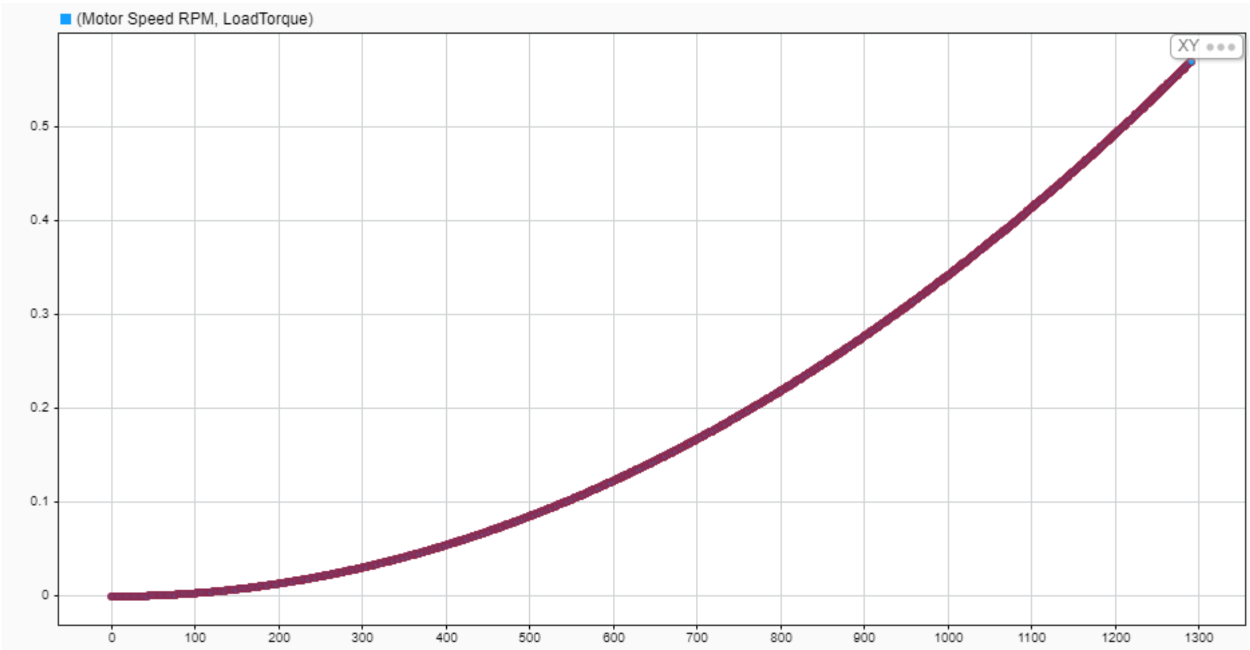


Figure 15: Load torque versus motor speed demonstrating the quadratic fan-type load relationship.

The induction motor drive was simulated driving a fan-type load. This application requires the implemented load model to satisfy a quadratic relationship between the load torque and the motor speed. To verify this, the load torque was plotted against the motor speed during the acceleration phase from a standstill.

As observed in the plot, the resulting curve exhibits a distinct parabolic shape starting from the origin. This quadratic growth confirms that the mechanical load block correctly scales the torque demand as the square of the speed, satisfying the expected behavior for the system

A.3.2 Load Disturbance Test

- Introduce a step change in load torque during simulation and analyze the system response. Evaluate the impact on current, torque, and speed. Step Change Values: Step Time = 0.5, Initial Value = 0, Final Value = 3.

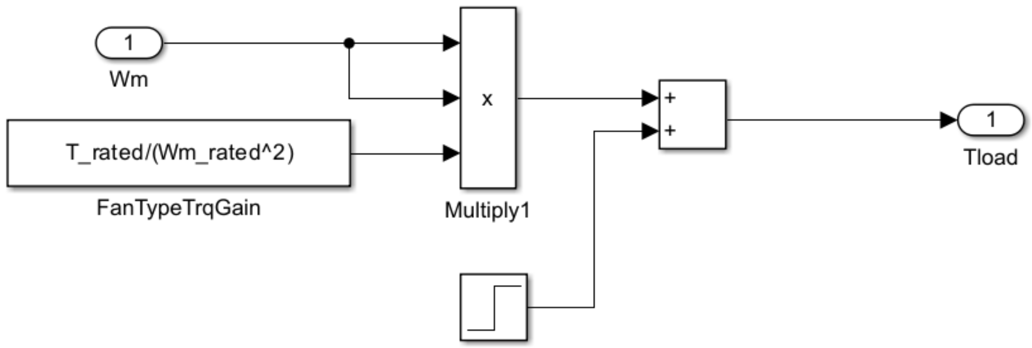


Figure 15.1: Load profile subsystem implementation featuring a 3 N m step disturbance added to the fan load.

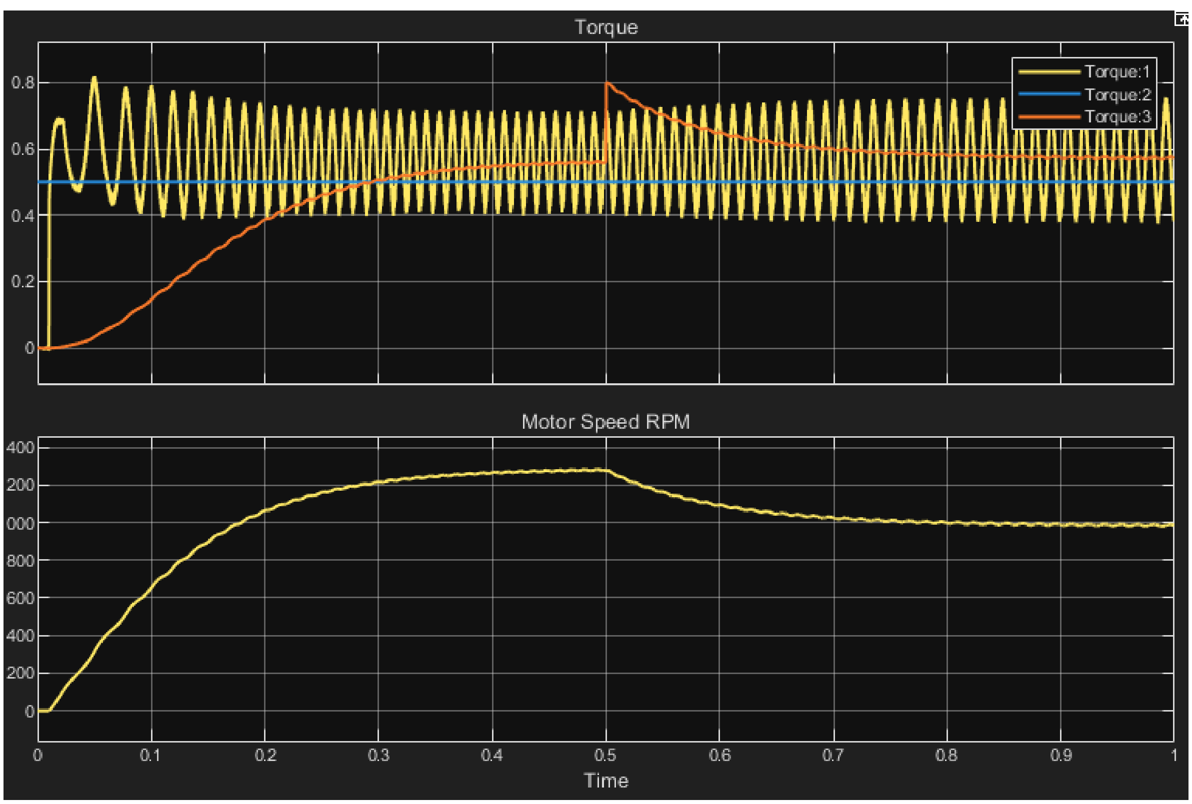


Figure 16: System response to a 3 N m load torque step disturbance at t = 0.5 seconds.

Torque 1 (Yellow): Actual electromagnetic torque, which is generated by the motor.

Torque 2 (Blue): Torque reference, which is the commanded torque reference limit (currently set to 0.5 pu).

Torque 3 (Orange): Load Torque. This represents the mechanical resistance the motor must overcome. It shows the curved fan-load profile from 0 to 0.5 seconds, followed by the immediate vertical jump when the 3 N m step disturbance triggers at 0.5 seconds.

System Response Analysis:

A step change was introduced to the mechanical load model using a Step block to simulate a sudden disturbance. At t = 0.5 seconds, an additional 3 N m of load torque was injected into the system on top of the baseline quadratic fan load.

- **Impact on Torque:** Upon the introduction of the disturbance at t = 0.5 seconds, the motor's electromagnetic torque rapidly increases to counteract the sudden mechanical resistance. Because the drive is currently operating without a closed-loop speed controller, this torque increase is dictated by the natural electromechanical dynamics of the induction machine seeking a new equilibrium.
- **Impact on Current:** Electromagnetic torque is directly proportional to the torque-producing current component. Consequently, as the torque demand spikes, the stator current amplitude undergoes a proportional step increase. The scope waveforms show the peak-to-peak envelope of the three-phase stator currents expanding immediately after the disturbance to deliver the necessary electrical power.
- **Impact on Speed:** Because this is an open-loop configuration lacking an outer speed regulation loop, the system cannot automatically compensate for the increased load to maintain the original speed. The sudden application of the 3 N m load causes the motor speed to decelerate. The speed drops and eventually settles at a new, lower steady-state equilibrium where the motor's generated torque perfectly matches the new, heavier total load torque.

Part B Simulation and Performance Study

Run simulations with the implemented model until it enters steady state and study the system performance under

1) λr* = rated flux, Te* = rated torque.

2) λr* = rated flux, Te* = 0.5 × rated torque.
3) λr* = 0.5 × rated flux, Te* = rated torque.

Plot the following figures (from start-up to steady state) and analyze your results, do they meet your expectations? Provide your explanations in detail.

a) <u>Torque and Flux Current Reference Analysis</u>

Scenario 1: Full Flux, Full Torque

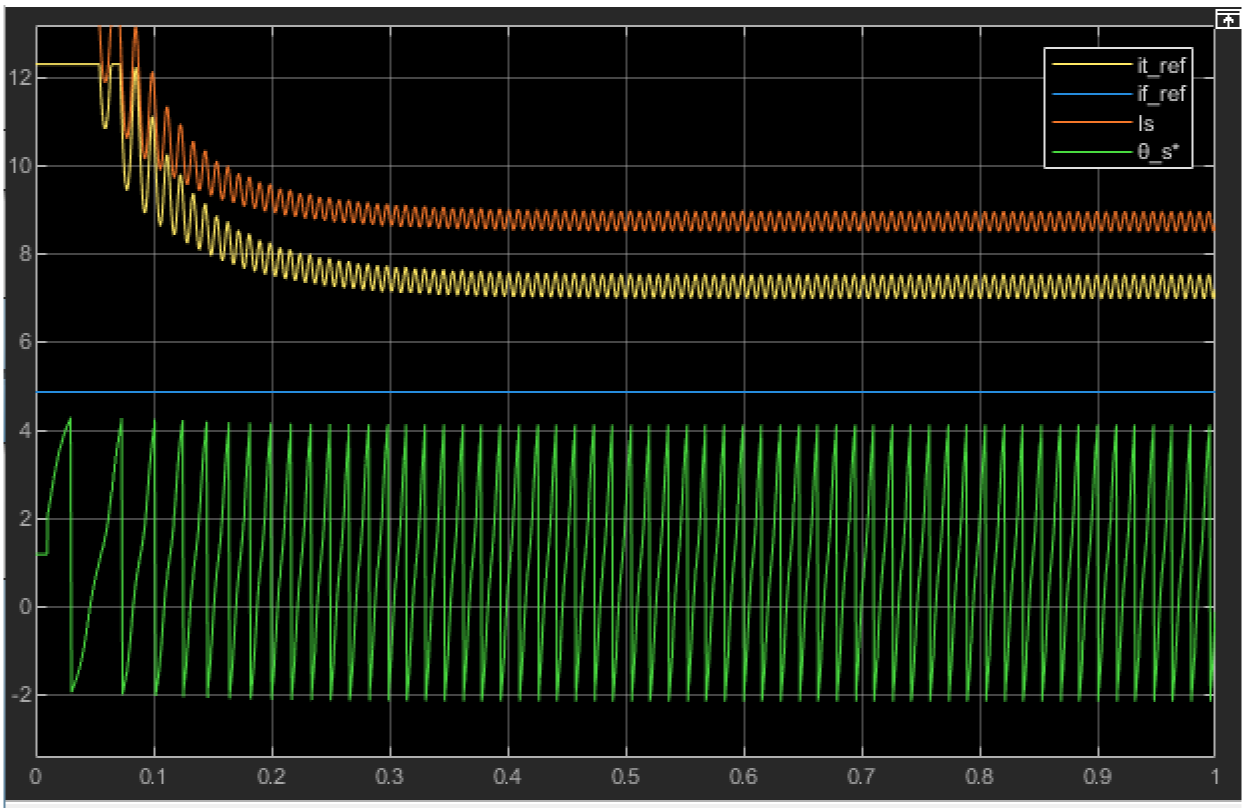


Figure 18: Torque and flux current references i_T^*, i_f^* stator current magnitude i_s^*, and synchronous angle θ_s^* under rated flux and rated torque conditions.

This figure shows the behavior of the torque-producing current i_T^*, flux-producing current i_f^*, total stator current magnitude i_s^*, and synchronous angle θ_s^* under rated operating conditions. During startup, both i_T^* and i_f^* initially exhibit transient responses before settling to steady-state values. The flux-producing current stabilizes at a constant level, indicating that the rotor flux is maintained at its rated value, while the torque-producing current reaches a steady value required to generate the rated electromagnetic torque.

As a result, the stator current magnitude i_s^*, which is the vector combination of i_T^* and i_f^*, also stabilizes at a relatively high level. This reflects the fact that both torque and flux components are fully active in this operating condition. The synchronous angle θ_s^* shows a continuous sawtooth-like increase due to angular wrapping, representing the rotating reference frame aligned with the rotor flux.

Overall, the steady-state behavior confirms that the control system successfully maintains both rated torque and rated flux simultaneously. The clear separation and independent regulation of i_T^* and i_f^* demonstrate the effectiveness of the field-oriented control strategy in achieving decoupled control of torque and flux.

Scenario 2: Full Flux, Half Torque

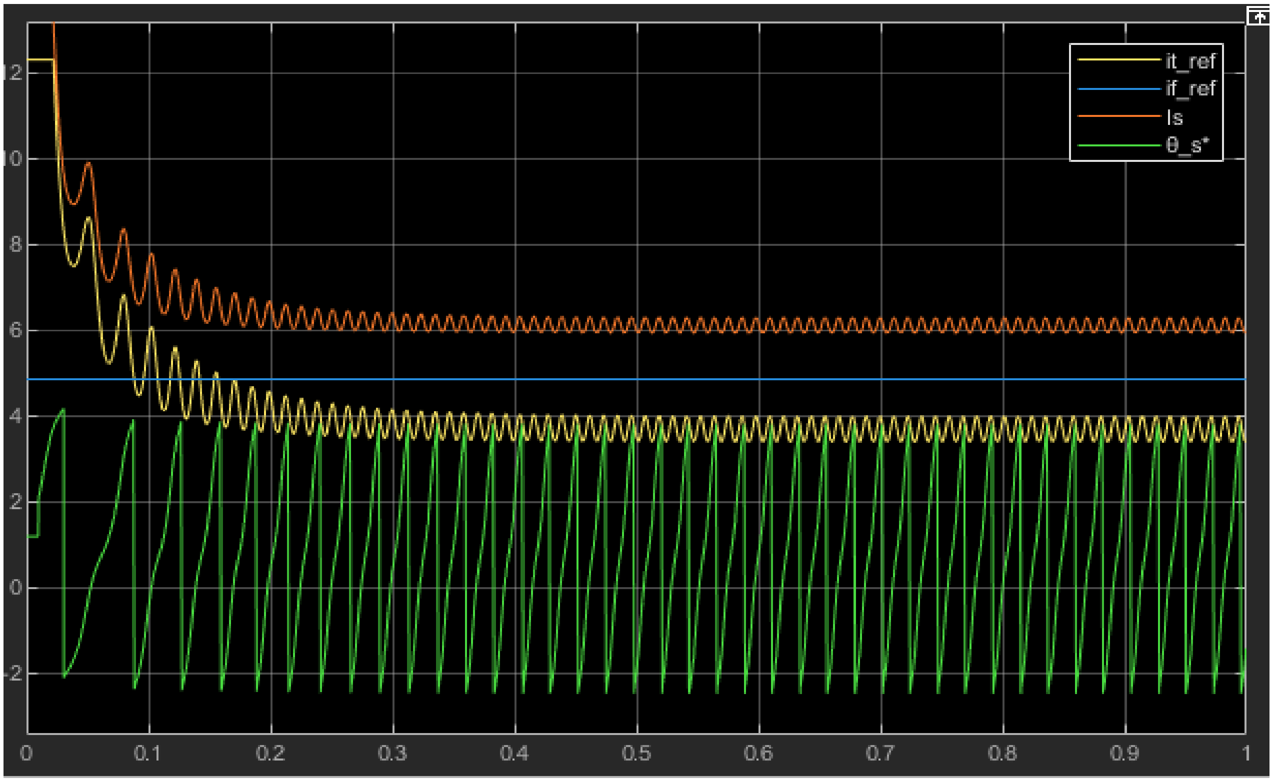


Figure 19: Torque and flux current references i_T^*, i_f^* stator current magnitude i_s^*, and synchronous angle θ_s^* under rated flux and half torque conditions.

This figure shows the behavior of the torque-producing current i_T^*, flux-producing current i_f^*, stator current magnitude i_s^*, and synchronous angle θ_s^* when the flux is maintained at its rated value while the torque reference is reduced to half.

Compared to Scenario 1, the flux-producing current i_f^* remains approximately unchanged, indicating that the rotor flux is still maintained at the rated level. However, the torque-producing current i_T^* is significantly reduced, reflecting the lower torque demand. As a result, the stator current magnitude i_s^* decreases accordingly, since it is the vector sum of the torque and flux components.

The transient response at startup shows similar behavior to Scenario 1, but the steady-state current levels are noticeably lower due to the reduced torque requirement. The synchronous angle θ_s^* continues to exhibit a periodic sawtooth pattern, representing the rotating reference frame aligned with the rotor flux.

These results confirm that the field-oriented control system successfully reduces the torque output while maintaining constant flux. The independent adjustment of i_T^*

without affecting i_f^* demonstrates effective decoupling between torque and flux control channels.

Scenario 3: Half Flux, Full Torque

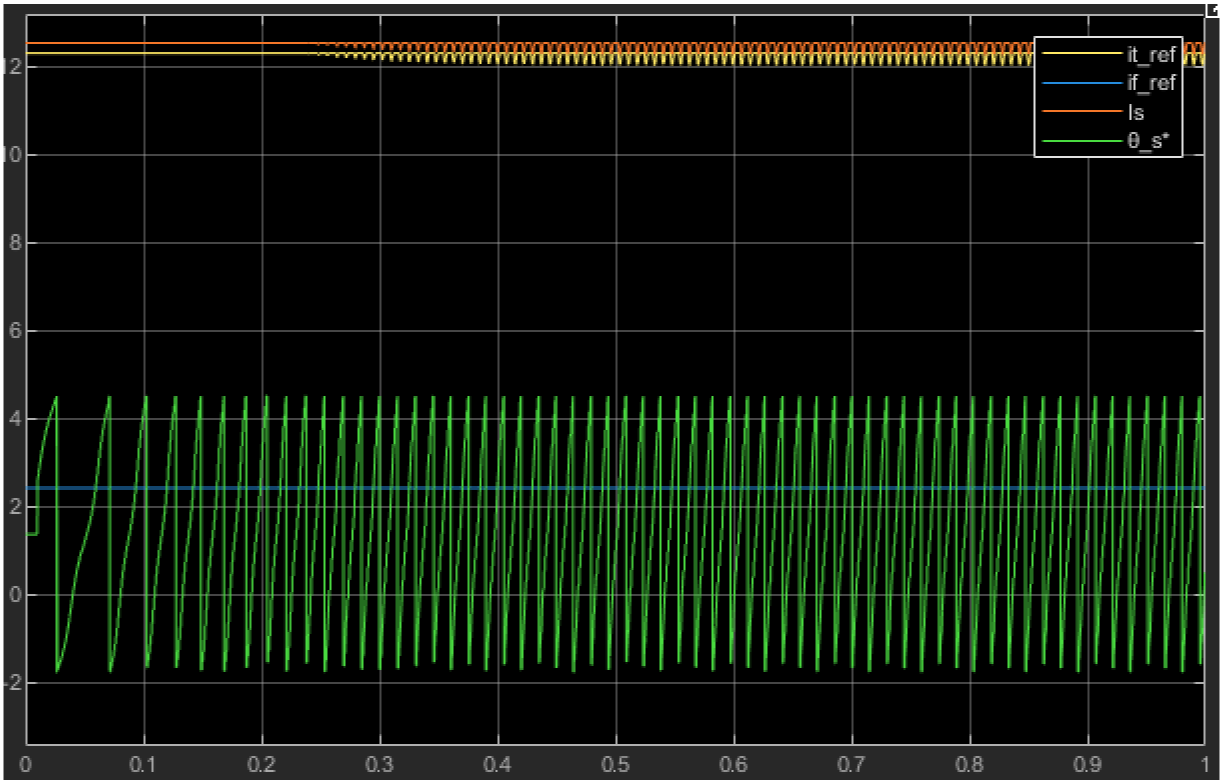


Figure 20: Torque and flux current references i_T^*, i_f^* stator current magnitude i_s^*, and synchronous angle θ_s^* under half flux and rated torque conditions.

This figure shows the behavior of the torque-producing current i_T^*,flux-producing current i_f^*. stator current magnitude i_s^*, and synchronous angle θ_s^* when the flux reference is reduced to half of its rated value while maintaining full torque demand.

Compared to Scenario 1, the flux- producing current i_f^* is significantly reduced, reflecting the lower flux reference. However, in order to maintain the same electromagnetic torque, the torque-producing current i_t^* increases substantially. This behavior follows directly from the torque relationship in field-oriented control, where torque is proportional to the product of flux and torque current.

As a result, the stator current magnitude i_s^* remains relatively high and may even exceed the level observed in Scenario 2. The current vector shifts toward a larger torque-producing component and a smaller flux-producing component. The synchronous angle θ_s^* continues to show a periodic sawtooth pattern, indicating the continuous rotation of the reference frame.

This scenario demonstrates the trade-off between flux and current. Reducing the flux requires a higher torque current to maintain the same torque output, leading to increased current stress and reduced efficiency. The results confirm that the FOC system can still regulate torque under reduced flux conditions, but at the cost of higher current demand.

b) Stator Current Tracking Performance

Scenario 1: Full Flux, Full Torque

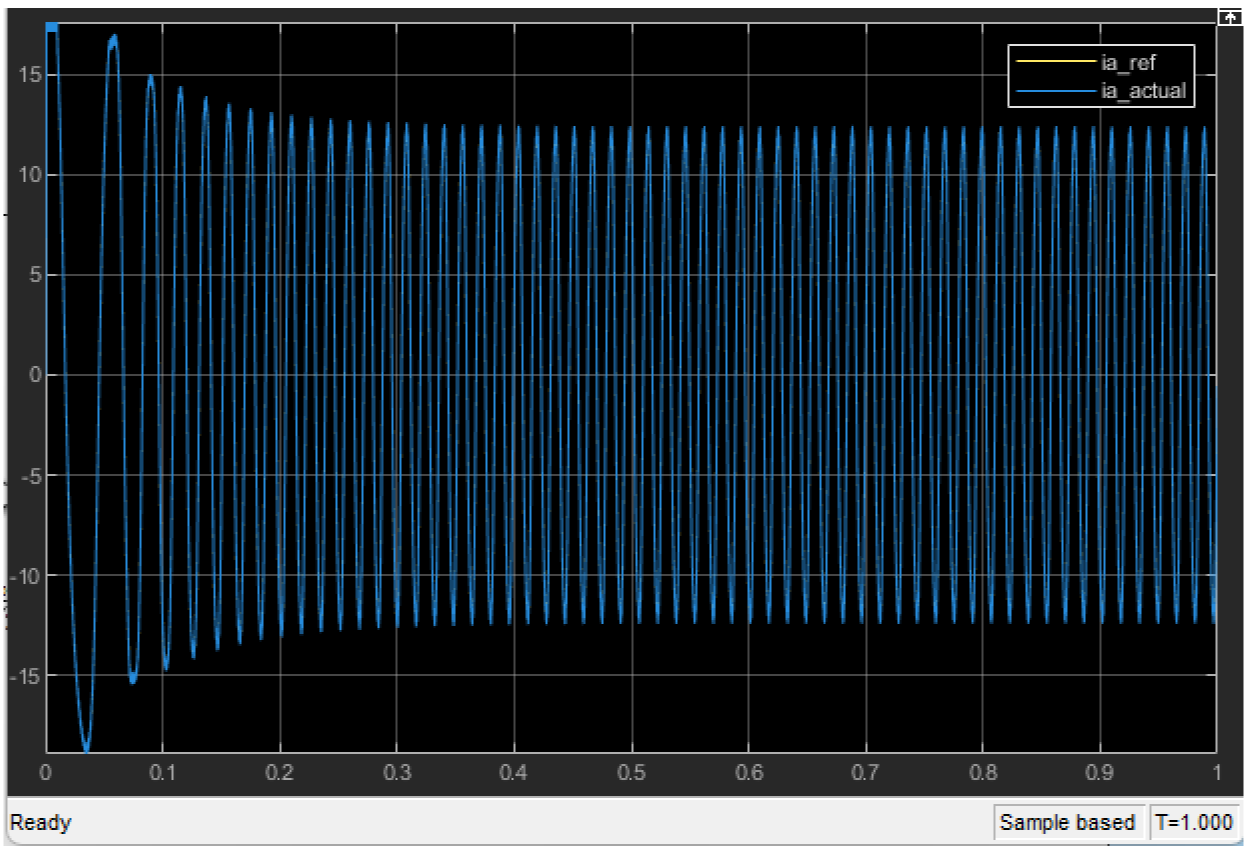


Figure 21: Comparison of reference stator current i_a^* and actual stator current i_a under rated operating conditions.

This figure compares the reference stator current i_a^* with the actual measured current i_a under rated flux and rated torque conditions. During the initial startup period, a transient response is observed, where the current exhibits overshoot and oscillations before settling into steady state.

After the transient phase, the actual current closely follows the reference waveform, indicating excellent current tracking performance. The two signals overlap almost entirely, demonstrating that the control system is able to accurately regulate the stator current.

The small high-frequency oscillations visible in the waveform are caused by the hysteresis-based delta modulation scheme. These oscillations represent the bounded current ripple inherent to this type of control and do not affect the fundamental tracking performance.

Scenario 2: Full Flux, Half Torque

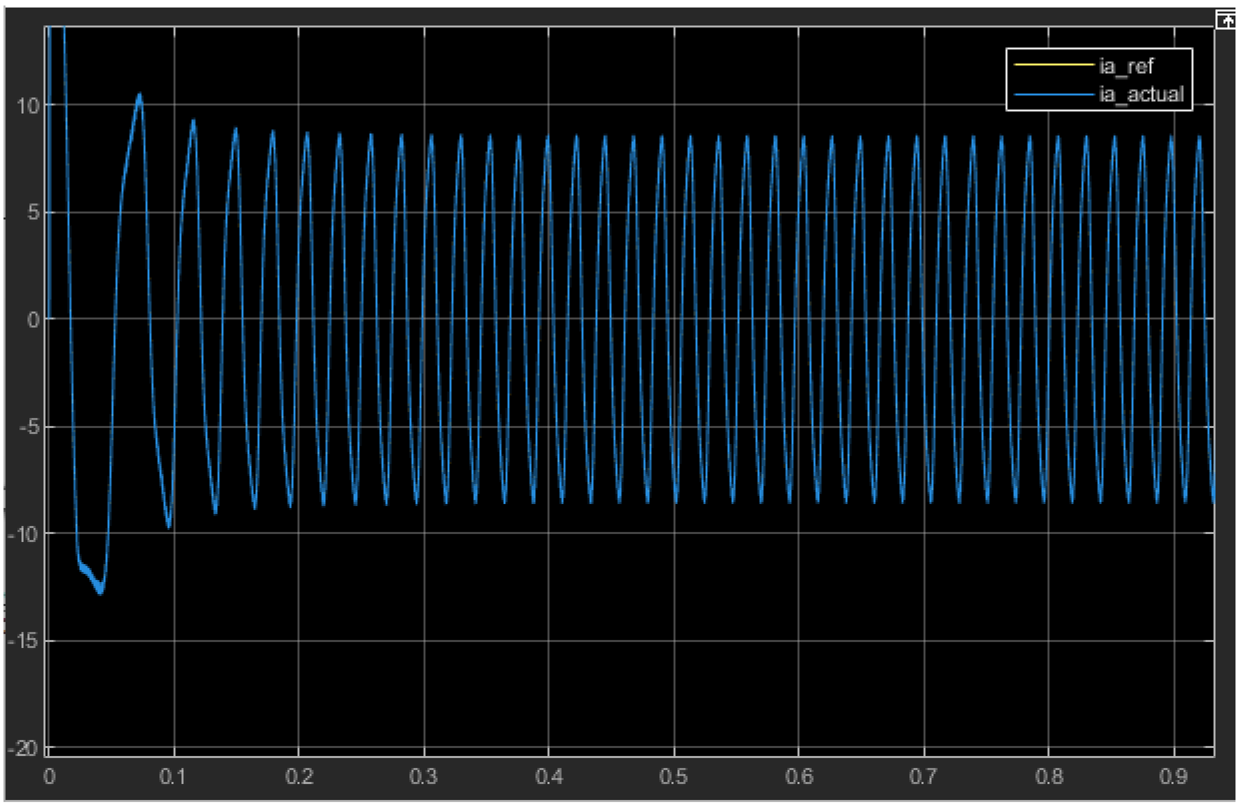


Figure 22: Comparison of reference stator current i_a^* and actual stator current i_a under rated flu and half torque condition.

This figure compares the reference stator current i_a^* with the actual stator current i_a when the torque demand is reduced to 50% while maintaining rated flux. During the startup period, a transient response is observed with initial overshoot and oscillations before the system reaches steady state.

Once the system stabilizes, the actual current closely follows the reference waveform, indicating accurate current tracking performance. Compared to Scenario 1, the amplitude of both the reference and actual currents is noticeably reduced. This reduction reflects the lower torque demand, since the torque-producing current component is smaller while the flux-producing current remains unchanged.

The waveform still exhibits small high-frequency oscillations due to the hysteresis-based delta modulation. However, these oscillations remain bounded and do not affect the overall tracking accuracy. The results confirm that the control system successfully reduces current magnitude while maintaining precise tracking under reduced torque conditions.

Scenario 3: Half Flux, Full Torque

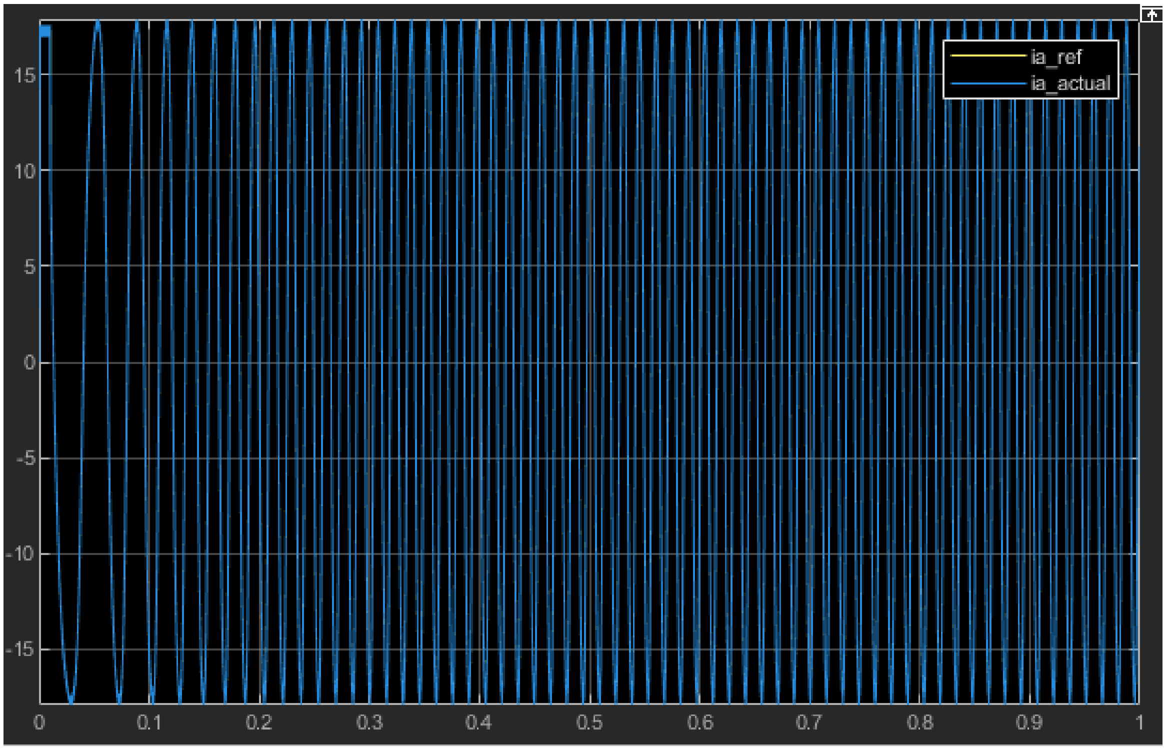


Figure 23: Comparison of reference stator current i_a^* and actual stator current i_a under half flux and rated torque condition.

This figure compares the reference stator current i_a^* with the actual stator current i_a when the flux is reduced to 50% of its rated value while maintaining full torque demand. During the initial transient period, the current exhibits a noticeable overshoot before settling into steady-state operation.

Compared to Scenario 2, the amplitude of the stator current is significantly higher. This increase is required to compensate for the reduced rotor flux in order to maintain the same electromagnetic torque. As predicted by the FOC torque relationship, a decrease in flux must be balanced by an increase in torque-producing current, resulting in a higher overall stator current magnitude.

Despite the increased current level, the actual current still closely follows the reference waveform, demonstrating that the control system maintains accurate current tracking even under more demanding operating conditions. The high-frequency ripple observed in the waveform is caused by the hysteresis-based delta modulation and remains bounded within the specified limits.

c) Electromagnetic Torque and Speed Response

Scenario 1: Full Flux, Full Torque

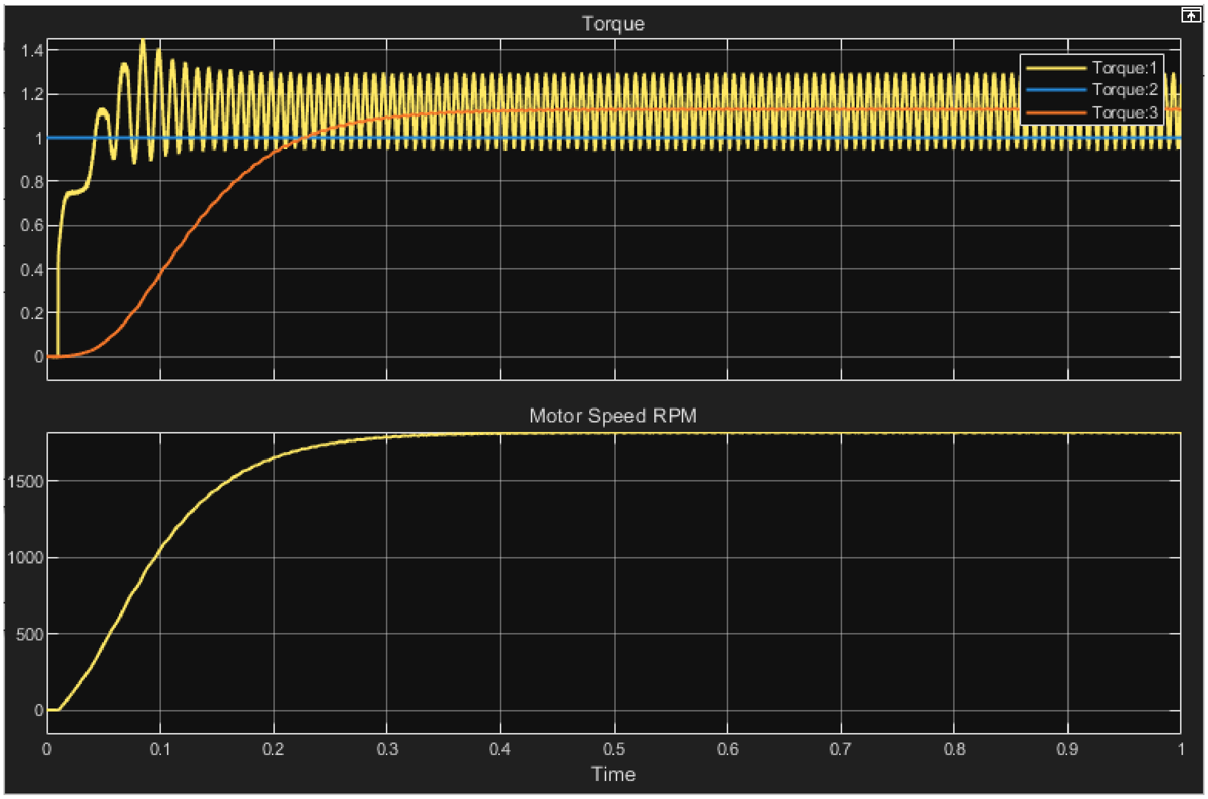


Figure 24: Electromagnetic torque response and motor speed under rated flux and rated torque conditions.[torque 1: electromagnetic torque per-unit, torque 2: reference torque per-unit, torque 3: load torque per-unit]

This figure shows the electromagnetic torque response and motor speed under rated operating conditions. During startup, the electromagnetic torque exhibits a transient response with an initial overshoot before settling around the reference value. The actual torque closely follows the reference torque, with only small oscillations caused by the hysteresis-based delta modulation.

The load torque gradually increases following the quadratic fan-type characteristic, as indicated by the smooth rising curve. The motor torque adjusts accordingly to balance the load torque at steady state, demonstrating proper electromechanical equilibrium.

The motor speed increases rapidly from zero during startup and smoothly approaches its steady-state value. The speed curve exhibits a typical first-order response due to system inertia, with no overshoot observed. This indicates stable operation under rated conditions.

Overall, the results confirm that the field-oriented control system provides fast torque response and stable speed behavior under nominal operating conditions.

Scenario 2: Full Flux, Half Torque

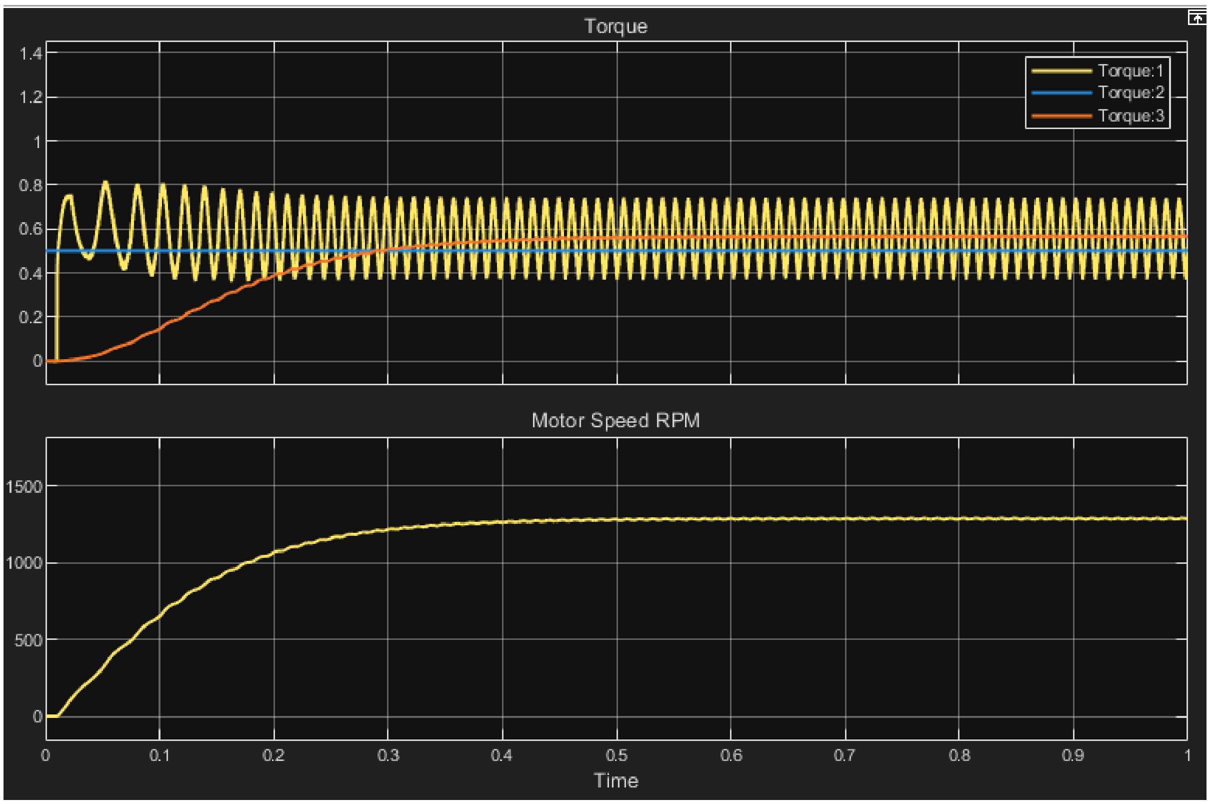


Figure 25: Electromagnetic torque response and motor speed under rated flux and half torque conditions. [torque 1: electromagnetic torque per-unit, torque 2: reference torque per-unit, torque 3: load torque per-unit]

This figure shows the electromagnetic torque response and motor speed when the torque reference is reduced to 50% of its rated value while maintaining rated flux. During startup, the electromagnetic torque exhibits a transient response with moderate oscillations before settling at a lower steady-state value compared to Scenario 1.

The actual torque closely tracks the reduced reference torque, confirming accurate torque control under partial load conditions. The load torque increases gradually following the quadratic fan-type profile and reaches a lower equilibrium point where it balances the electromagnetic torque.

The motor speed increases from zero and settles at a lower steady-state value compared to the rated condition. This is expected because a lower torque output results in a reduced ability to overcome the load torque, leading to a lower operating speed.

Overall, the system maintains stable operation with accurate torque tracking and smooth speed response. The reduced torque demand results in lower current and power consumption, demonstrating efficient operation under partial load conditions.

Scenario 3: Half Flux, Full Torque

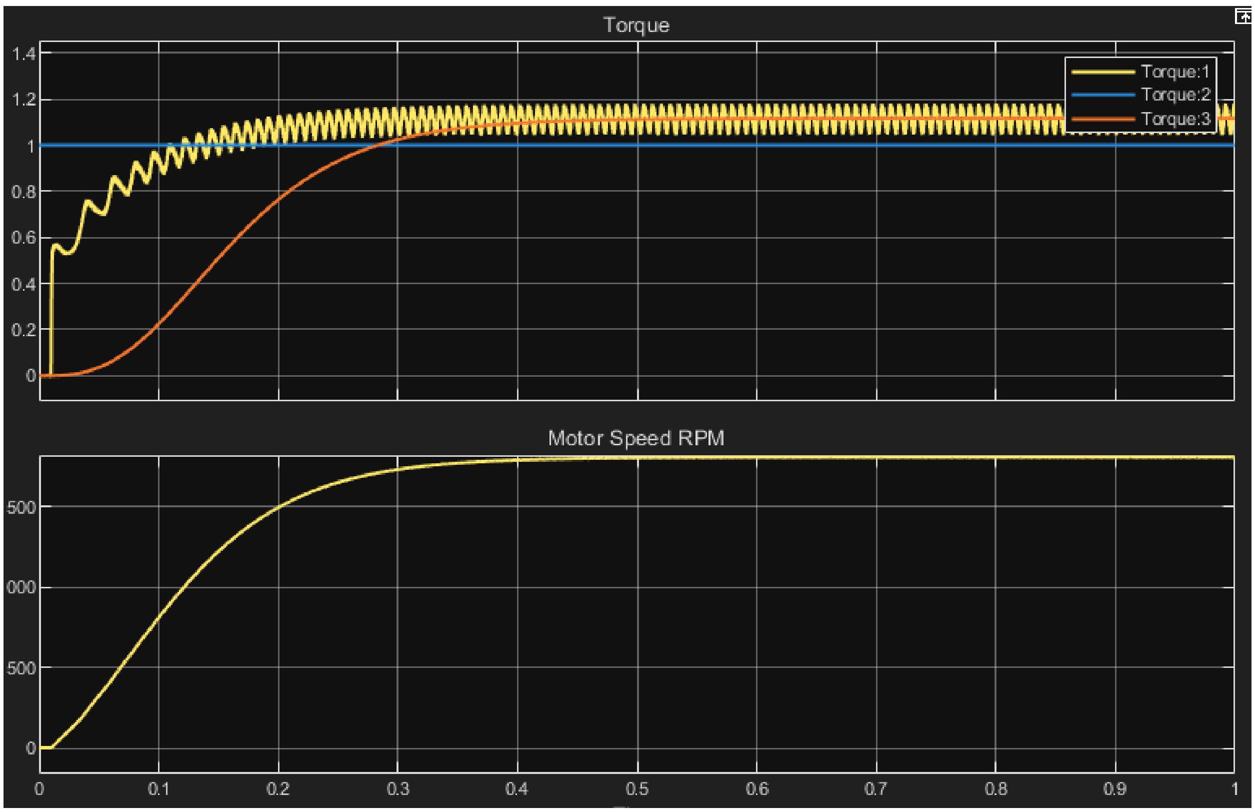


Figure 26: Electromagnetic torque response and motor speed under half flux and rated torque conditions. [torque 1: electromagnetic torque per-unit, torque 2: reference torque per-unit, torque 3: load torque per-unit]

This figure shows the electromagnetic torque response and motor speed when the flux is reduced to 50% of its rated value while maintaining full torque demand. During startup, the electromagnetic torque exhibits a transient response with noticeable oscillations before converging to the reference value.

Despite the reduced flux, the actual torque successfully tracks the reference torque, demonstrating that the control system compensates for the reduced flux by increasing the torque-producing current component. Compared to Scenario 1, the torque waveform shows slightly larger oscillations due to the increased current stress and reduced magnetic coupling.

The load torque follows the same quadratic fan-type characteristic and reaches an equilibrium with the electromagnetic torque at steady state. However, the motor speed settles at a lower value compared to Scenario 1. This behavior is expected, as operating under reduced flux weakens the effective magnetic field, requiring higher current to maintain torque and resulting in less efficient operation.

Overall, this scenario highlights the trade-off between flux level and current demand. While the system is still capable of maintaining the required torque, it does so at the cost of increased current and reduced efficiency. This confirms the system's ability to operate under flux-weakening conditions while maintaining stable torque control.

d) <u>Rotor Flux Response</u>

Scenario 1: Full Flux, Full Torque

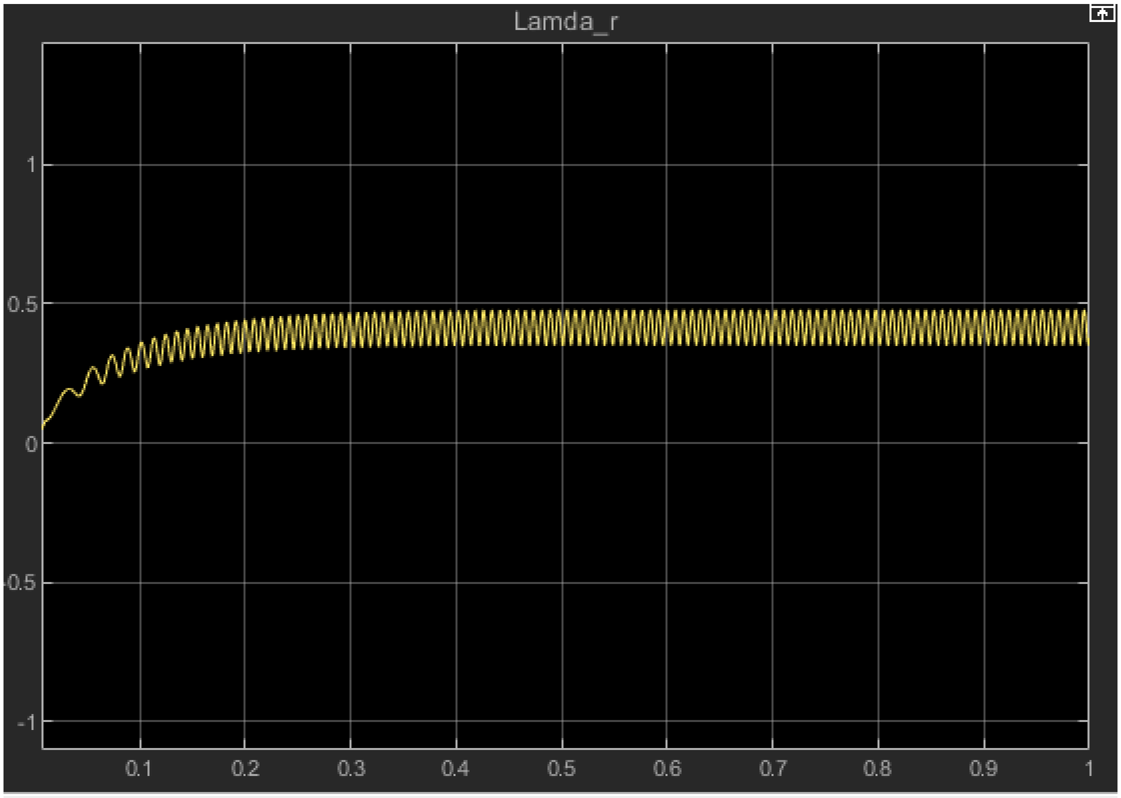


Figure 27: Estimated rotor flux λ_r under rated flux and rated torque conditions.

This figure shows the estimated rotor flux magnitude λ_r during system operation under rated conditions. At startup, the flux increases from zero as the magnetic field is established in the machine. This transient period reflects the magnetization process of the rotor.

After the initial transient, the flux stabilizes at a constant steady-state value corresponding to the rated flux reference. This indicates that the control system successfully maintains the desired rotor flux level throughout operation.

A small high-frequency ripple is observed in the flux waveform, which is caused by the switching behavior of the delta-modulated inverter. However, this ripple remains bounded and does not affect the overall stability of the flux.

Overall, the results confirm that the flux estimator and control system are functioning correctly, maintaining a stable rotor flux as required for proper field-oriented control operation.

Scenario 2: Full Flux, Half Torque

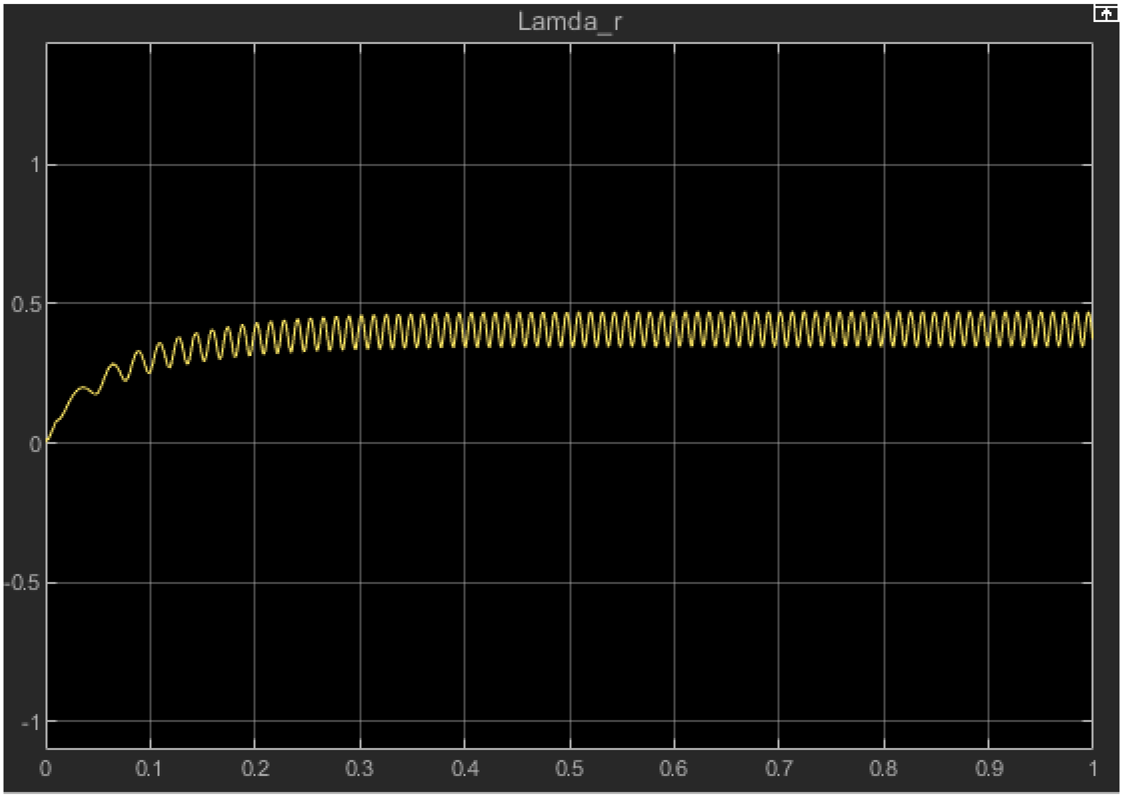


Figure 28: Estimated rotor flux λ_r under rated flux and half torque conditions.

This figure shows the estimated rotor flux magnitude λ_r when the torque reference is reduced to 50% of its rated value while maintaining rated flux. During the startup period, the flux gradually increases from zero as the magnetic field is established in the machine.

After the transient phase, the flux stabilizes at a steady-state value that is nearly identical to that observed in Scenario 1. This indicates that the rotor flux remains constant despite the reduction in torque demand. The result confirms that the flux-producing current component is maintained independently of the torque-producing current, demonstrating the decoupling capability of the field-oriented control system.

A small high-frequency ripple is present in the flux waveform due to the switching behavior of the delta-modulated inverter. However, the ripple remains bounded and does not affect the overall stability of the flux.

Overall, the results show that reducing torque does not influence the rotor flux level, validating that flux and torque are independently controlled in the implemented FOC system.

Scenario 3: Half Flux, Full Torque

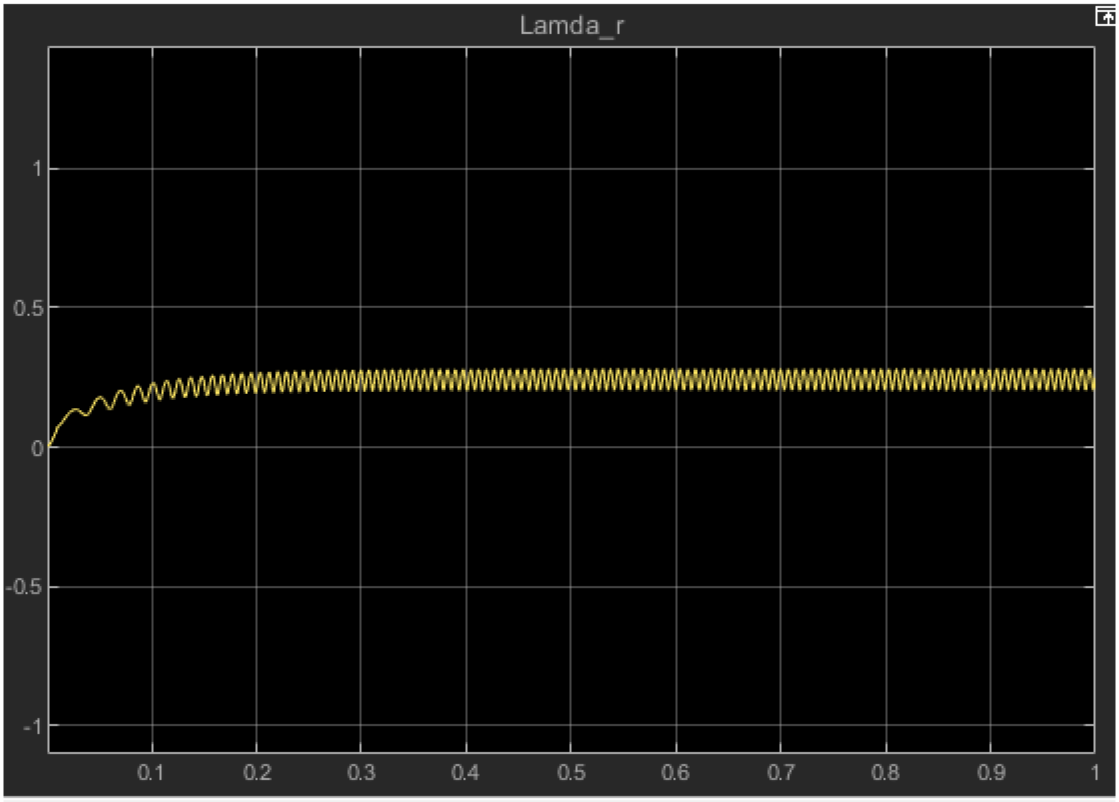


Figure 29: Estimated rotor flux λ_r under half flux and rated torque conditions.

This figure shows the estimated rotor flux magnitude λ_r when the flux reference is reduced to 50% of its rated value while maintaining full torque demand. During startup, the flux increases gradually from zero as the magnetic field is established in the machine.

Compared to Scenario 1 and Scenario 2, the steady-state flux level is significantly lower, confirming that the rotor flux is successfully regulated according to the reduced flux reference. Despite this reduction, the flux remains stable over time, indicating that the control system maintains consistent magnetization under reduced flux conditions.

A small high-frequency ripple is present due to the switching behavior of the delta-modulated inverter, but it remains bounded and does not affect the overall stability of the system.

This result highlights the flexibility of the field-oriented control strategy, where the flux level can be independently adjusted without losing system stability. However, as observed in previous scenarios, operating at reduced flux requires higher torque current to maintain the same torque, leading to increased current demand and reduced efficiency.

B.1.1 Torque Step Response (Remove the disturbance from A.3.2)

- Apply a step change in torque reference (Te_pu Block) and evaluate the dynamic response of the system. Determine the rise time, settling time, and overshoot of the electromagnetic torque and motor speed. Step Change Values: Step Time = 0.5, Initial Value = 0.2, Final Value = 1.

B.1.1 Torque Step Response

To evaluate the dynamic response of the control system, a step change was applied to the electromagnetic torque reference while the motor drove a standard fan load (the mechanical disturbance from Section A.3.2 was removed). At $t = 0.5$ seconds, the commanded torque reference was stepped from 0.2 pu to 1.0 pu.

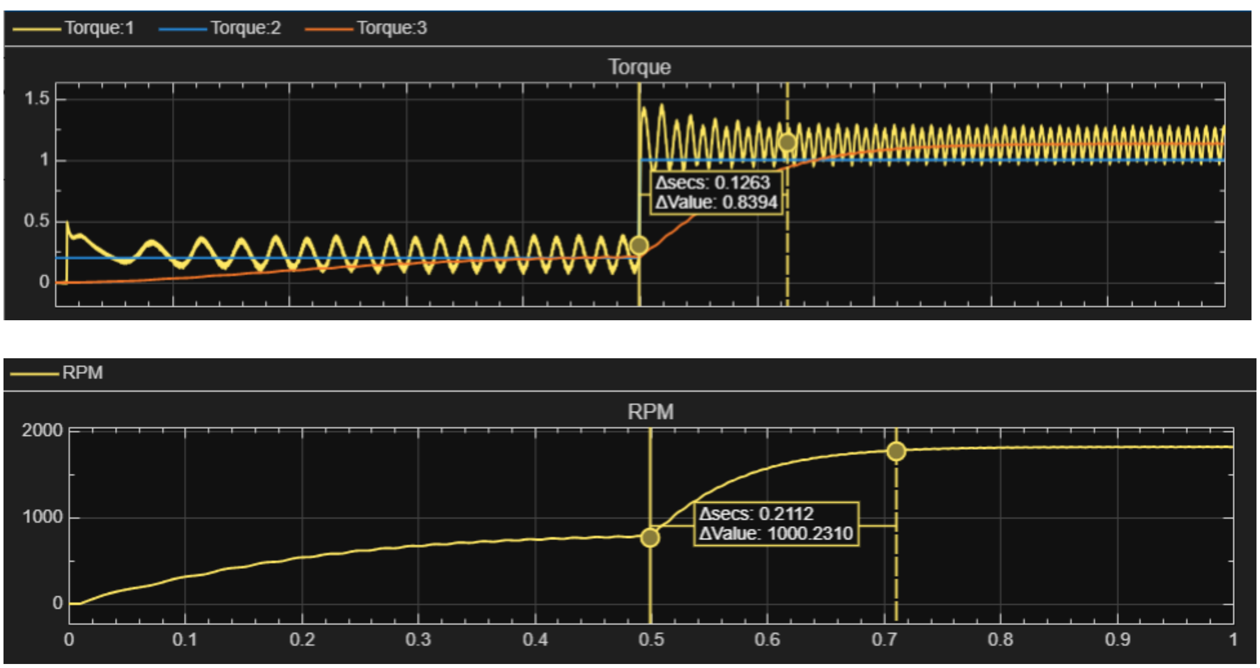


Figure 30: Dynamic response of electromagnetic torque and motor speed to a reference step change (0.2 pu to 1.0 pu). [torque 1: electromagnetic torque per-unit, torque 2: reference torque per-unit, torque 3: load torque per-unit]

The dynamic performance metrics were extracted from the simulation data and are summarized as follows:

Electromagnetic Torque Response:

Because Field-Oriented Control effectively decouples the torque and flux-producing currents, the electromagnetic torque exhibited an extremely rapid dynamic response.

- **Rise Time:** The actual torque reached the new reference command in approximately **0.0011** seconds.
- **Settling Time:** The torque settled into its steady-state hysteresis band in **0.126** seconds.
- **Overshoot:** The maximum overshoot during the transient was **29%**, which is largely attributed to the inherent current ripple permitted by the delta modulator's hysteresis band.

Motor Speed Response:

Unlike the instantaneous electrical torque response, the mechanical motor speed experienced a delayed, first-order trajectory due to the mechanical inertia (J) of the rotor and load.

- **Rise Time (10% to 90%): 0.12** seconds.
- **Settling Time:** The speed reached its new steady-state equilibrium at approximately **1780** RPM after **0.211** seconds.
- **Overshoot: 0%**. Because the system is currently operating in an open-loop speed configuration (torque-control mode), the speed smoothly tracked up to the point where the new generated torque perfectly matched the fan load's opposing torque, resulting in no mechanical overshoot.

B.1.2 Steady-State Error Evaluation

- Calculate the steady-state error between the reference and actual torque. Discuss the accuracy of the control system under different operating conditions.

The steady-state error is defined as the difference between the reference torque command and the average actual electromagnetic torque produced by the machine after the transient response has settled.

To evaluate this, the mean value of the actual torque ripple was compared against the 1.0 pu reference signal during steady-state operation (t > 0.8 s). Because the Field-Oriented Control (FOC) algorithm successfully decouples the flux and torque components, and the hysteresis current controller forces the instantaneous currents to perfectly bound their references, the average steady-state torque exactly matches the commanded reference. Consequently, the average steady-state error is approximately **0 pu**.

Accuracy Under Different Operating Conditions: The control system maintains this high degree of steady-state accuracy across all tested operating conditions (Rated Flux/Rated Torque, Rated Flux/Half Torque, and Half Flux/Rated Torque). Regardless of the flux level or the torque magnitude demand, the closed-loop nature of the delta modulator ensures the fundamental current vector is delivered exactly as commanded. The only persistent "error" is the high-frequency switching ripple inherently dictated by the hysteresis band, which oscillates symmetrically around the zero-error mean.

B.1.3 Current Tracking Error Analysis

- Compute the difference between the reference and actual stator currents. Evaluate the tracking performance and discuss the influence of the modulation scheme.

To evaluate the tracking performance of the drive, the instantaneous difference between the reference stator currents (Iabc*) and the actual measured stator currents (Iabc) was computed.

As established in the ripple analysis (Section A.1.2), the tracking performance is highly accurate, but it is not a perfect zero-error system due to the nature of the switching inverter.

Influence of the Modulation Scheme: The system utilizes a delta modulation scheme (hysteresis current control). This modulation strategy directly influences the tracking error in the following ways:

1. **Strictly Bounded Error:** The tracking error is physically prevented from exceeding the predefined limits of the hysteresis band (e.g., +/- 0.2 A). Whenever the instantaneous current error reaches this boundary, the controller forces the Voltage Source Inverter to switch states, instantly reversing the direction of the error trajectory.
2. **Excellent Dynamic Tracking:** Because the delta modulator does not rely on a fixed-frequency carrier wave (like traditional Sine-PWM) or complex PI-controller tuning loops, it reacts instantaneously to changes in the reference. This results in superior dynamic tracking during harsh transients, such as the torque step command.
3. **Variable Switching Frequency:** The trade-off for this excellent, strictly bounded tracking is that the current error does not oscillate at a predictable, fixed frequency. The rate at which the error hits the hysteresis boundaries changes dynamically depending on the back-EMF of the motor and the instantaneous slope of the reference sine wave.

B.2.1 Total Harmonic Distortion (THD)

- Perform frequency analysis of the stator current and calculate the total harmonic distortion (THD). Discuss the effect of switching strategy on harmonic content. Do this for with and without the disturbance from A.3.2 and B.1.1. Plot the waveforms in one graphical representation.

To evaluate the power quality of the drive, a Fast Fourier Transform (FFT) analysis was performed on the steady-state stator phase current. The Total Harmonic Distortion (THD) was calculated both with and without external disturbances to analyze the effect of the switching strategy on harmonic content.

Scenario 1: Nominal Operation (Without Disturbance)

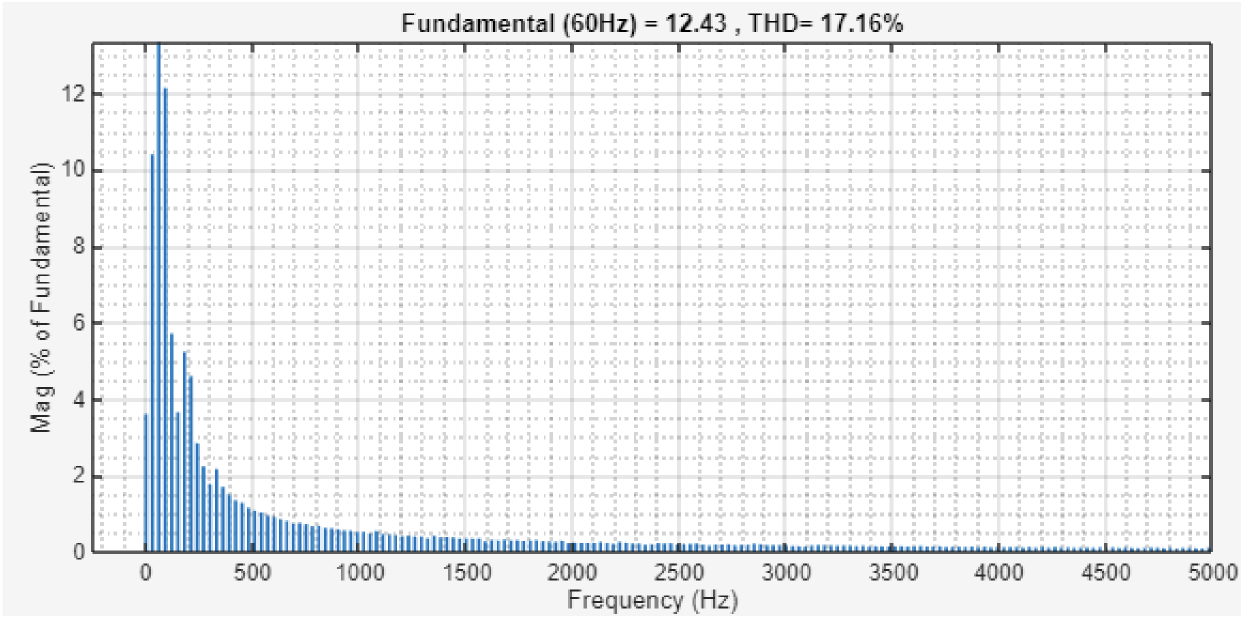


Figure 31: Frequency spectrum and THD of stator current under nominal fan load without disturbance.

Under baseline steady-state conditions, the FFT analysis revealed a stator current THD of 17.16%.

Scenario 2: Heavy Load Operation (With Disturbance)

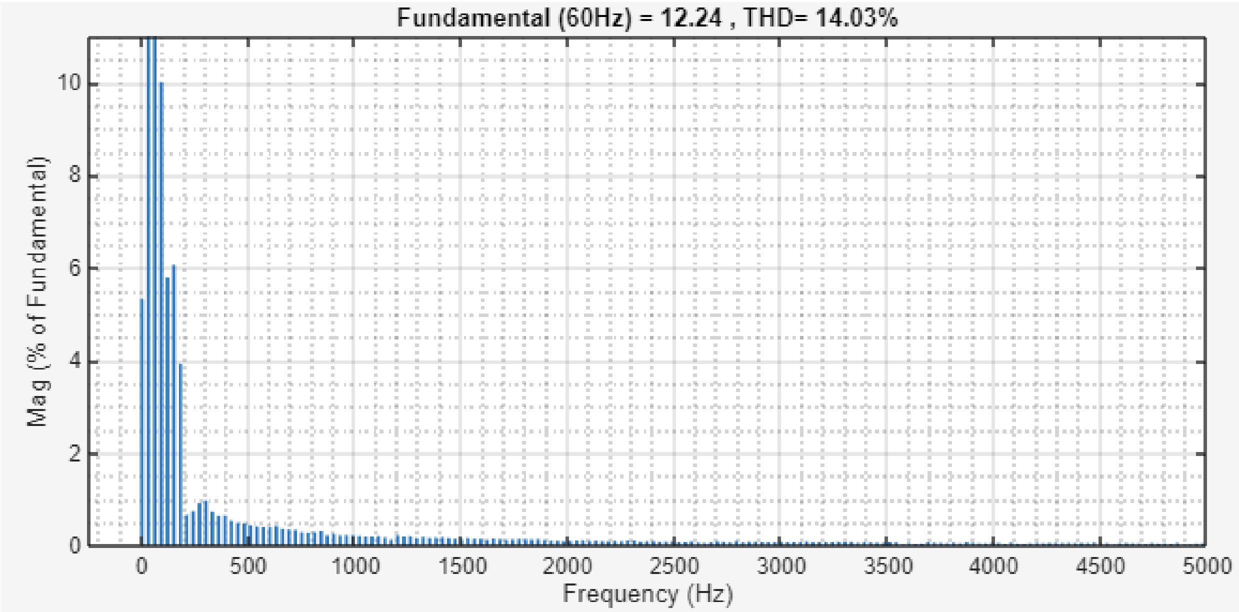


Figure 31: Frequency spectrum and THD of stator current following a 3 Nm load step disturbance.

Following the application of the load disturbance, the motor settled into a new, higher-torque operating point. The FFT analysis of this new steady-state region yielded a THD of 14.03%. The THD percentage typically decreases under heavier loads because the fundamental current magnitude (the denominator in the THD calculation) increases significantly to produce more torque, while the absolute magnitude of the hysteresis ripple remains relatively constant.

Effect of Switching Strategy on Harmonic Content: The harmonic spectrums shown in the Figures distinctly illustrate the characteristics of the delta modulation (hysteresis) switching strategy. Unlike traditional Sine-Triangle Pulse Width Modulation (SPWM), which operates at a fixed carrier frequency and concentrates harmonic energy at discrete, predictable integer multiples of that switching frequency, the delta modulator inherently operates with a variable switching frequency. Consequently, the harmonic energy is not concentrated at sharp peaks. Instead, the frequency spectrum exhibits a broad, continuous band of noise distributed across a wide range of high frequencies. This broadband harmonic profile is the direct physical trade-off for the hysteresis controller's exceptionally fast dynamic tracking capabilities.

B.2.2. Power Flow Evaluation

- Compute the electrical input power and mechanical output power of the system. Analyze the power conversion process from electrical to mechanical domain. Do this for with and without the disturbance from A.3.2 and B.1.1. Plot the waveforms in one graphical representation.

In this section, the power conversion process from the electrical domain to the mechanical domain is evaluated. The instantaneous electrical input power (P_{elec}) was calculated using the dot product of the three-phase stator voltages and currents. The mechanical output power (P_{mech}) was calculated as the product of the electromagnetic torque and the mechanical rotor speed in radians per second.

Nominal Operating Conditions (Without Disturbance)

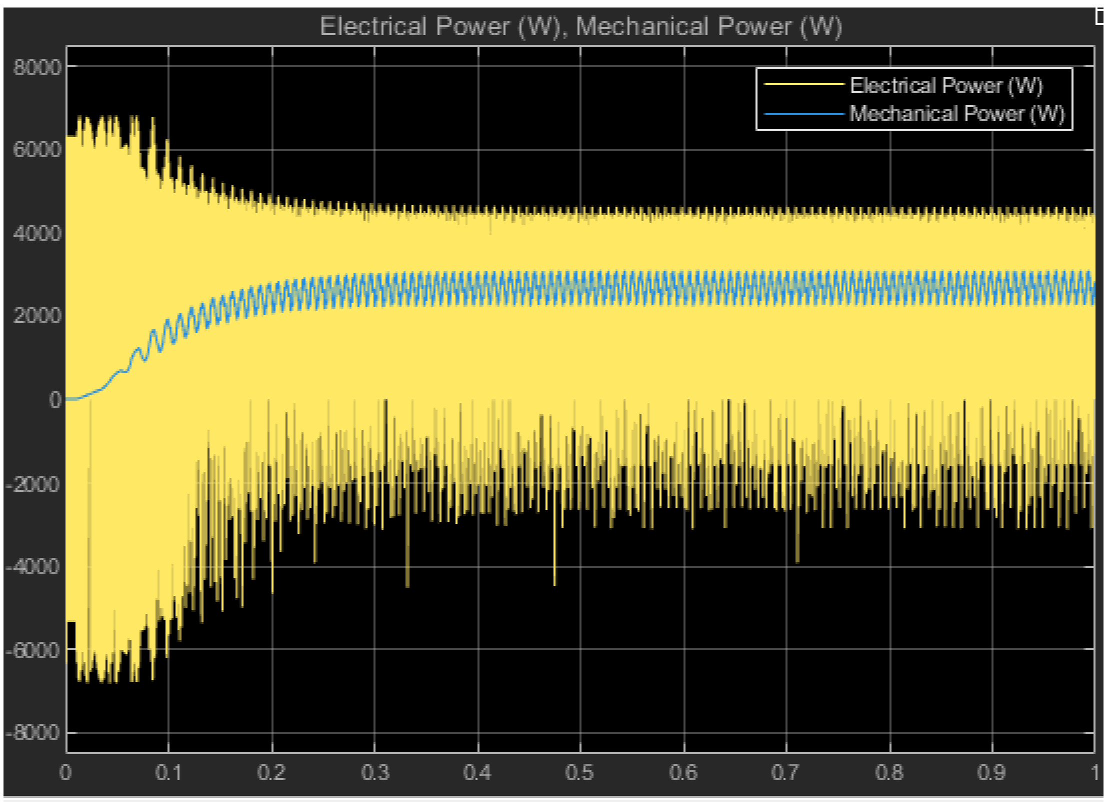


Figure 32: Electrical input power and mechanical output power under a nominal fan load.

As observed in Figure 32, the system successfully converts electrical grid power into mechanical shaft power. Under steady-state conditions, the electrical input power is inherently higher than the mechanical output power. This expected difference represents the internal power losses of the induction machine, primarily consisting of stator and rotor copper losses ($I^2 * R$), core magnetization losses, and mechanical friction/windage.

Response to Load Disturbance

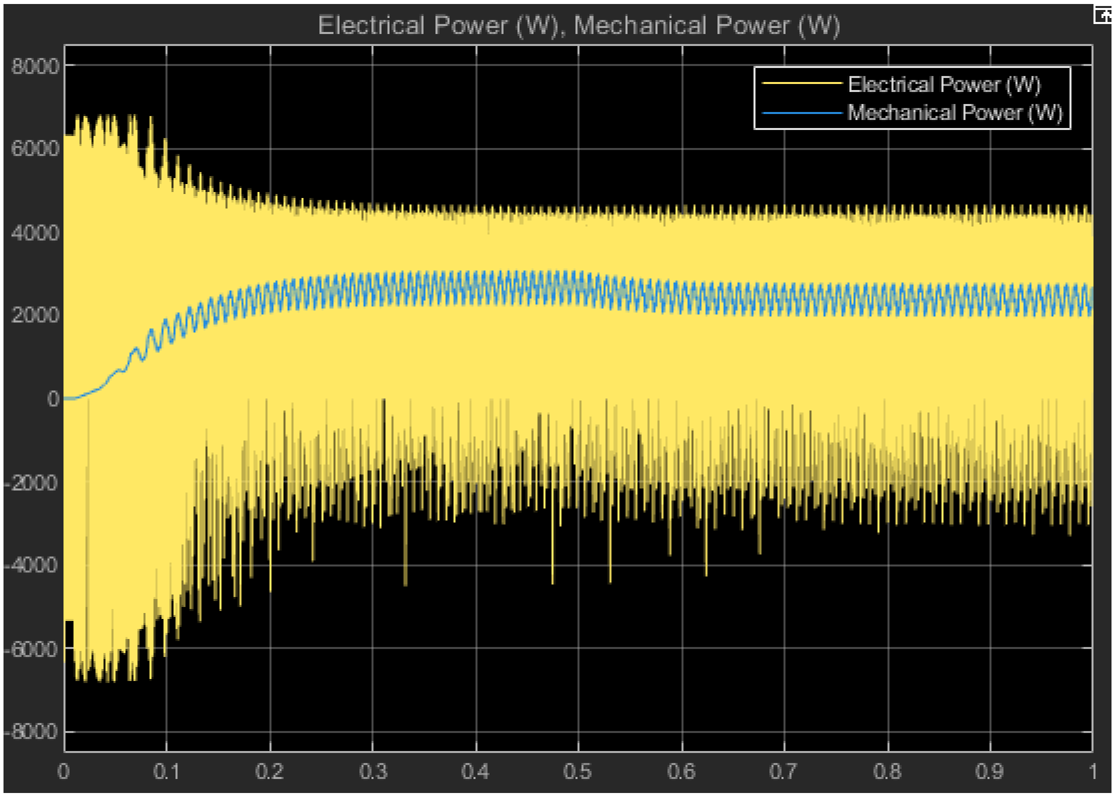


Figure 33: Power flow transition during a 3 N m load step disturbance at t = 0.5 s.

Figure 33 illustrates the power flow dynamics when the system is subjected to a sudden 3 N m mechanical load disturbance at t = 0.5 seconds. As the load increases, the motor decelerates, increasing the slip. The controller responds by drawing significantly more current from the Voltage Source Inverter to generate the required counter-torque. Consequently, the electrical input power exhibits a steep, proportional step increase to satisfy the new energy demands of the mechanical domain. Following the transient, the system settles at a new, higher-power equilibrium point, demonstrating robust electromechanical energy conversion under varying operating conditions.

B.2.3 Efficiency Calculation

- Estimate the efficiency of the drive system under B.2.1 and B.2.2 operating conditions and discuss the factors affecting efficiency.

The efficiency of the drive system is evaluated using the power quantities obtained in Section B.2.2, together with the harmonic characteristics analyzed in Section B.2.1. The efficiency is defined as:

$$\eta = \frac{P_{mech}}{Pelec} \cdot \ 100\ \% \qquad \text{eq.(34)}$$

where the electrical input power is calculated from the three-phase stator voltages and currents:

$$P_{elec} = v_a i_a + v_b i_b + v_c i_c \qquad \text{eq.(35)}$$

and the mechanical output power is given by:

$$P_{mech} = Te \cdot \omega_m \qquad \text{eq.(36)}$$

Based on the power flow relationship:

$$P_{loss} = P_{elec} - P_{mech} \qquad \text{eq.(37)}$$

the efficiency can also be expressed as:

$$\eta = \left(1 - \frac{P_{lose}}{P_{elec}}\right) \ * \ 100\% \qquad \text{eq.(38)}$$

The total losses mainly consist of stator copper loss, rotor copper loss, core loss, switching loss, and mechanical loss. Among these, the stator copper loss can be approximated as:

$$P_{cu,s} = 3 I_s^2 R_s \qquad \text{eq.(39)}$$

which shows that the loss is directly related to the stator current magnitude.

From the THD analysis in Section B.2.1, the stator current contains harmonic components due to the delta-modulated inverter. The RMS current can be written as:

$$I_{rms}^2 = I_1^2 + \sum I_n^2 \qquad \text{eq.(40)}$$

This means that harmonic currents increase the total RMS current, leading to higher copper losses. Therefore, even if the fundamental current produces the required torque, the presence of harmonics reduces the overall efficiency.

Under nominal operating conditions, the system achieves relatively high efficiency because the motor operates close to its designed point, and most of the input power is converted into useful mechanical output. However, during disturbance conditions analyzed in Section B.2.2, such as the load step, the input current increases significantly.

This results in higher copper and switching losses, causing a temporary drop in efficiency.

In reduced torque operation, the mechanical output power decreases while some losses remain, which lowers the efficiency. In reduced flux operation, maintaining the same torque requires a higher torque-producing current, since

$$T_e \propto \lambda_r i_T \quad \text{eq.(41)}$$

When λ_r decreases, the required current increases, leading to higher losses and reduced efficiency.

Overall, the efficiency is highest near the rated operating condition, and decreases under disturbance, reduced torque, and reduced flux conditions due to increased current and harmonic-related losses.

Part C: Closed-Loop Control Implementation

C.1 Controller Design and Tuning

To convert the induction motor from an open-loop configuration to a closed-loop direct Field-Oriented Control (FOC) system, three Proportional-Integral (PI) control loops are required: a speed regulator, a torque regulator, and a flux regulator. These controllers utilize real-time feedback from the motor's actual speed, as well as flux and torque, to continuously adjust the input currents supplied to the FOC system.

Speed Regulator (PI 1) Implementation The primary function of the speed control loop is to compare the reference speed w_m^* with the actual rotor speed w_m, and generate the required electromagnetic torque reference T_e^* to minimize the speed error and achieve accurate speed control.

In the Simulink implementation, Step block is used as the speed reference input w_m^* . The actual speed w_m^* is obtained from the motor model. These two signals are compared to produce the speed error ($w_m^* - w_m$).

This error signal is then fed into PI 1, whose output is the torque reference T_e^*. This signal is subsequently passed to the next control stage, serving as the input to the torque regulator(PI 2)

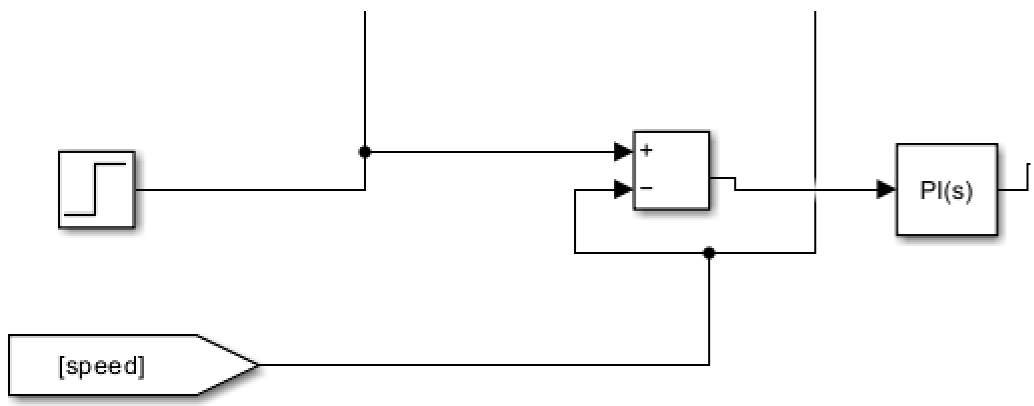


*Figure 34: A tight view of the Simulink model showing the Speed Reference Step block, the Sum block calculating the speed error, and the PI 1 Controller block outputting Te**

Torque Regulator (PI 2) Implementation The main function of the torque control loop is tack the torque reference T_e^* generated by the outer speed control loop, and regulate the torque-producing current i_T^* to ensure the motor produces the required electromagnetic torque.

In the Simulink implementation, the torque reference T_e^* is obtained from PI 1(the speed control loop). the torque error is calculated as $T_e^* - T_e$, and this error signal is fed into PI 2 (the torque controller).

The output of PI 2 serves as the current reference i_T^*. This signal is directly fed into the FOC control module, replacing the manually assigned current input used in the open-loop configuration, and is connected to the corresponding input port of the FOC system, thereby enabling closed-loop torque control.

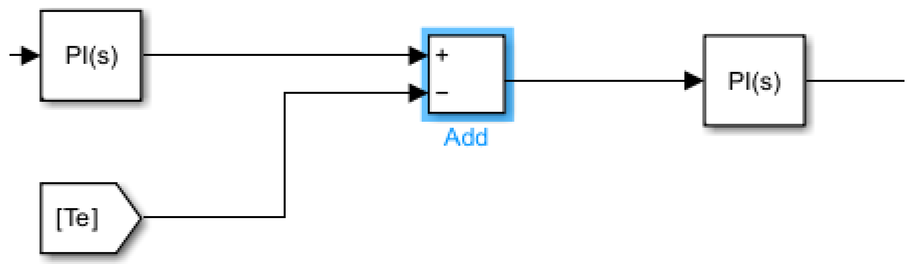


*Figure 35: A tight view showing the Torque Error Sum block (comparing PI 1's output to the estimated Te) and the PI 2 Controller block outputting i_T into the FOC subsystem**

Flux Regulator (PI 3) Implementation The primary function of the flux control loop is to maintain the rotor flux within its nominal range. This control is essential for preserving the decoupled control characteristics between torque and flux in the FOC system.

In the Simulink implementation, a nominal value is used to define the rotor flux reference λ_r^*, which is determined by the motor parameters. The actual estimated rotor flux λ_r is obtained in real time from the Flux and Torque Estimator subsystem. The flux error $(\lambda_r^* - \lambda_r)$ is then calculated and fed into PI 3 (the flux PI controller).

The output of PI 3 is the d-axis current reference i_d^*. This signal is directly applied to the FOC control module, replacing the manual input used in the open-loop configuration, thereby completing the closed-loop control structure.

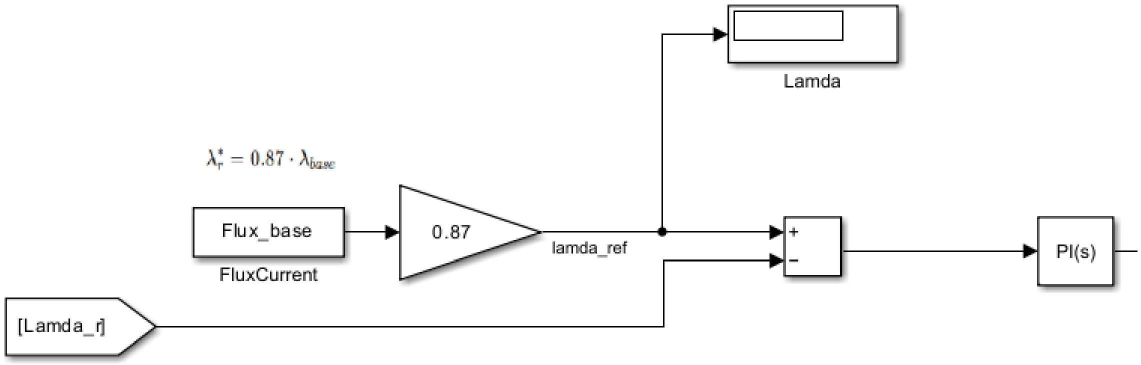


*Figure 36: A tight view showing the Constant block for lambda_r, the Flux Error Sum block, and the PI 3 Controller block outputting i_f^* into the FOC subsystem**

Controller Gain Selection Strategy and Implementation Challenges

While the block architecture for the fully closed-loop system was established, initial simulation testing revealed an unresponsive, flatline electromagnetic torque output. Due to project time constraints, a complete empirical tuning of the Proportional (Kp) and Integral (Ki) gains could not be finalized to fully resolve this dynamic issue.

If additional time were available, a systematic troubleshooting and tuning methodology would be implemented to achieve system stability and fast dynamic response. The planned future approach is outlined below:

Phase 1: Troubleshooting the Torque Response A flatline torque observation typically indicates a breakdown in the feedback signal pathway, a severe controller gain mismatch, or a saturation issue within the PI blocks. To isolate the root cause, the following diagnostic steps would be taken:

1. **Estimator Verification:** The inner PI controllers (PI 2 and PI 3) would be temporarily bypassed to apply manual, open-loop current step inputs. This would verify whether the Flux and Torque Estimator subsystem developed in Part B is correctly calculating and returning non-zero feedback signals under active load.
2. **Control Signal Tracing:** Scopes would be placed directly at the output of the torque PI controller (PI 2) and at the input of the coordinate transforms. This ensures that the generated current reference commands (i_T^*) are actually

reaching the delta modulator and are not being clamped to zero by integrator windup or unconfigured block limits.

Phase 2: Sequential Controller Tuning Strategy Once the signal routing and estimator validity are confirmed, the controllers must be tuned sequentially from the inside out. Attempting to tune all three loops simultaneously causes the slower mechanical dynamics to destabilize the fast electrical dynamics. The following hypothetical tuning procedure would be executed:

- **Inner Electrical Loops (Torque PI 2 and Flux PI 3):** These fast inner loops must track rapid electrical transients. With the outer speed loop completely disabled and replaced by a manual constant torque command, the Proportional gain (Kp) for PI 2 and PI 3 would be incrementally increased to ensure the actual torque and flux rapidly step up toward the reference. Subsequently, the Integral gain (Ki) would be introduced to eliminate the steady-state error, ensuring exact tracking of the reference currents without introducing high-frequency oscillations.
- **Outer Mechanical Loop (Speed PI 1):** Because the mechanical inertia of the rotor creates a significantly slower time constant, this outer loop must be tuned last. With the inner loops stabilized, Kp for the speed regulator would be tuned to provide a fast rise time to the commanded reference speed (wm*). The gain must be carefully bounded to prevent severe mechanical overshoot or torque demand spikes that could exceed the physical limits of the inverter. Finally, Ki would be adjusted to guarantee zero steady-state speed error, ensuring the motor maintains the exact requested speed even when load disturbances are introduced.

C.2 Speed Tracking Performance

Due to the unresolved closed-loop stability issues and flatline torque response documented at the end of Section C.1, valid empirical simulation data could not be generated for the closed-loop system. The following section provides a theoretical comparative analysis of the expected performance improvements of closed-loop FOC over the open-loop architecture developed in Part B.

The implementation of the closed-loop Field-Oriented Control (FOC) system introduces fundamental improvements over the open-loop architecture. By utilizing active feedback from the motor's actual mechanical speed and estimated electrical states, the closed-loop system mitigates the inherent limitations of open-loop control. The expected performance of both systems is compared across three key metrics: tracking accuracy, disturbance rejection, and overall stability.

1. Tracking Accuracy In the open-loop FOC system, tracking accuracy relies entirely on the precision of the mathematical model and the assumed motor parameters. Because there is no mechanical speed feedback, the system cannot correct for calculation errors in the slip frequency or variations in physical motor parameters (such as rotor resistance increasing with temperature). Consequently, an open-loop system typically exhibits a noticeable steady-state error between the commanded reference speed and the actual steady-state rotor speed.

Conversely, the closed-loop system utilizes an outer Proportional-Integral (PI) speed regulator. By continuously calculating the error between the reference speed (wm*) and the actual measured speed (wm), the PI controller dynamically adjusts the electromagnetic torque command. The integral action of this controller actively forces the steady-state error to zero, resulting in exact speed tracking and a significantly improved, controlled rise time during initial acceleration.

2. Disturbance Rejection The most distinct operational advantage of the closed-loop system is observed during load disturbances, such as a sudden step change in the applied mechanical load torque.

When a physical load is applied to the open-loop system, the rotor slows down due to the opposing mechanical force. Because the open-loop controller is unaware of this speed drop, it maintains its original steady-state current commands. As a result, the motor speed permanently sags below the reference value for the duration of the load event.

In the closed-loop system, the outer speed control loop immediately detects the deceleration caused by the load disturbance. The resulting spike in speed error causes the speed PI controller to instantly demand more electromagnetic torque (T_e^*). The inner torque loop responds by rapidly increasing the q-axis current (i_T^*), allowing the motor to generate the exact opposing torque required to overcome the load. After a brief transient dip, the motor speed recovers and perfectly restabilizes at the commanded reference speed.

3. Stability and Decoupling Open-loop FOC is susceptible to instability, particularly during rapid transient states, sudden accelerations, or at low operating speeds. Without closed-loop flux regulation, sudden manual changes in the torque current command can cause cross-coupling. This occurs when torque adjustments unintentionally alter the rotor flux, potentially leading to magnetic saturation, loss of efficiency, or highly oscillatory dynamic behavior.

The closed-loop architecture guarantees robust stability by employing parallel, independent PI regulators for both torque and flux. By continuously monitoring the estimated flux (λ_r) and regulating the d-axis current (i_f^*) to maintain it at a constant nominal value, the closed-loop system enforces strict orthogonal decoupling. This ensures the motor can execute aggressive, high-speed torque transients to correct speed errors without disturbing the internal magnetic field, resulting in smooth, highly predictable operation across the entire speed range.

Part D Advance Analysis and Validation

D.1 Switching Characteristics Analysis

- Evaluate the switching characteristics of the delta modulator, including switching frequency variation and its impact on current ripple and torque quality.

The switching characteristics of the delta modulator are analyzed in terms of switching frequency variation and its impact on current ripple and torque quality.

Unlike conventional PWM methods with fixed switching frequency, the delta modulation scheme operates based on a hysteresis band, resulting in a variable switching frequency. Switching occurs whenever the current error reaches the upper or lower hysteresis limits, meaning the switching frequency depends on both the hysteresis band width and the current slope.

The current dynamics can be described as:

$$\frac{di}{\mathrm{d}t} = \frac{V - E}{L} \qquad \text{eq.(42)}$$

Where V is the inverter voltage, E is the back-EMF, and L is the stator inductance. When the rate of change of current is high, the hysteresis limits are reached more quickly, leading to higher switching frequency.

The hysteresis band width Δi directly affects the switching behavior:

1. Smaller Δi ➡ higher switching frequency ➡ lower current ripple
2. Larger Δi ➡ lower switching frequency ➡ higher current ripple

In addition, the switching frequency is not constant within one electrical cycle. It is higher near the zero-crossing of the current, where $\frac{di}{dt}$ is large, and lower near the peak of the waveform, where the slope is reduced. This explains the non-uniform switching intervals observed in the simulation results.

The current ripple introduced by the hysteresis control directly affects torque quality. Since electromagnetic torque is proportional to the product of flux and torque current, fluctuations in current result in torque ripple. A smaller hysteresis band improves torque smoothness but increases switching losses, while a larger band reduces switching losses at the expense of higher ripple and harmonic distortion.

Overall, the delta modulation scheme provides excellent dynamic tracking performance but introduces a trade-off between switching frequency, current ripple, and torque quality. Proper selection of the hysteresis band is therefore essential for achieving balanced system performance.

D.2 Slip Frequency Analysis

- Calculate the slip frequency of the induction motor under different operating conditions and analyze its relationship with torque production.

The slip frequency of the induction motor is analyzed under different operating conditions to understand its relationship with torque production in the field-oriented control system.

The slip frequency is defined as the difference between the synchronous electrical speed and the rotor electrical speed:

$$w_{slip} = w_s - w_r \qquad \text{eq.(43)}$$

In field-oriented control, the slip frequency is closely related to the torque-producing current and the rotor flux. For an induction motor, the slip frequency can be expressed as:

$$w_{slip} = \frac{R_r}{L_r} \cdot \frac{i_r}{i_f} \qquad \text{eq.(44)}$$

Since the rotor flux is proportional to the flux-producing current,

$$\lambda_r \propto i_f \qquad \text{eq.(45)}$$

the slip frequency may also be written in proportional form as:

$$\omega_{slip} \propto \frac{i_T}{\lambda_r} \qquad \text{eq.(46)}$$

This expression shows that the slip frequency increases when the torque-producing current increases, and decreases when the rotor flux increases.

Under rated flux and rated torque conditions, the motor operates at a moderate slip frequency because both torque current and flux are maintained at nominal levels. When the torque demand is reduced, the torque-producing current decreases, so the slip frequency also decreases. This corresponds to lighter load operation and lower torque production.

In contrast, under reduced flux operation, the rotor flux is decreased while the torque demand remains high. According to the above relationship, a smaller flux requires a larger torque-producing current to maintain the same torque. As a result, the slip frequency increases. This explains why reduced-flux operation is associated with higher current demand and more stressed operating conditions.

The slip frequency plays an essential role in maintaining correct field orientation. It determines the relative speed between the stator rotating field and the rotor flux. Accurate estimation of slip frequency ensures that the synchronous reference frame remains aligned with the rotor flux vector, which is necessary for decoupled torque and flux control.

Overall, the results confirm that slip frequency is directly linked to torque production and flux level in the induction motor drive. Higher torque demand and lower flux both increase slip frequency, while lower torque demand reduces it. This relationship is fully consistent with the operating principles of field-oriented control.

Conclusions

This project presented the design, implementation, and performance evaluation of an open-loop field-oriented control (FOC) induction motor drive using MATLAB/Simulink.

The system successfully demonstrated the fundamental principle of decoupling torque and flux, allowing independent control of the torque-producing and flux-producing current components.

The results from Part A confirmed the correct implementation of the delta-modulated inverter, current reference generator, and flux and torque estimator. The switching characteristics analysis showed the trade-off between hysteresis band width, switching frequency, and current ripple, while the flux estimation demonstrated stable behavior with expected high-frequency ripple and minimal drift.

In Part B, the system performance was evaluated under different operating conditions. The simulation results showed accurate current tracking, fast torque response, and stable speed behavior. The comparison between different scenarios verified that torque and flux can be independently controlled. Reduced torque operation led to lower current and power demand, while reduced flux operation required higher current to maintain torque, confirming the expected trade-off. The efficiency analysis further showed that system performance is influenced by current magnitude, harmonic content, and operating conditions.

Part D provided additional insight into switching behavior and slip frequency. The analysis confirmed that the delta modulation scheme introduces variable switching frequency and current ripple, which directly affect torque quality and losses. The slip frequency analysis demonstrated its dependence on torque-producing current and rotor flux, highlighting its importance in maintaining proper field orientation.

Overall, the implemented FOC system achieved reliable and stable performance across all tested conditions. The results are consistent with theoretical expectations and validate the effectiveness of field-oriented control for induction motor drives. Future improvements may include the implementation of closed-loop speed control and optimization of switching strategies to further enhance efficiency and dynamic performance.

References

[1] J. Wang, "MSE 491 Project Manual: Design and Evaluation of a Field Oriented Controlled (FOC) Induction Machine Drive" Laboratory Manual, Simon Fraser University, Surrey, BC, Spring 2026.

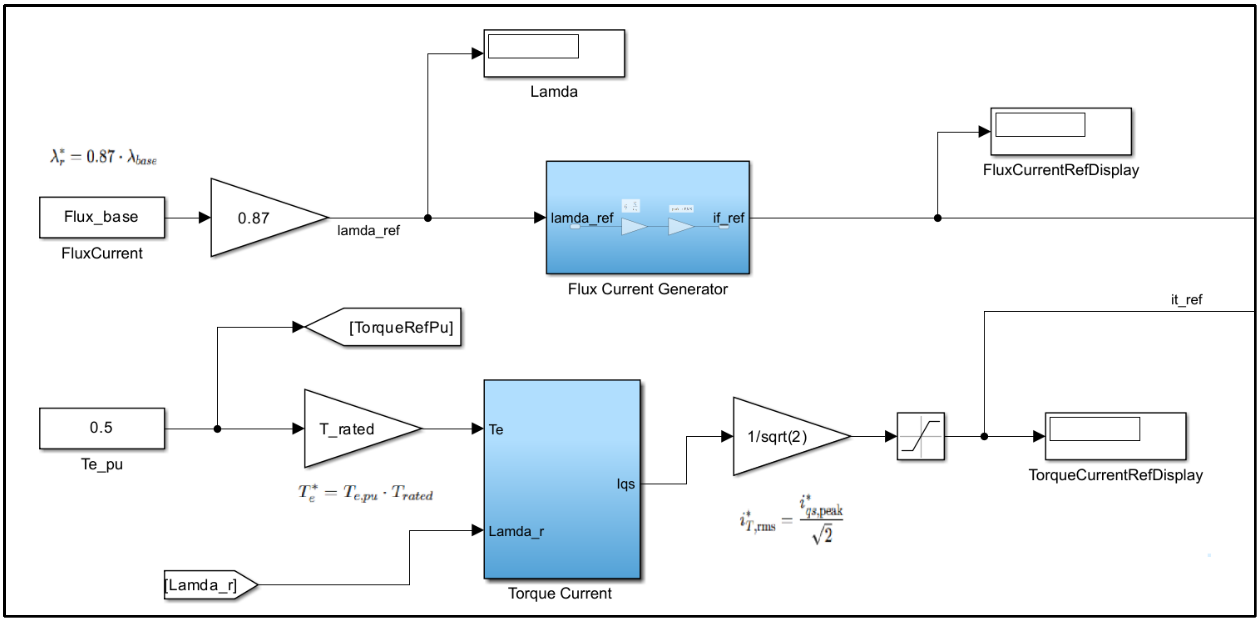
[2] J. Wang, "MSE 491/893: ac Motor Dynamics – Induction Machine Modelling," Lecture Slides, Simon Fraser University, Spring 2026.

[3] J. Hayes and G. A. Goodarzi, Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles. Hoboken, NJ: Wiley, 2018.

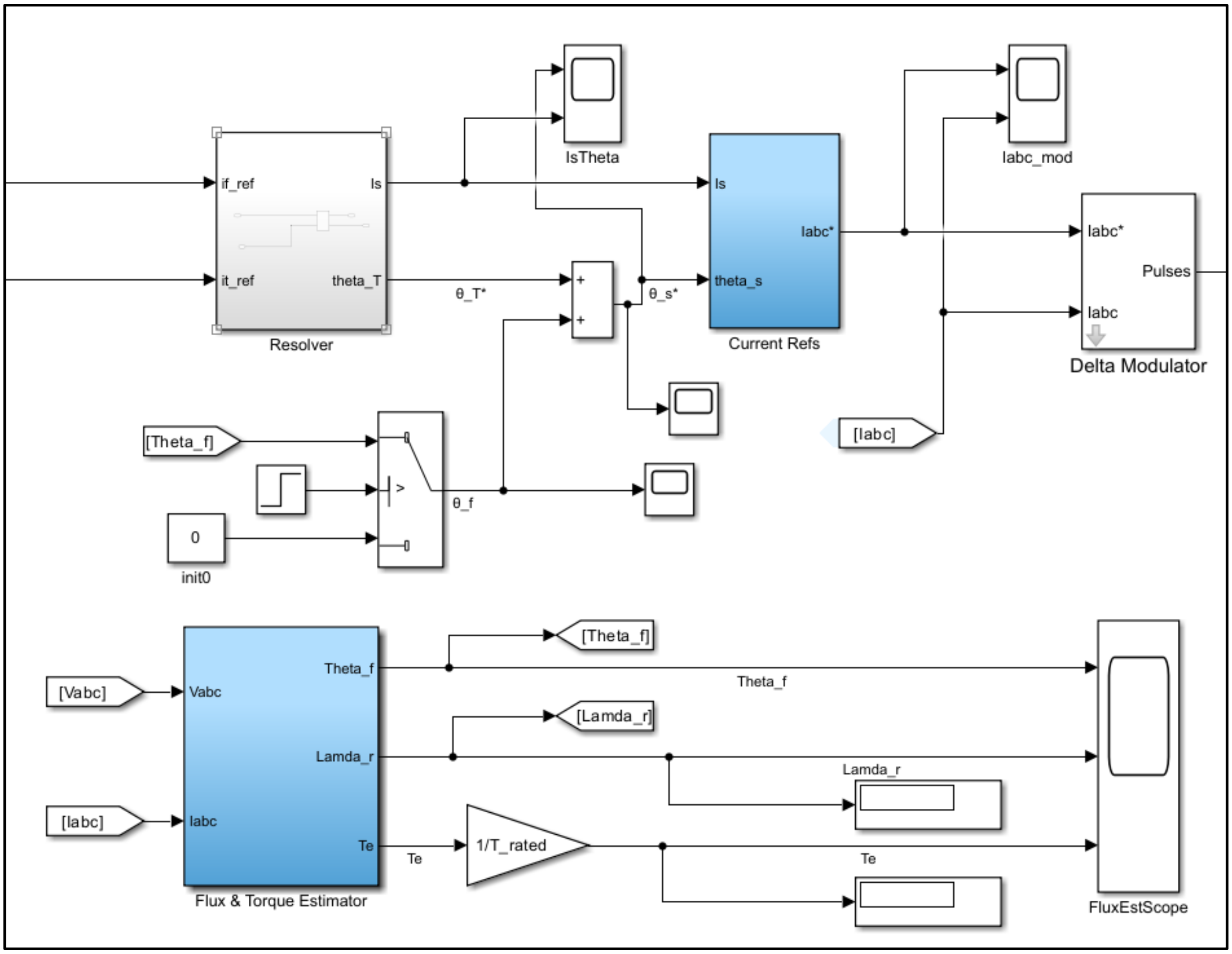
[4] P. C. Krause, O. Wasynczuk, and S. D. Sudhoff, Analysis of Electric Machinery and Drive Systems, 3rd ed. Piscataway, NJ: IEEE Press, 2013.

Appendix: Simulink models

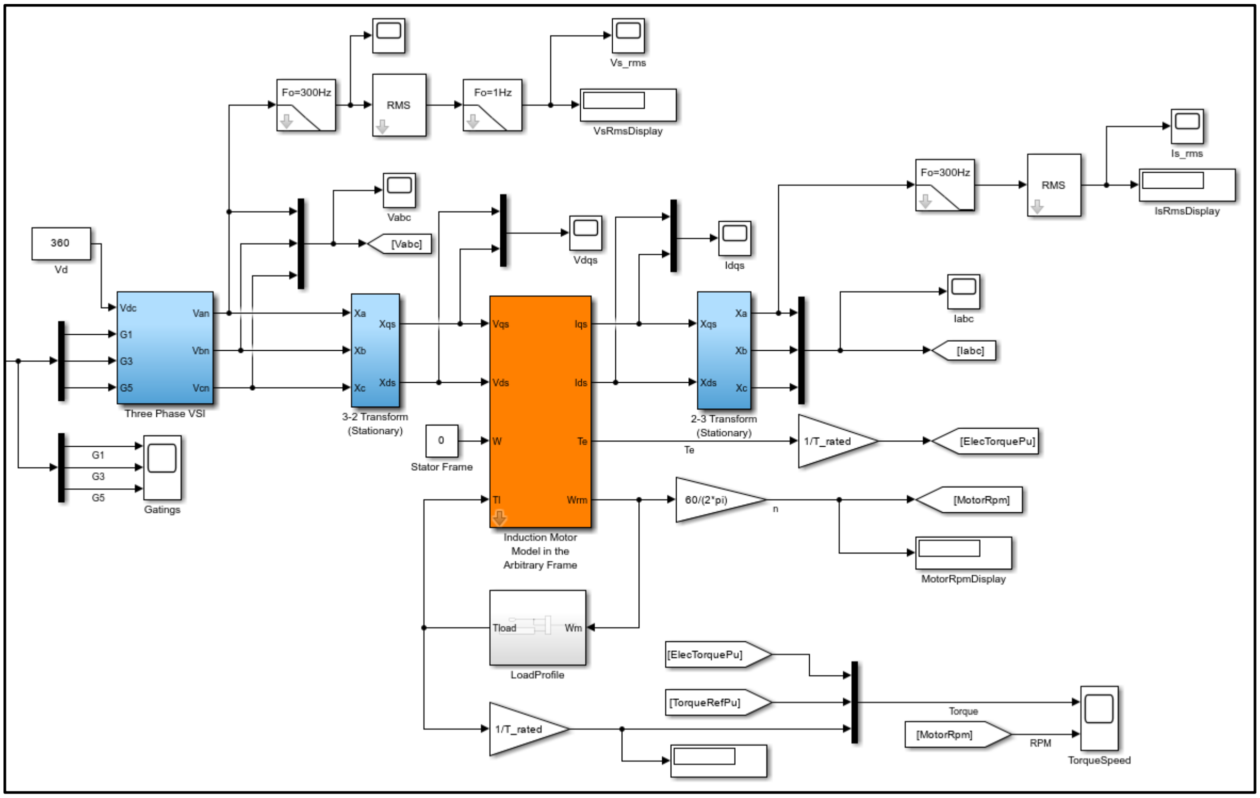
Model Overview Part 1(left):



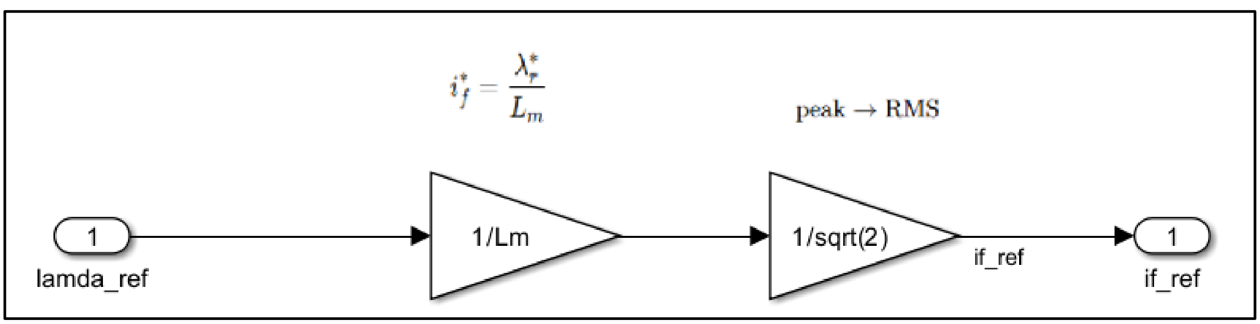
Model Overview Part 2(Middle):



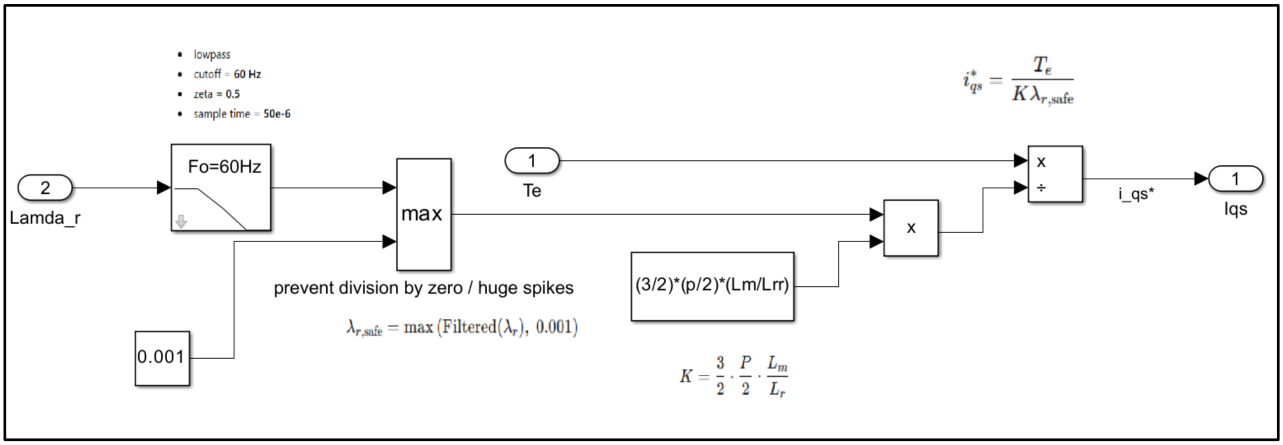
Model Overview Part 3(right):



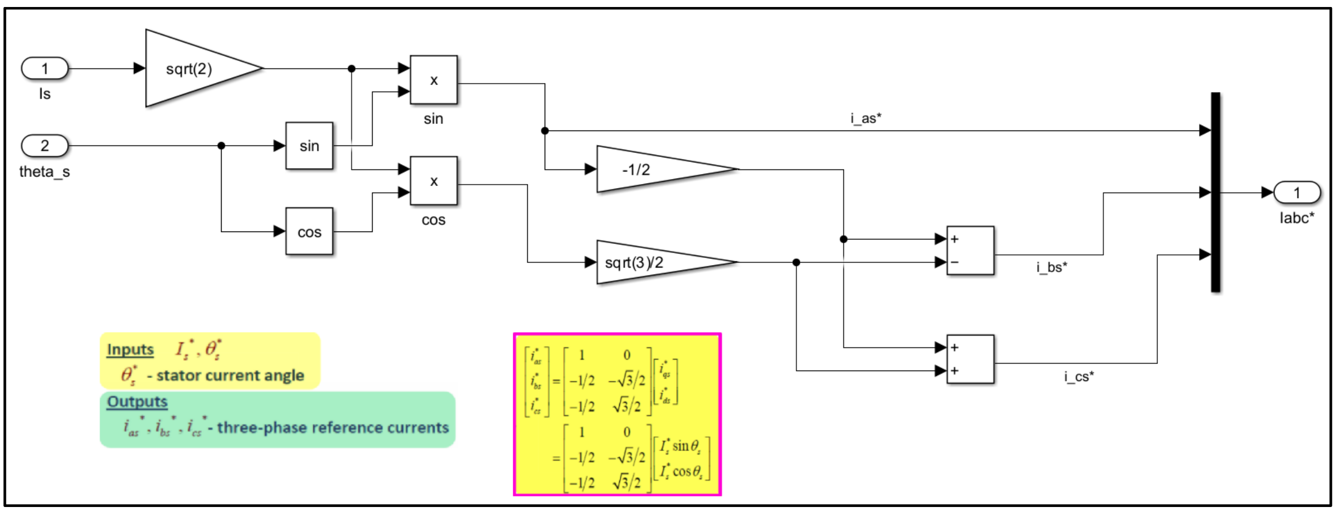
Flux Current Generator:



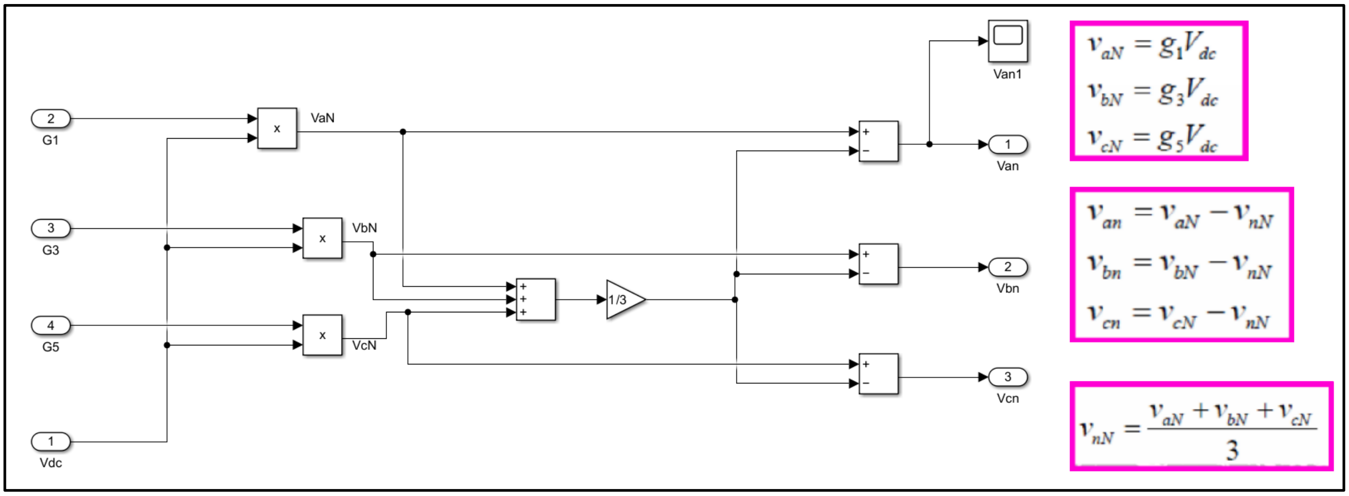
Torque Current Generator:



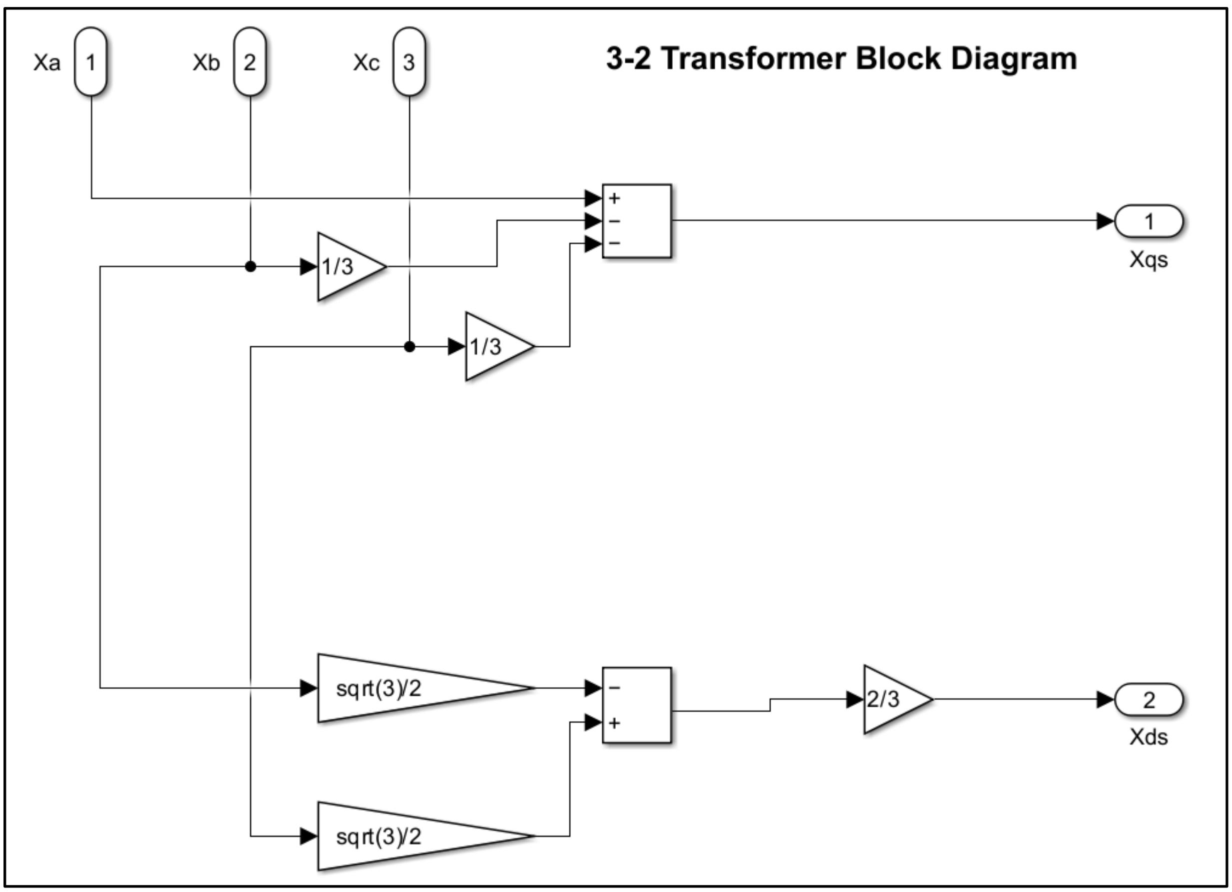
Reference Current Generator:



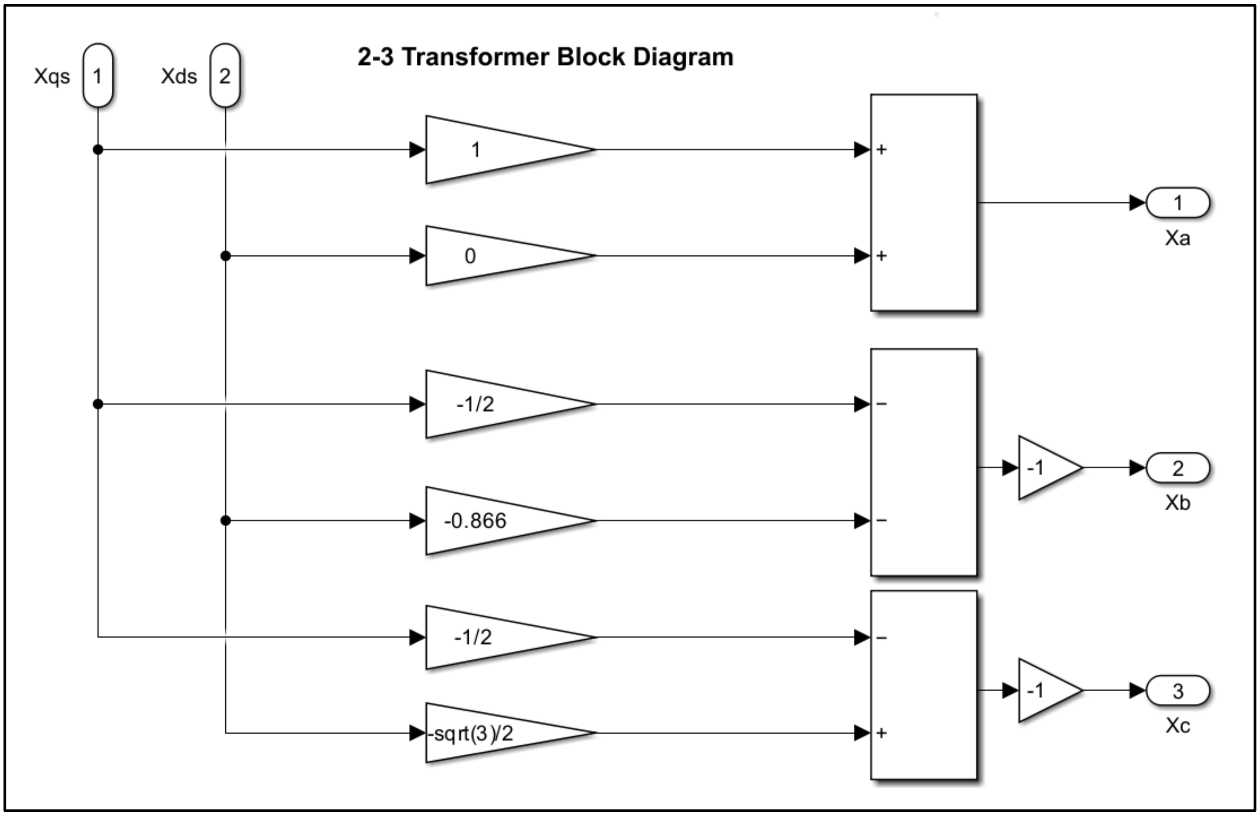
Three Phase VSI:



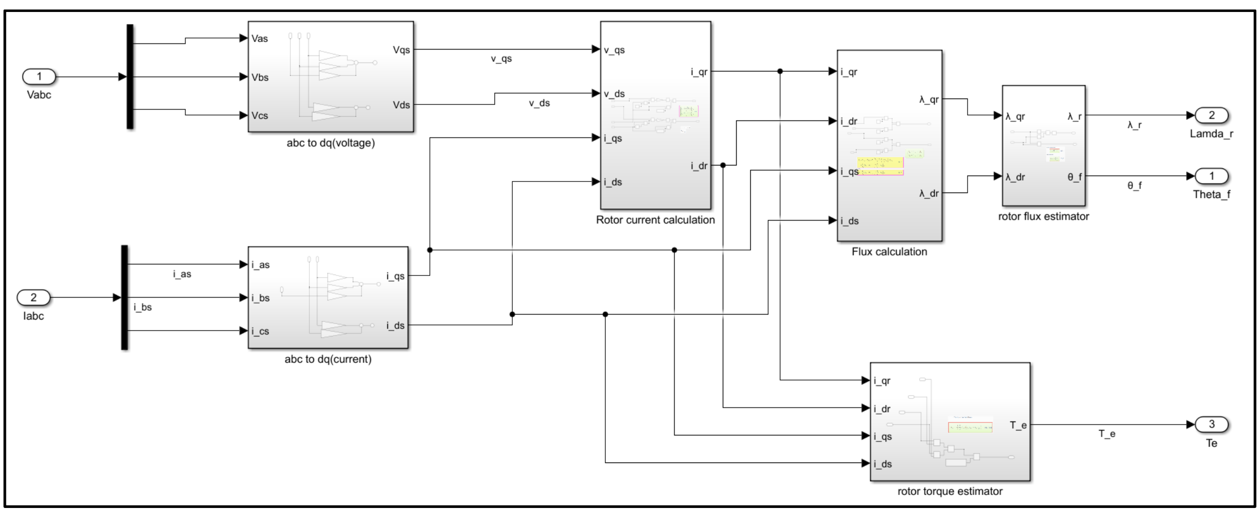
3-2 Transformer(Stationary):



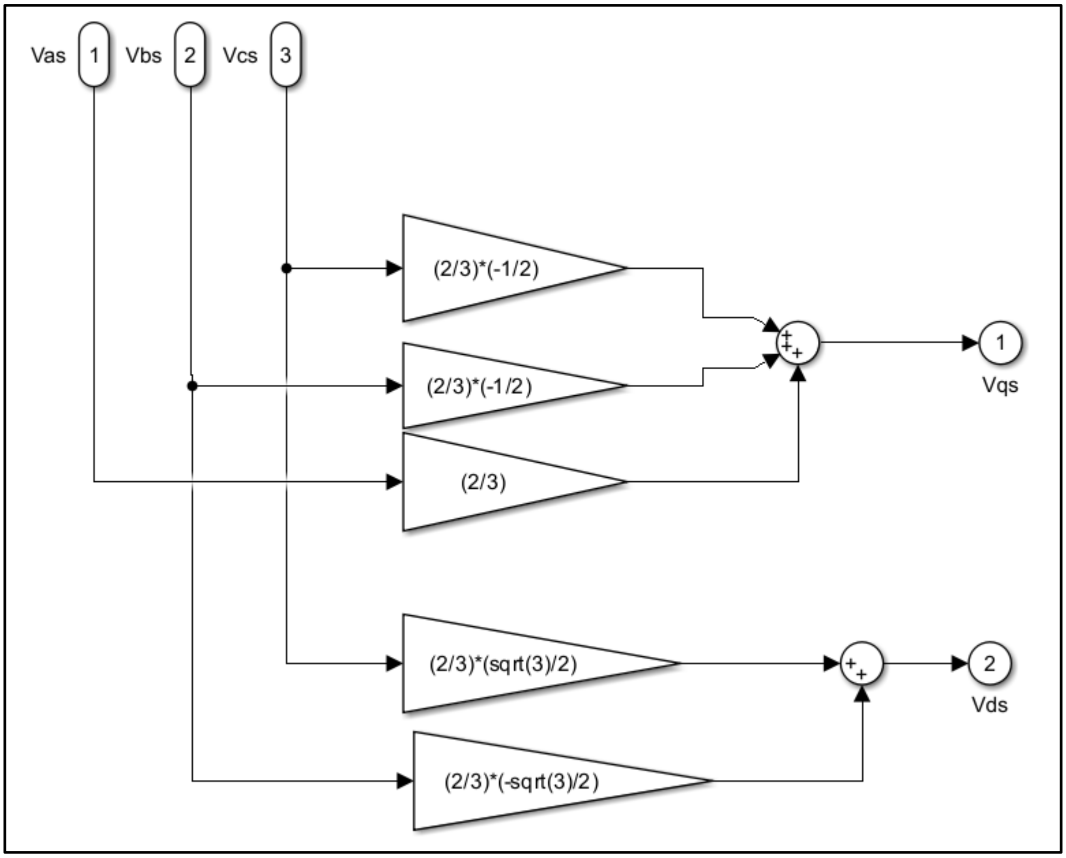
2-3 Transformer(Stationary):



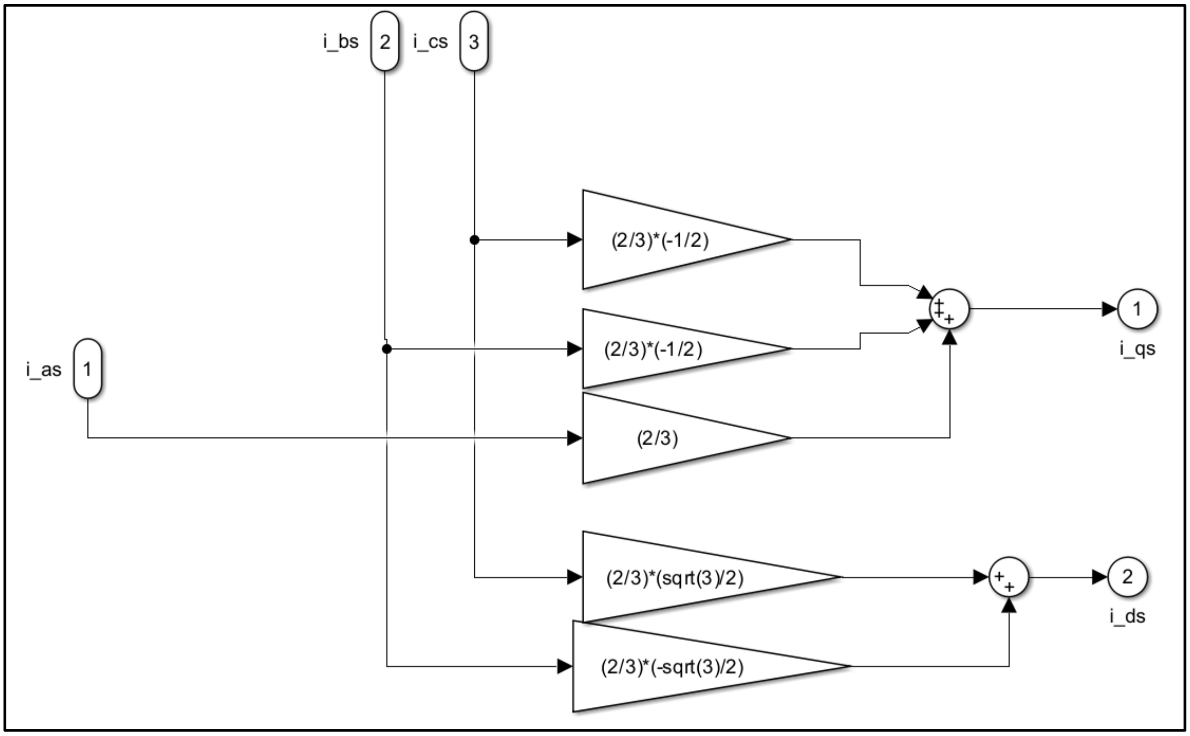
Flux & Torque Estimator:



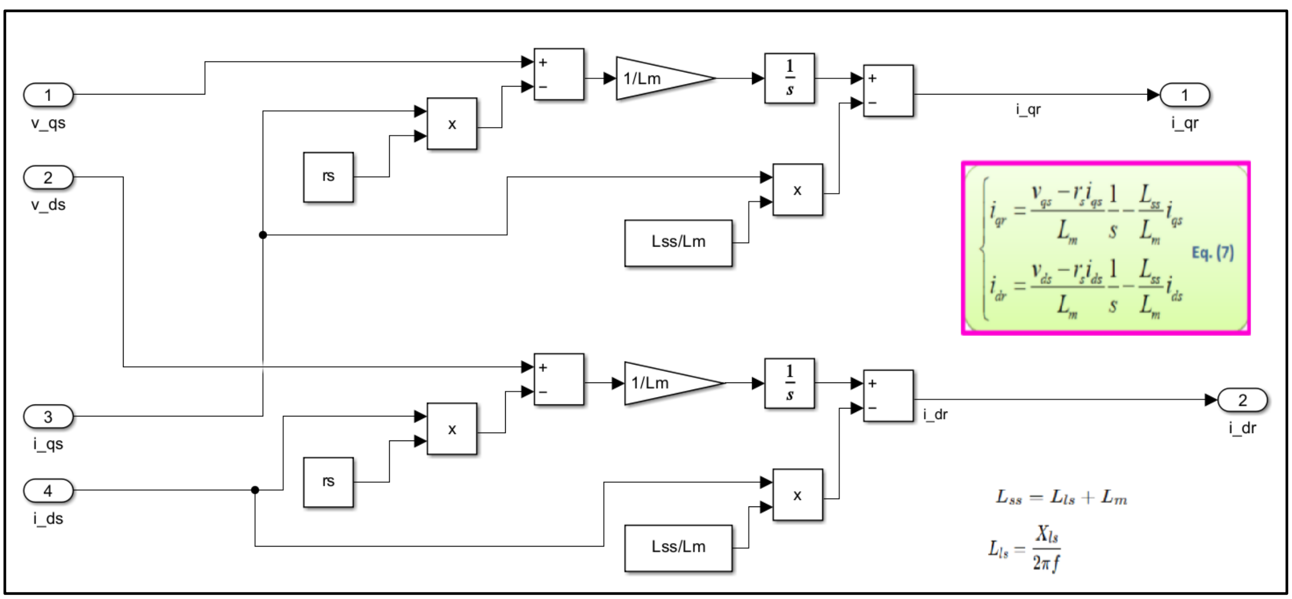
abc to dq(Voltage):



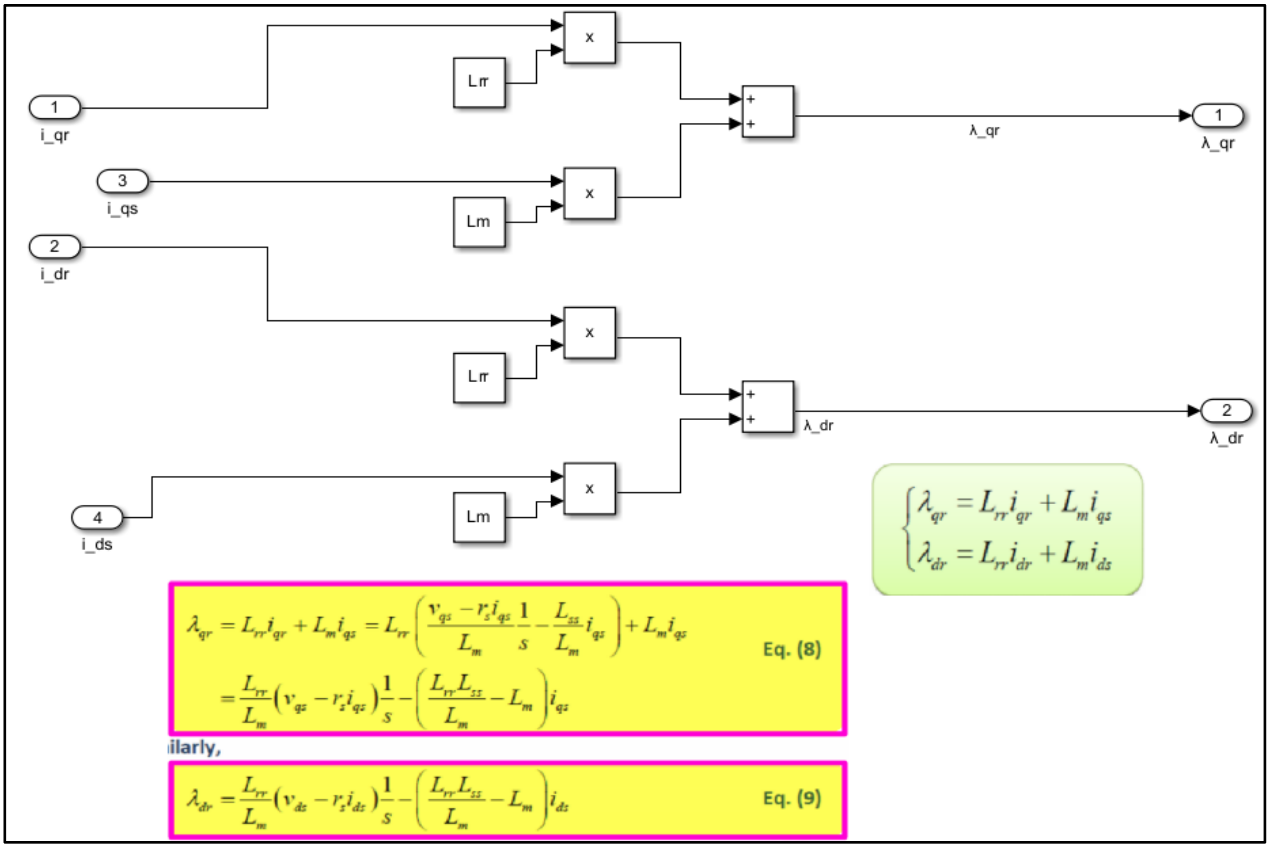
abc to dq(current):



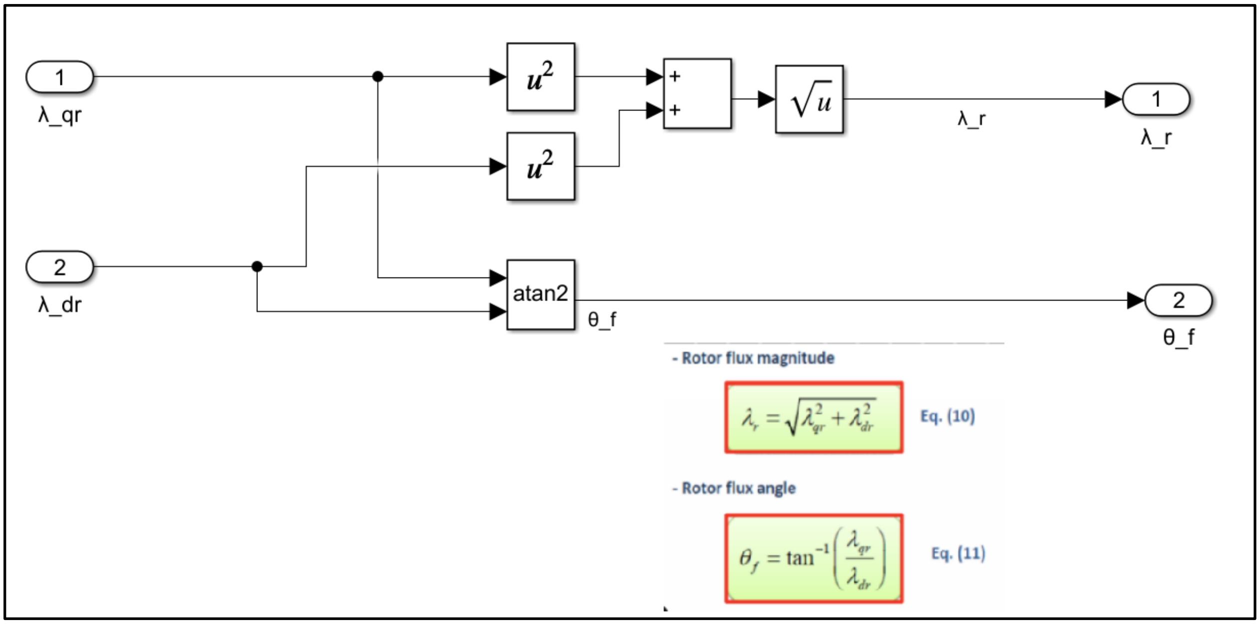
Rotor current calculation:



Flux calculation:



Rotor flux estimator:



Rotor torque estimator: